Chapter Three

FERMAT'S PRINCIPLE AND ITS APPLICATIONS

Now in the further development of science, we want more than just a formula. First we have an observation, then we have numbers that we measure, then we have a law which summarizes all the numbers. But the real glory of science is that we can find a way of thinking such that the law is evident. The first way of thinking that made the law about the behavior of light evident was discovered by Fermat in about 1650, and it is called the principle of least time, or Fermat's principle.

—Richard Feynman in *Feynman Lectures on Physics*, Vol. I

Learning Objectives

After reading this chapter, the reader should be able to:

LO 1: state Fermat's principle and its modified version.
LO 2: derive laws of reflection and refraction from Fermat's principle.
LO 3: describe various phenomena by solving the ray equation.
LO 4: obtain ray paths in inhomogeneous media.
LO 5: calculate the time taken by the rays in propagating through a parabolic index medium.

Important Milestones

140 AD *Greek physicist Claudius Ptolemy measured the angle of refraction in water for different angles of incidence in air and tabulated it.*

1621 *Although the above-mentioned numerical table was made in 140 AD, it was only in 1621 that Willebrord Snell, a Dutch mathematician, discovered the law of refraction which is now known as Snell's law.*

1637 *Descartes derived Snell's law; his derivation assumed corpuscular model of light.*

1657 *Pierre de Fermat enunciated his principle of 'least time' and derived Snell's law of refraction and showed that if the velocity of light in the second medium is less, the ray would bend towards the normal, contrary to what is predicted by the 'corpuscular theory'.*

3.1 INTRODUCTION LO 1

The study of the propagation of light in the realm of geometrical optics employs the concept of rays. To understand what a ray is, consider a circular aperture in front of a point source P as shown in Fig. 3.1. When the diameter of the aperture is quite large (~ 1 cm), then on the screen SS', one can see a patch of light with well-defined boundaries. When we start decreasing the size of the aperture, then at first the size of the patch starts decreasing, but when the size of the aperture becomes very small ($\lesssim$ 0.1 mm) then the pattern obtained on SS' ceases to have well-defined boundaries. This phenomenon is known as diffraction and is a direct consequence of the finiteness of the wavelength (which is denoted by λ). In Chapters 18, 19 and 20 we will discuss the phenomenon of diffraction in great detail and will show that the diffraction effects become smaller with the decrease in wavelength and indeed in the limit of $\lambda \to 0$, the diffraction effects will be absent, and even for extremely small diameters of the aperture, we will obtain a well-defined

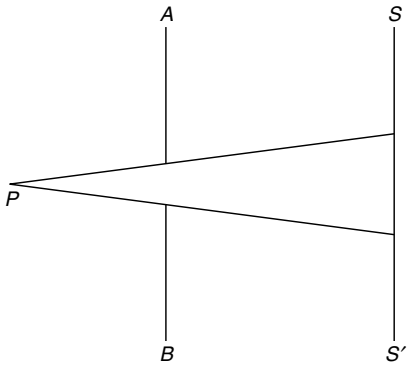

Fig. 3.1 The light emitted by the point source P is allowed to pass through a circular hole and if the diameter of the hole is very large compared to the wavelength of light, then the light patch on the screen SS' has well-defined boundaries.

shadow on the screen SS'. Therefore, in the zero wavelength limit one can obtain an infinitesimally thin pencil of light; this is called a *ray*. Thus, a ray defines the path of propagation of the energy in the limit of the wavelength going to zero. Since light has a wavelength of the order of 10^{-5} cm, which is small compared to the dimensions of normal optical instruments like lenses, mirrors, etc., one can, in many applications, neglect the finiteness of the wavelength. The field of optics under such an approximation (i.e., the neglect of the finiteness of the wavelength) is called geometrical optics.

The field of geometrical optics can be studied by using Fermat's principle which determines the path of the rays. According to this principle, *the ray will correspond to that path for which the time taken is an extremum in comparison to nearby paths*, i.e., it is either a minimum or a maximum or stationary*. Let $n(x, y, z)$ represent the position-dependent refractive index. Then

$$\frac{ds}{c/n} = \frac{n\,ds}{c}$$

will represent the time taken to traverse the geometric path ds in a medium of refractive index n. Here, c represents the speed of light in free space. Thus, if τ represents the total time taken by the ray to traverse the path AB along the curve C (see Fig. 3.2) then

$$\tau = \frac{1}{c}\sum_i n_i\ ds_i = \frac{1}{c}\int_{A\xrightarrow[C]{}B} n\,ds \qquad \textbf{(3.1)}$$

where ds_i represents the i^{th} arc length and n_i the corresponding refractive index; the symbol $A\xrightarrow[C]{}B$ below the integral represents the fact that the integration is from

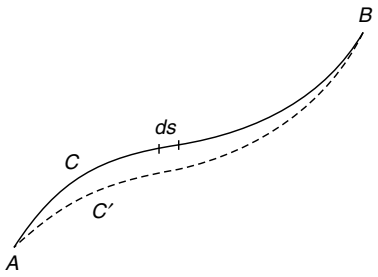

Fig. 3.2 If the path ACB represents the actual ray path then the time taken in traversing the path ACB will be an extremum in comparison to any nearby path $AC'B$.

the point A to B through the curve C. Let τ' be the time taken along the nearby path $AC'B$ (shown as the dashed curve in Fig. 3.2), and if ACB indeed represents the path of a ray, then τ will be either less than, greater than or equal to τ' for *all* nearby paths like $AC'B$. Thus according to Fermat's principle, out of the many paths connecting the two points, the light ray would follow that path for which the time taken is an extremum. Since c is a constant, one can alternatively define a ray as the path for which

$$\int_{A\xrightarrow[C]{}B} n\,ds \qquad \textbf{(3.2)}$$

is an extremum**; the above integral represents the optical path from A to B along C; i.e., the ray would follow the path for which

$$\delta \int_{A\to B} n\,ds = 0 \qquad \textbf{(3.3)}$$

where the left-hand side represents the change in the value of the integral due to an infinitesimal variation of the ray path. We may mention here that according to the original statement of Fermat:

> *The actual path between two points taken by a beam of light is the one which is traversed in the least time.*

The above statement is incomplete and slightly incorrect. The correct form is:

> *The actual ray path between two points is the one for which the optical path length is stationary with respect to variations of the path.*

This is expressed by Eq. (3.3) and in this formulation, the ray paths may be maxima, minima or stationary.

From the above principle one can immediately see that in a homogenous medium (i.e., in a medium whose refractive

* The entire field of classical optics (both geometrical and physical) can be understood from Maxwell's equations, and of course, Fermat's principle can be derived from Maxwell's equations (see Refs. 3.1 and 3.2).

** A nice discussion on the extremum principle has been given in Chapter 26 of Ref. 3.3.

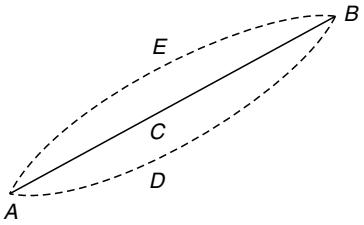

Fig. 3.3 Since the shortest distance between two points is along a straight line, light rays in a homogenous medium are straight lines; all nearby paths like AEB or ADB will take longer times.

index is constant at each point), the rays will be straight lines because a straight line will correspond to a minimum value of the optical path connecting two points in the medium. Thus referring to Fig. 3.3, if A and B are two points in a homogenous medium, then the ray path will be along the straight line ACB because any nearby path like ADB or AEB will correspond to a longer optical path.

3.2 LAWS OF REFLECTION AND REFRACTION FROM FERMAT'S PRINCIPLE LO 2

To obtain the laws of reflection and refraction from Fermat's principle, consider a plane mirror MN as shown in Fig. 3.4. We have to determine the path from A to B (via the mirror) which has the minimum optical path length. Since the path would lie completely in a homogenous medium, we need to minimize only the path length. Thus we have to find that path APB for which $AP + PB$ is a minimum. To find the position of P on the mirror, we drop a perpendicular from A on the mirror

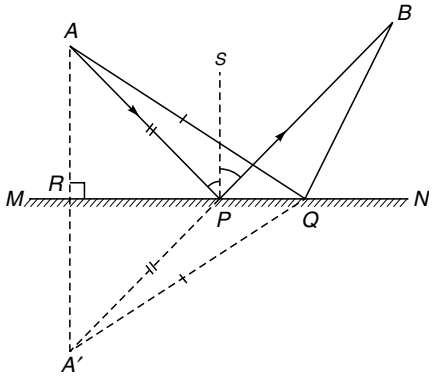

Fig. 3.4 The shortest path connecting the two points A and B via the mirror is along the path APB where the point P is such that AP, PS and PB are in the same plane and $\angle APS = \angle SPB$; PS being the normal to the plane of the mirror. The straight line path AB is *also* a ray.

and let A' be a point on the perpendicular such that $AR = RA'$; thus $AP = PA'$ and $AQ = A'Q$ where AQB is another path adjacent to APB. Thus we have to minimize the length $A'PB$. Clearly, for $A'PB$ to be a minimum, P must be on the straight line $A'B$. Thus, the points A, A', P and B will be in the same plane and if we draw a normal PS at P then this normal will also lie in the same plane. Simple geometric considerations show that

$$\angle APS = \angle SPB$$

Thus for minimum optical path length, the angle of incidence $i(= \angle APS)$ and the angle of reflection r $(= \angle SPB)$ must be equal and the incident ray, the reflected ray and the normal to the surface at the point of incidence on the mirror must be in the same plane. These form the laws of reflection. It should be pointed out that, in the presence of the mirror there will be two ray paths which will connect the points A and B; the two paths will be AB and APB. Fermat's principle tells us that whenever the optical path length is an extremum, we will have a ray, and thus, in general, there may be more than one ray path connecting two points.

To obtain the laws of refraction, let PQ be a surface separating two media of refractive indices n_1 and n_2 as shown in Fig. 3.5. Let a ray starting from the point A, intersect the interface at R and proceed to B along RB. Clearly, for minimum optical path length, the incident ray, the refracted ray and the normal to the interface must all lie in the same plane. To determine that point R for which the optical path length from A to B is a minimum, we drop perpendiculars AM and BN from A and B respectively on the interface PQ. Let $AM = h_1$, $BN = h_2$ and $MR = x$. Then since A and B are fixed, $RN = L - x$, where $MN = L$ is a fixed quantity. The optical path length from A to B, by definition, is

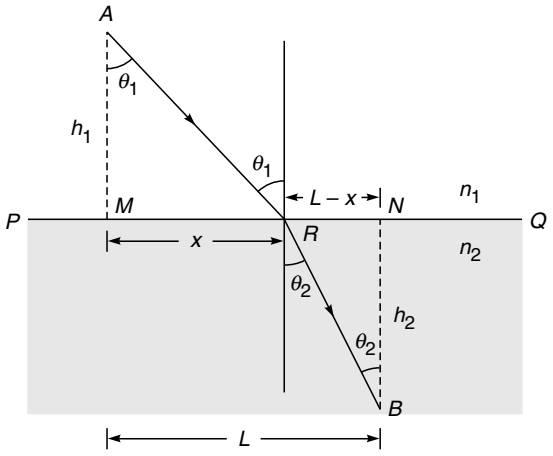

Fig. 3.5 A and B are two points in media of refractive indices n_1 and n_2. The ray path connecting A and B will be such that $n_1 \sin \theta_1 = n_2 \sin \theta_2$.

$$L_{op} = n_1 AR + n_2 RB$$
$$= n_1 \sqrt{x^2 + h_1^2} + n_2 \sqrt{(L-x)^2 + h_2^2} \quad \textbf{(3.4)}$$

To minimize this, we must have

$$\frac{dL_{op}}{dx} = 0$$

i.e., $$\frac{n_1 x}{\sqrt{x^2 + h_1^2}} - \frac{n_2 (L-x)}{\sqrt{(L-x)^2 + h_2^2}} = 0 \quad \textbf{(3.5)}$$

Further, as can be seen from Fig. 3.5

$$\sin \theta_1 = \frac{x}{\sqrt{x^2 + h_1^2}}$$

and $$\sin \theta_2 = \frac{(L-x)}{\sqrt{(L-x)^2 + h_2^2}}$$

Thus, Eq. (3.5) becomes

$$n_1 \sin \theta_1 = n_1 \sin \theta_2 \quad \textbf{(3.6)}$$

which is the Snell's law of refraction.

The laws of reflection and refraction form the basic laws for tracing light rays through simple optical systems, like a system of lenses and mirrors, etc.

Example 3.1 Consider a set of rays, parallel to the axis, incident on a paraboloidal reflector (see Fig. 3.6). Show by using Fermat's principle, that all the rays will pass through the focus of the paraboloid; a paraboloid is obtained by rotating a parabola about its axis. This is the reason why a paraboloidal reflector is used to focus parallel rays from a distant source, like in radio astronomy (see Figs. 3.7 and 3.8).

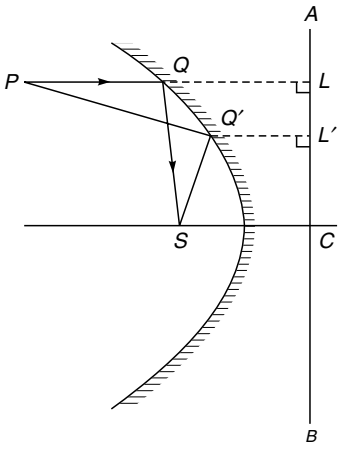

Fig. 3.6 All rays parallel to the axis of a paraboloidal reflector pass through the focus after reflection (the line *ACB* is the directrix). It is for this reason that antennas (for collecting electromagnetic waves) or solar collectors are often paraboloidal in shape.

Fig. 3.7 A paraboloidal satellite dish. [*Photograph courtesy: McGraw-Hill Digital Access Library. A color photograph of the above figure appears as Fig. 1 in the prelim pages*].

Fig. 3.8 Fully steerable 45 m paraboloidal dishes of the Giant Metrewave Radio Telescope (GMRT) in Pune, India. The GMRT consists of 30 dishes of 45 m diameter with 14 antennas in the Central Array. [*Photograph courtesy: Professor Govind Swarup, GMRT, Pune. A color photograph of the above figure appears as Fig. 2 in the prelim pages*].

Solution: Consider a ray PQ, parallel to the axis of the parabola, incident at the point Q (see Fig. 3.6). In order to find the reflected ray, one has to draw a normal at the point Q and then draw the reflected ray. It can be shown from geometrical considerations that the reflected ray QS will always pass through the focus S. However, this procedure will be quite cumbersome and as we will show, the use of Fermat's principle leads us to the desired results immediately.

To use Fermat's principle, we try to find out the ray connecting the focus S and an arbitrary point P (see Fig. 3.6). Let the ray path be $PQ'S$. According to Fermat's principle, the ray path will correspond to a minimum value of $PQ' + Q'S$. From the point Q' we drop a perpendicular $Q'L'$, on the directrix AB. From the definition of the parabola it follows that $Q'L' = Q'S$. Thus,

$$PQ' + Q'S = PQ' + Q'L'$$

Let L be the foot of the perpendicular drawn from the point P on AB. Then, for $PQ' + Q'L'$ to be a minimum, the point Q should lie on the straight line PQL, and thus the actual ray which connects the points P and S will be $PQ + QS$ where PQ is parallel to the axis. Therefore, all rays parallel to the axis will pass through S and conversely, all rays emanating from the point S will become parallel to the axis after suffering a reflection.

Example 3.2 Consider an elliptical reflector whose foci are the points S_1 and S_2 (see Fig. 3.9). Show that all rays emanating from the point S_1 will pass through the point S_2 after undergoing a reflection.

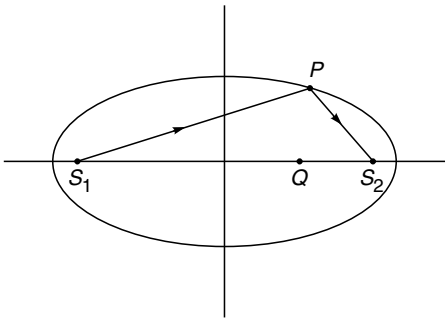

Fig. 3.9 All rays emanating from one of the foci of an ellipsoidal reflector will pass through the other focus.

Solution: Consider an arbitrary point P on the ellipse (see Fig. 3.9). It is well known that $S_1P + S_2P$ is a constant and therefore, all rays emanating from the point S_1 will pass through S_2. (Notice that here we have an example where the time taken by the ray is stationary, i.e., it is neither a maximum nor a minimum but has a constant value for all points lying on the mirror.) As a corollary, we may note the following two points:

(a) Excepting the rays along the axis, no other ray (emanating from either of the foci) will pass through an arbitrary point Q which lies on the axis.
(b) The above considerations will remain valid even for an ellipsoid of revolution obtained by rotating the ellipse about its major axis.

Because of the above-mentioned property of elliptical reflectors, they are often used in laser systems. For example, in a ruby laser (see Sec. 27.3) one may have a configuration in which the laser rod and the flash lamp coincide with the focal lines of a cylindrical reflector of elliptical cross section; such a configuration leads to an efficient transfer of energy from the lamp to the ruby rod.

Example 3.3 Consider a spherical refracting surface SPM separating two media of refractive indices n_1 and n_2 (see Fig. 3.10). The point C represents the center of the spherical surface SPM. Consider two points O and Q such that the points O, C and Q are

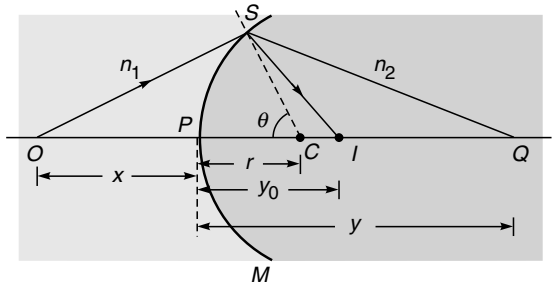

Fig. 3.10 SPM is a spherical refracting surface separating two media of refractive indices n_1 and n_2. C represents the center of the spherical surface.

in a straight line. Calculate the optical path length OSQ in terms of the distances x, y, r and the angle θ (see Fig. 3.10). Use Fermat's principle to find the ray connecting the two points O and Q. Also, assuming the angle θ to be small, determine the paraxial image of the point O.

[*Note*: We reserve the symbol R to represent the radius of curvature of a spherical surface which will be positive (or negative) depending upon whether the center of curvature lies on the right (or left) of the point P. The quantity r represents the magnitude of the radius of curvature which, for Fig. 3.10, happens to be R. Similarly, the quantities x and y are the magnitudes of the distances; the sign convention is discussed later on in this problem.]

Solution: From the triangle SOC, we have

$$OS = [(x + r)^2 + r^2 - 2(x + r)\, r \cos\theta]^{1/2}$$

$$\approx \left[x^2 + 2rx + 2r^2 - 2(xr + r^2)\left(1 - \frac{\theta^2}{2}\right)\right]^{1/2}$$

$$\approx x\left[1 + \frac{rx + r^2}{x^2}\theta^2\right]^{1/2} \approx x + \frac{1}{2}r^2\left(\frac{1}{r} + \frac{1}{x}\right)\theta^2$$

where we have assumed θ (measured in radians) to be small so that we may use the expression

$$\cos\theta \approx 1 - \frac{\theta^2}{2}$$

and also make a binomial expansion. Similarly, by considering the triangle SCQ, we would have

$$SQ \approx y - \frac{1}{2}r^2\left(\frac{1}{r} - \frac{1}{y}\right)\theta^2$$

Thus the optical path length OSQ is given by

$$L_{op} = n_1\, OS + n_2\, SQ$$

$$\approx (n_1 x + n_2 y) + \frac{1}{2}r^2\left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r}\right]\theta^2 \qquad (3.7)$$

For the optical path to be an extremum we must have

$$\frac{dL_{op}}{d\theta} = 0 = r^2\left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r}\right]\theta \qquad (3.8)$$

quantity and the image distance v and the radius of curvature R are positive quantities. For u to be positive, we must have a situation like the one shown in Fig. 4.1(b); in the absence of a refracting surface, the rays converge to a point to the *right* of P.

3. The angle that the ray makes with the axis is positive if the axis has to be rotated in the anticlockwise direction (through the acute angle) to coincide with the ray. Conversely, if the axis has to be rotated in the clockwise direction (through the acute angle) to coincide with the ray, then the slope angle is negative. Thus in Fig. 4.1(a), if θ_1 and θ_2 are the angles that the rays OS and SI make with the axis, then $\theta_1 = \alpha_1$ and $\theta_2 = -\alpha_2$; α_1, α_2 and β represent the magnitudes of the angles. (If the final result does not depend on the angles, then it is more convenient to use the magnitude of the angles as has indeed been done in Sec. 4.2). In Fig. 4.1(b), both θ_1 and θ_2 are negative quantities.
4. The angle that a ray makes with the normal to the surface is positive if the normal has to be rotated in the anticlockwise direction (through the acute angle) to coincide with the ray, and conversely. Thus, in Fig. 4.1 (a), ϕ_1 and ϕ_2 are positive quantities.
5. All distances measured upward from the axis (along a perpendicular to the axis) are positive, and all distances measured in the downward direction are negative.

4.2.2 The Gaussian Formula for a Single Spherical Surface

If we now use the sign convention discussed above, then for the ray diagram shown in Fig. 4.1(a), $u = -x$, $v = y$ and $R = r$. Thus Eq. (4.4) becomes

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \qquad \textbf{(4.5)}$$

which gives the image point due to refraction at a spherical surface (see also Eq. 3.10 of the previous chapter). Equation (4.5) is known as the **Gaussian formula for a single spherical surface.** It should be noted that, corresponding to Fig. 4.1(a), u is negative and v positive, whereas for Fig. 4.1(b) u and v are both positive.

Example 4.1 Consider a medium of refractive index 1.5 bounded by two spherical surfaces $S_1P_1M_1$ and $S_2P_2M_2$ as shown in Fig. 4.2. The radii of curvature of the two surfaces are 15 and 25 cm with their centers at C_1 and C_2, respectively. There is an object at a distance of 40 cm (from P_1) on the line joining C_1 and C_2. Determine the position of the paraxial image.

Solution: We first consider refraction by $S_1P_1M_1$. Obviously $u = -40$ cm, $R = +15$ cm, $n_1 = 1.0$ and $n_2 = 1.5$. Thus,

$$\frac{1.5}{v} + \frac{1}{40} = \frac{0.5}{15} \quad \Rightarrow v = +180 \text{ cm}$$

In the absence of the second surface, the image is formed at O' at a distance of 150 cm from P_2. O' now acts as a virtual object and since it is to the right of $S_2P_2M_2$ we have, while considering refraction by the second surface, $u = +150$ cm, $R = -25$ cm, $n_1 = 1.5$ and $n_2 = 1.0$. Thus,

$$\frac{1.0}{v} - \frac{1.5}{150} = +\frac{0.5}{25}$$

giving

$$v = +33\tfrac{1}{3} \text{ cm}$$

and a real image is formed on the right of P_2 at a distance of $33\frac{1}{3}$ cm.

We should point out that while considering refraction by a single surface (as in Fig. 4.1), the axis of the system is defined by the line joining the object point O and the center of curvature C; thus *any* ray from the point O (like OS) will be in the plane containing the axis and the normal at the point S and consequently, the refracted ray will *always* intersect the axis. On the other hand, if there is second refracting surface (like in Fig. 4.2 or as in a lens) then the line joining the two

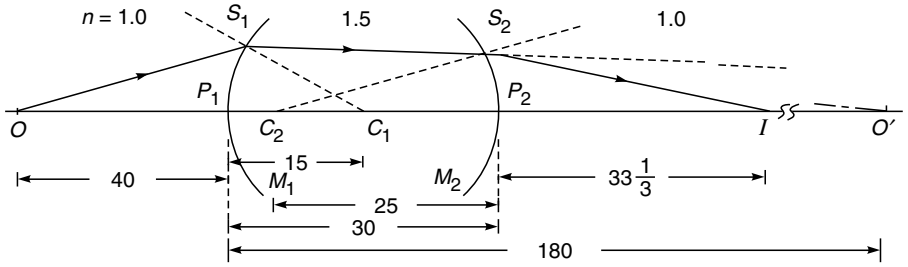

Fig. 4.2 Paraxial image formation by a medium of refractive index 1.5 bounded by two spherical surfaces $S_1P_1M_1$ and $S_2P_2M_2$. All distances are measured in centimeters.

centers of curvature is defined as the axis. In the latter case, as can be readily seen, not all rays from an off-axis point will intersect the axis and after refraction at the second surface will, in general, not remain confined to a single plane; these rays are known as *skew rays*. Rays which remain confined to a plane (containing the axis) are known as *meridional rays*; obviously, all rays emanating from a point on the axis are *meridional rays*.

4.3 REFLECTION BY A SINGLE SPHERICAL SURFACE

We next consider the imaging of a point object O by a spherical mirror SPM (see Fig. 4.3) in the paraxial approximation; the point C represents the center of curvature. We proceed in a manner exactly similar to that in Sec. 4.2 and we refer to Fig. 4.3 to obtain

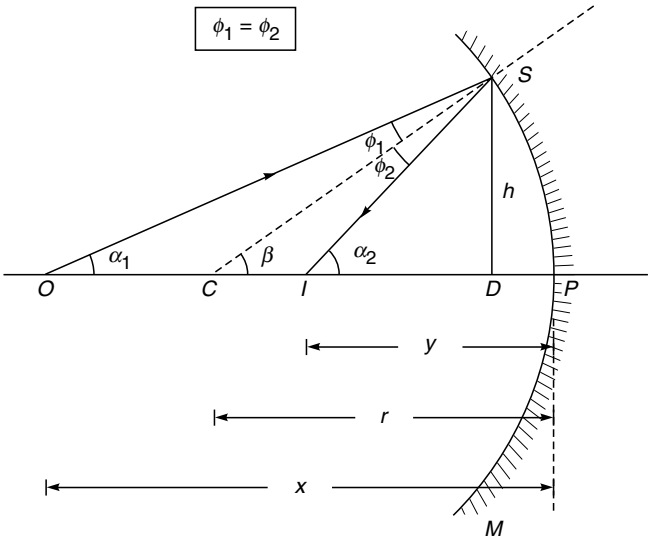

Fig. 4.3 Paraxial image formation by a spherical reflecting surface SPM.

$$\phi_1 = \beta - \alpha_1 \approx \frac{h}{r} - \frac{h}{x}$$

and

$$\phi_2 = \alpha_2 - \beta \approx \frac{h}{y} - \frac{h}{r}$$

where the distances x, y, h and r are defined in Fig. 4.3. Since $\phi_1 = \phi_2$ (the law of reflection), we get

$$\frac{1}{x} + \frac{1}{y} = \frac{2}{r} \qquad \textbf{(4.6)}$$

If we again use the sign convention that all the distances to the right of P are positive and those to its left negative, then $u = -x$, $v = -y$ and $R = -r$; thus we obtain the mirror equation

$$\frac{1}{u} + \frac{1}{v} = \frac{2}{R} \qquad \textbf{(4.7)}$$

which is the same as was derived by using Fermat's principle (see Problem 3.1). It is interesting to note that if we set $n_2 = -n_1$ in Eq. (4.5), we get Eq. (4.7). This follows from the fact that Snell's law of refraction Eq. (4.1) becomes the law of reflection if we have $n_2 = -n_1$.

We illustrate the use of Eq. (4.7) through an example.

Example 4.2 Consider an optical system consisting of a concave mirror $S_1P_1M_1$ and convex mirror $S_2P_2M_2$ of radii of curvatures 60 cm and 20 cm, respectively (see Fig. 4.4). We would like to determine the final image position of the object point O which is at a distance of 80 cm from the point P_1, the two mirrors being separated by a distance of 40 cm.

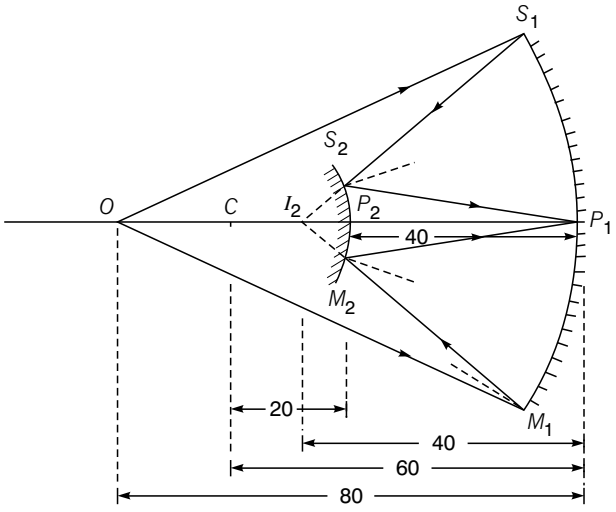

Fig. 4.4 Paraxial image formation by an optical system comprising a concave mirror $S_1P_1M_1$ and a convex mirror $S_2P_2M_2$.

We first consider the imaging by $S_1P_1M_1$; since $u = -80$ cm and $R = -60$ cm (because both O and C are on the left of P_1), we have

$$-\frac{1}{80} + \frac{1}{v} = -\frac{2}{60} \quad \Rightarrow \quad v = -48 \text{ cm}$$

In the absence of the mirror $S_2P_2M_2$, a real image will be formed at I_1 which now acts as a virtual object for $S_2P_2M_2$. Since I_1 is to the left of P_2, we have (considering imaging by $S_2P_2M_2$), $u = -8$ cm and $R = -20$ cm, giving

$$\frac{1}{v} - \frac{1}{8} = -\frac{2}{20} \quad \Rightarrow \quad v = +40 \text{ cm}$$

Thus, the final image is formed on the right of $S_2P_2M_2$ at a distance of 40 cm, which happens to be the point P_1.

4.4 THE THIN LENS LO 2

A medium bounded by two spherical refracting surfaces is referred to as a spherical lens. If the thickness of such a lens (shown as t in Fig. 4.5) is very small compared to object and image distances, and to the radii of curvature of the refracting surfaces then the lens is referred to as a thin spherical lens. In general, a lens may have non-spherical refracting surfaces (e.g., it may have cylindrical surfaces). However, most lenses employed in optical systems have spherical refracting surfaces. Therefore, we will simply use the term 'lens' to imply a spherical lens. Different types of lenses are shown in Fig. 4.6. The line joining the centers of curvature of the spherical refracting surfaces is referred to as the *axis* of the lens.

In this section, we will consider the paraxial image formation by a thin lens. The corresponding considerations for a thick lens will be discussed in Problem 4.6.

We consider a point object O on the axis of a (thin) lens as shown in Fig. 4.5. The lens is placed in a medium of refractive index n_1 and the refractive index of the material of the lens is n_2. Let R_1 and R_2 be the radii of curvature of the left and right surfaces of the lens; for the lens shown in Fig. 4.5, R_1 is positive and R_2 is negative. In order to determine the position of the image, we will consider successive refractions at the two surfaces; the image formed by the first surface is considered as the object (which may be real or virtual) for the second surface. Thus, if the second refracting surface had not been there, the image of the point O would have been formed at Q whose position (given by v') is determined from Eq. (4.8) (see also Eq. (4.5))

$$\frac{n_2}{v'} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} \qquad \textbf{(4.8)}$$

where u is the object distance which is negative for the object point O shown in the figure. Obviously if v' is positive, then the point Q lies to the right of the surface and if v' is negative then Q lies to the left of the surface. The point Q now acts as the (virtual) object for the second refracting surface and the final image is formed at I whose position is determined from the equation

$$\frac{n_1}{v} - \frac{n_2}{v'} = \frac{n_1 - n_2}{R_2} \qquad \textbf{(4.9)}$$

In Eqs. (4.8) and (4.9) the distances are measured from the center of the lens P; this is justified because the lens has been assumed to be thin. Adding Eqs. (4.8) and (4.9), we get

$$\frac{1}{v} - \frac{1}{u} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad \textbf{(4.10)}$$

where

$$n \equiv \frac{n_2}{n_1}$$

Equation (4.10) is known as the *thin lens formula* and is usually written in the form

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \qquad \textbf{(4.11)}$$

where f, known as the focal length of the lens, is given by

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad \textbf{(4.12)}$$

For a lens placed in air (which is usually the case), $n > 1$ and if $[(1/R_1) - (1/R_2)]$ is a positive quantity then the focal length is positive and the lens acts as a converging lens [see Fig. 4.7(a)]. Similarly, if $(1/R_1) - (1/R_2)$ is a negative quantity then the lens acts as a diverging lens (see Fig. 4.7(b)). However, if the double convex lens is placed in a medium whose refractive index is greater than that of the material of the lens, then the focal length becomes negative and the lens acts as a diverging lens [see Fig. 4.7(c)]; similarly for the double concave lens [see Fig. 4.7(d)].

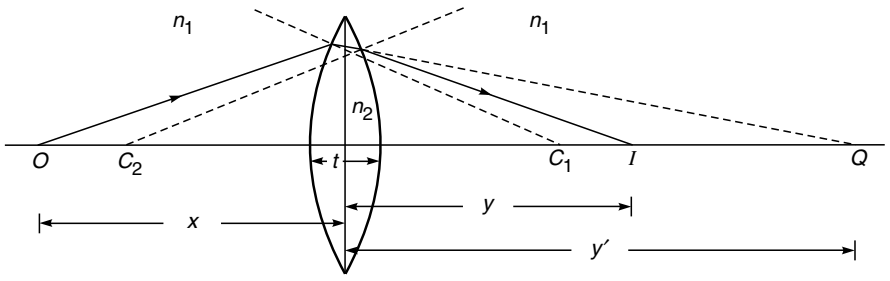

Fig. 4.5 Image formation by a thin lens. The line joining the two centers of curvature is known as the axis of the lens [$u = -x$, $v' = y'$ $v = y$].

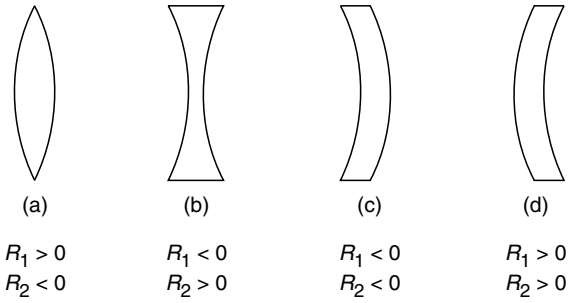

Fig. 4.6 Signs of R_1 and R_2 for different lens types.

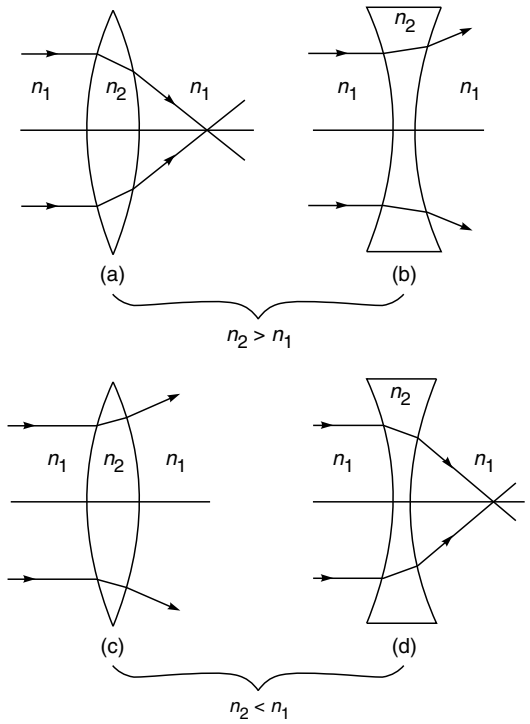

Fig. 4.7 (a) and (b) correspond to the situation when the refractive index of the material of the lens is greater than that of the surroundings and therefore a biconvex lens acts as a converging lens and a biconcave lens acts as a diverging lens. (c) and (d) correspond to the situation when the refractive index of the material of the lens is smaller than that of the surrounding medium and therefore a biconvex lens acts as a diverging lens and a biconcave lens as a converging lens.

4.5 THE PRINCIPAL FOCI AND FOCAL LENGTHS OF A LENS

For a converging lens, the *first principal focus* is defined as the point (on the axis) such that a ray passing through that point will, after refraction through the lens, emerge parallel to the axis—see ray 1 in Fig. 4.8(a); the point F_1 is the first principal focus. For a diverging lens, the ray which (in the absence of the lens) would have passed through the first principal focus emerges, after refraction by the lens, as a ray parallel to the axis—see ray 1 in Fig. 4.8(b). The point F_1 is the **first principal focus** and its distance from the lens (denoted by f_1) is known as the **first focal length** of the lens. Obviously, f_1 is negative for a converging lens and positive for a diverging lens.

We next consider a ray which travels parallel to the axis [see ray 2 in Figs. 4.8(a) and (b)]. For a converging lens the point at which the ray will intersect the axis [shown as F_2 in Fig. 4.8(a)] is known as the **second principal focus** of the lens. Similarly, for a diverging lens, the point at which the ray would have intersected the axis (if produced backwards) is the second principal focus [see the point F_2 in Fig. 4.8(b)]. The distance of the second principal focus from the lens is known as the **second focal length** and is denoted by f_2. As can be seen from Fig. 4.8, f_2 is positive for a converging lens and negative for a diverging lens.

For a thin lens placed in a medium such that the refractive indices on both sides of the lens are the same ($n_3 = n_1$ in Fig. 4.8), the values of f_1 and f_2 can be readily obtained by considering the thin lens formula [see Eq. (4.10)] and one gets

$$\frac{1}{f_2} = -\frac{1}{f_1} = \left[(n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)\right] = \frac{1}{f} \qquad \textbf{(4.13)}$$

However, if $n_3 \neq n_1$ then the thin lens formula assumes the form as given in Eq. (4.14) (see also Problem 4.2):

$$\frac{n_3}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2} \qquad \textbf{(4.14)}$$

Now, when $v = \infty$, $u = f_1$ (ray 1 in Fig. 4.8), we have

$$\frac{1}{f_1} = -\frac{1}{n_1}\left[\frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2}\right] \qquad \textbf{(4.15)}$$

Similarly, when $u = -\infty$, $v = f_2$ (ray 2 in Fig. 4.8) we have

$$\frac{1}{f_2} = \frac{1}{n_3}\left[\frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2}\right] \qquad \textbf{(4.16)}$$

Once we know f_1 and f_2 (and therefore the positions of the first and second principal foci) the (paraxial) image can be graphically constructed from the following rules:

(i) A ray passing through the first principal focus will, after refraction, emerge parallel to the axis [see ray 1 in Figs. 4.8(a) and (b)]

(ii) A ray parallel to the axis will, after refraction, either pass through or appear to come from (depending on the sign of f_2) the second principal focus [see ray 2 in Figs. 4.8(a) and (b)]

(iii) A ray passing through the center of the lens P will pass through undeviated* [see ray 3 in Figs. 4.8(a) and (b)]

4.6 THE NEWTON FORMULA

LO 3

Let x_1 be the distance of the object from the first principal focus F_1 (x_1 will be positive if the object point is on the right of F_1 and conversely) and let x_2 be the distance of the image from the second principal focus F_2 as shown in Figs. 4.8(a) and (b). Considering similar triangles in Fig. 4.8(a), we have

$$\frac{-y'}{y} = \frac{-f_1}{-x_1} \qquad \textbf{(4.17)}$$

and

$$\frac{-y'}{y} = \frac{x_2}{f_2} \qquad \textbf{(4.18)}$$

where the vertical distances are positive if measured above the line and negative if measured below the line (see Sec. 4.2.1). Equations (4.17) and (4.18) give

$$f_1 f_2 = x_1 x_2 \qquad \textbf{(4.19)}$$

which is known as the *Newtonian lens formula*. It may be noted that for a diverging lens [see Fig. 4.8(b)], Eqs. (4.17) and (4.18) would be

$$\frac{y'}{y} = \frac{f_1}{-x_1} = \frac{x_2}{-f_2}$$

which are identical to Eqs. (4.17) and (4.18).

When the thin lens has the same medium on the two sides, then using Eq. (4.13) we have

$$x_1 x_2 = -f^2 \qquad \textbf{(4.20)}$$

showing that x_1 and x_2 must be of opposite signs. Thus, if the object lies on the left of the first principal focus, then the image will lie on the right of the second principal focus, and vice versa.

4.7 LATERAL MAGNIFICATION

The lateral magnification m is the ratio of the height of the image to that of the object. Considering either Fig. 4.8(a) or (b) we readily get

$$m = \frac{y'}{y} = \frac{v}{u} = \frac{f_2 + x_2}{f_1 + x_1} = -\frac{f_1}{x_1} = -\frac{x_2}{f_2} \qquad \textbf{(4.21)}$$

where we have made use of Eqs. (4.17) and (4.18). Obviously, if m is positive, the image is erect [as in Fig. 4.8(b) and conversely if m is negative, the image is inverted as in Fig. 4.8(a)].

The magnification can also be calculated as the product of the individual magnifications produced by each of the

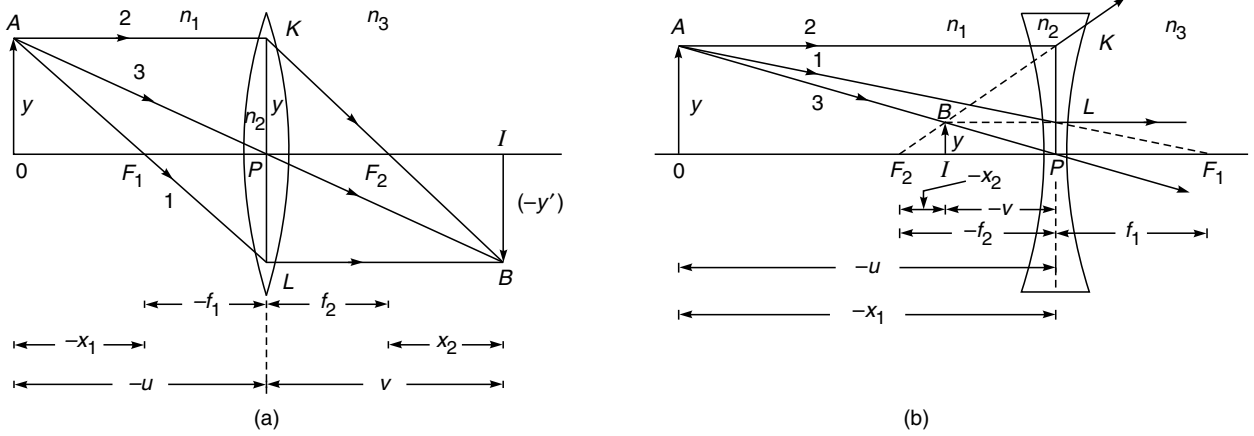

Fig. 4.8 (a) Paraxial imaging by a converging lens; x_1, f_1 and u are negative quantities and x_2, f_2, and v are positive quantities. (b) Paraxial imaging by a diverging lens; here x_1, f_2, u and v are negative quantities and x_2 and f_1 are positive quantities.

* This follows from the fact that, for a thin lens, when $u = 0$, v is also equal to zero [see Eqs. (4.10) and (4.14)].

refracting surfaces; referring to Fig. 4.9, the magnification produced by a single refracting surface is given by

$$m = \frac{y'}{y}$$

and considering triangles AOC and ICB, we get

$$\frac{-y'}{y} = \frac{v - R}{-u + R} = \frac{\frac{v}{R} - 1}{-\frac{u}{R} + 1} \qquad \textbf{(4.22)}$$

Now, Eq. (4.5) gives us

$$\frac{n_2}{n_1} - \frac{v}{u} = \frac{n_2 - n_1}{n_1}\frac{v}{R}$$

and

$$\frac{u}{v} - \frac{n_1}{n_2} = \frac{n_2 - n_1}{n_2}\frac{u}{R}$$

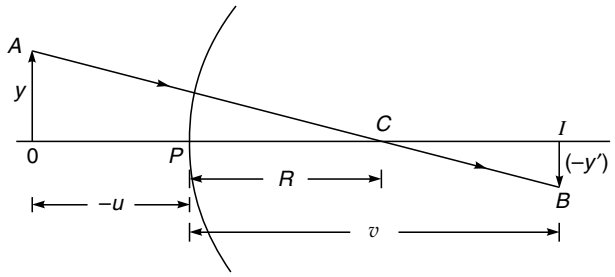

Fig. 4.9 Imaging of an object of height y by a spherical refracting surface.

Substituting for v/R and u/R in Eq. (4.22), we get

$$m = \frac{y'}{y} = \frac{n_1 v}{n_2 u} \qquad \textbf{(4.23)}$$

Thus, if m_1 and m_2 represent the magnifications produced by the two refracting surfaces in Fig. 4.8, then

$$m = \frac{n_1}{n_2}\frac{v'}{u}$$

and

$$m_2 = \frac{n_2}{n_1}\frac{v}{v'}$$

where v' represents the distance of the image formed by the first refracting surface. Thus,

$$m = m_1 m_2 = \frac{v}{u} \qquad \textbf{(4.24)}$$

consistent with Eq. (4.21).

Example 4.3 Consider a system of two thin lenses as shown in Fig. 4.10. The convex lens has a focal length of +20 cm and the concave lens has a focal length of –10 cm. The two lenses are separated by 8 cm. For an object of height 1 cm (at a distance of 40 cm from the convex lens), calculate the position and size of the image. (The same problem will be solved again in Chapter 5 by using the matrix method.)

Solution: Let us first calculate the position and size of the image formed by the first lens:

$$u = -40 \text{ cm}, f = +20 \text{ cm}$$

Therefore, using Eq. (4.11), we get

$$\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = -\frac{1}{40} + \frac{1}{20} = +\frac{1}{40}$$

Thus, $v = +40$ cm and $m_1 = -1$; the image is of the same size but inverted. This image acts as a virtual object for the concave lens with $u = +32$ cm and $f = -10$ cm. Thus,

$$\frac{1}{v} = \frac{1}{32} - \frac{1}{10} = -\frac{22}{320}$$

giving

$$v \simeq -14.5 \text{ cm}$$

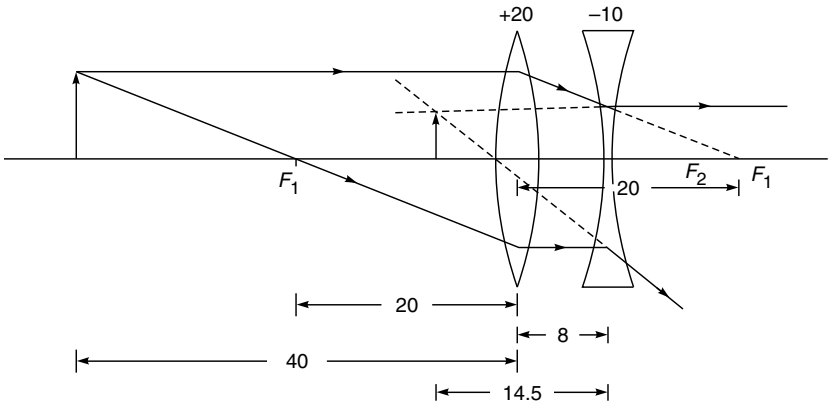

Fig. 4.10 Paraxial imaging by an optical system consisting of a converging lens of focal length 20 cm and a diverging lens of focal length –10 cm separated by 8 cm. All distances in the figure are in centimeters.

13.5 SUPERPOSITION OF TWO SINUSOIDAL WAVES

Let us consider the superposition of two sinusoidal waves (having the same frequency) at a particular point. Let

$$\left.\begin{aligned} x_1(t) &= a_1 \cos(\omega t + \theta_1) \\ x_2(t) &= a_2 \cos(\omega t + \theta_2) \end{aligned}\right\} \quad \text{and} \quad \textbf{(13.20)}$$

represent the displacements produced by each of the disturbances: we are assuming that the displacements are in the same direction*, however, they may have different amplitudes and different initial phases. In Sec. 17.5 now, according to the superposition principle the resultant displacement $x(t)$ would be given by

$$\begin{aligned} x(t) &= x_1(t) + x_2(t) \\ &= a_1 \cos(\omega t + \theta_1) + a_2 \cos(\omega t + \theta_2) \end{aligned} \quad \textbf{(13.21)}$$

which can be written in the form

$$x(t) = a \cos(\omega t + \theta) \quad \textbf{(13.22)}$$

where

$$a \cos\theta = a_1 \cos\theta_1 + a_2 \cos\theta_2 \quad \textbf{(13.23)}$$

and

$$a \sin\theta = a_1 \sin\theta_1 + a_2 \sin\theta_2 \quad \textbf{(13.24)}$$

Thus, the resultant disturbance is also simple harmonic in character having the same frequency but different amplitude and different initial phase. If we square and add Eqs. (13.23) and (13.24), we would obtain

$$a = [a_1^2 + a_2^2 + 2a_1a_2 \cos(\theta_1 - \theta_2)]^{1/2} \quad \textbf{(13.25)}$$

Further

$$\tan\theta = \frac{a_1 \sin\theta_1 + a_2 \sin\theta_2}{a_1 \cos\theta_1 + a_2 \cos\theta_2} \quad \textbf{(13.26)}$$

Though angle θ is not uniquely determined from Eq. (13.26); however, if we assume a to be always positive, then cos θ and sin θ can be determined from Eqs. (13.23) and (13.24) which will uniquely determine θ. From Eq. (13.25) we find that if

$$\theta_1 \sim \theta_2 = 0, 2\pi, 4\pi, \ldots \quad \textbf{(13.27)}$$

then

$$a = a_1 + a_2 \quad \textbf{(13.28)}$$

Thus, if the two displacements are in phase, then the resultant amplitude will be the sum of the two amplitudes; this is known as **constructive interference**. Similarly, if

$$\theta_1 \sim \theta_2 = \pi, 3\pi, 5\pi, \ldots \quad \textbf{(13.29)}$$

then

$$a = a_1 \sim a_2 \quad \textbf{(13.30)}$$

and the resultant amplitude is the difference of the two amplitudes. This is known as **destructive interference**. If we refer to Fig. 13.2, then we can see that constructive interference occurs at $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \ldots$ (i.e., at the points $P', Q', R', \ldots$) and destructive interference occurs at $x = 0, \lambda/2, \lambda, 3\lambda/2, \ldots$ (i.e., at points $A, P, Q, R, \ldots$). It may be mentioned that when constructive and destructive interferences occur, there is no violation of the principle of conservation of energy; the energy is just redistributed.

In general, if we have n displacements

$$\left.\begin{aligned} x_1 &= a_1 \cos(\omega t + \theta_1) \\ x_2 &= a_2 \cos(\omega t + \theta_2) \\ &\cdots\cdots\cdots\cdots\cdots\cdots \\ x_n &= a_n \cos(\omega t + \theta_n) \end{aligned}\right\} \quad \textbf{(13.31)}$$

then

$$x = x_1 + x_2 + \ldots + x_n = a \cos(\omega t + \theta) \quad \textbf{(13.32)}$$

where

$$a \cos\theta = a_1 \cos\theta_1 + \ldots + a_n \cos\theta_n \quad \textbf{(13.33)}$$

and

$$a \sin\theta = a_1 \sin\theta_1 + \ldots + a_n \sin\theta_n \quad \textbf{(13.34)}$$

13.6 THE GRAPHICAL METHOD FOR STUDYING SUPERPOSITION OF SINUSOIDAL WAVES

In this section, we will discuss the graphical method for adding displacements of the same frequency. This method is particularly useful when we have a large number of superposing waves as it indeed happens when we consider the phenomenon of diffraction.

* In Sec. 17.5, we will consider the superposition of waves having nearly equal frequencies which leads to the phenomenon of beats. Indeed in Sec. 13.2, while discussing stationary waves on a string, we had, at a particular value of x, two sinusoidal waves of the same frequency (but having different initial phases) superposing on each other. However, in general, one could have superposition of displacements which are in different directions; for example, the superposition of two linearly polarized waves to produce a circularly polarized wave (see Sec. 22.4).

Let us first try to obtain the resultant of the two displacements given by Eq. (13.20) using the graphical method. We draw a circle of radius a_1 and let the point P on the circle be such that OP makes an angle θ_1 with the x-axis* (see Fig. 13.8). We next draw a circle of radius a_2 and let the point Q on the circle be such that OQ makes an angle θ_2 with the x-axis. We use the law of parallelograms to find the resultant $\overrightarrow{OR}$ of the vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$. The length of the vector $\overrightarrow{OR}$ will represent the amplitude of the resultant displacement and if θ is the angle that OR makes with the x-axis, then the initial phase of the resultant will be θ. This can be easily seen by noting that

$$OR \cos\theta = OP \cos\theta_1 + PR \cos\theta_2 = a_1 \cos\theta_1 + a_2 \cos\theta_2 \qquad \textbf{(13.35)}$$

Similarly,

$$OR \sin\theta = a_1 \sin\theta_1 + a_2 \sin\theta_2 \qquad \textbf{(13.36)}$$

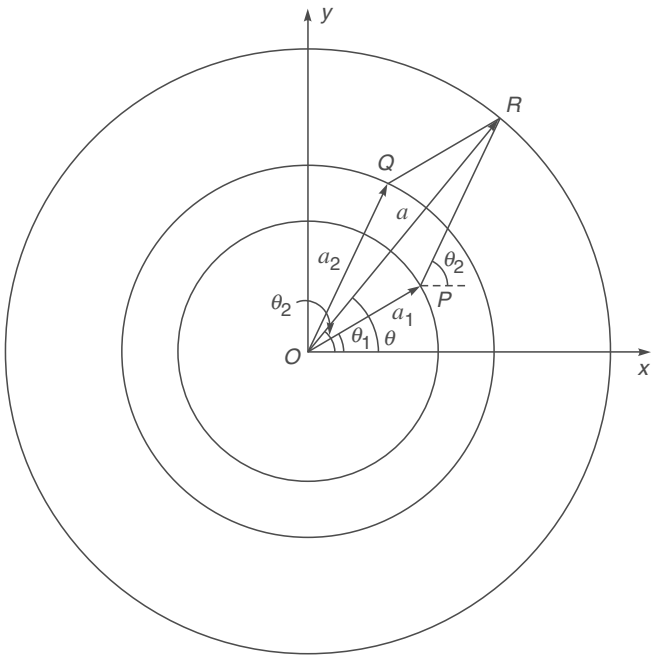

Fig. 13.8 The graphical method for determining the resultant of two simple harmonic motions along the same direction and having the same frequency.

consistent with Eqs. (13.23) and (13.24). Further, as the vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ rotate on the circumference of the circles of radii a_1 and a_2, the vector $\overrightarrow{OR}$ rotates on the circumference of the circle of radius OR with the same frequency.

Thus, if we wish to find the resultant of the two displacements given by Eq. (13.20) then we must first draw a vector ($\overrightarrow{OP}$) of length a_1 making an angle θ_1 with the axis; from the tip of this vector we must draw another vector ($\overrightarrow{PR}$) of length a_2 making an angle θ_2 with the axis. The length of the vector $\overrightarrow{OR}$ will represent the resultant amplitude and the angle that it makes with the axis will represent the initial phase of the resultant displacement. It can be easily seen that if we have a third displacement

$$x_3 = a_3 \cos(\omega t + \theta_3) \qquad \textbf{(13.37)}$$

then from the point R we must draw a vector $\overrightarrow{RR'}$ of length a_3 which makes an angle θ_3 with the axis; the vector $\overrightarrow{OR'}$ will represent the resultant of x_1, x_2 and x_3.

As an illustration of the above procedure we consider the resultant of N simple harmonic motions all having the same amplitude and with their phases increasing in arithmetic progression. Thus,

$$\begin{aligned} x_1 &= a \cos \omega t \\ x_2 &= a \cos[\omega t + \theta_0] \\ &\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\ x_N &= a \cos[\omega t + (N-1)\theta_0] \end{aligned} \qquad \textbf{(13.38)}$$

In Fig. 13.9, the vectors $\overrightarrow{OP_1}$, $\overrightarrow{P_1P_2}$, $\overrightarrow{P_2P_3}$, ..., correspond to x_1, x_2, x_3,..., respectively. The resultant is denoted by the

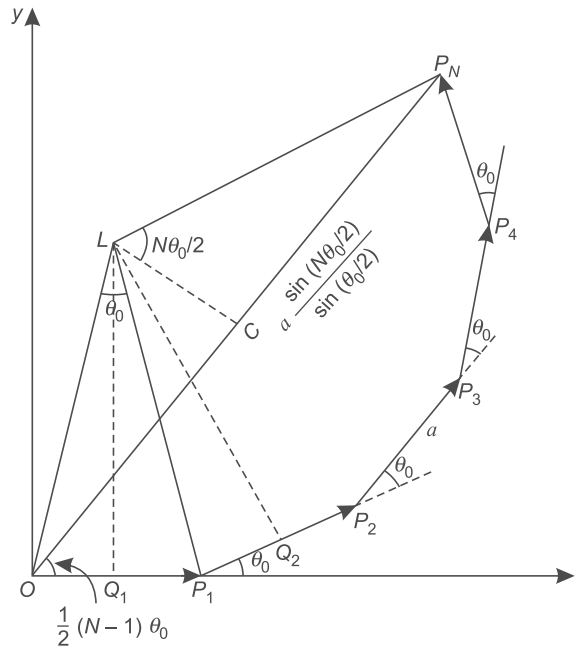

Fig. 13.9 The graphical method for determining the resultant of N simple harmonic motions along the same direction and having the same frequency.

* Clearly, if we assume the vector $\overrightarrow{OP}$ to rotate (in the anticlockwise direction) with angular velocity ω then the x-coordinate of the vector $\overrightarrow{OP}$ will be $a_1 \cos(\omega t + \theta_1)$ where $t = 0$ corresponds to the instant when the rotating vector is at the point P.

Chapter Fourteen

TWO BEAM INTERFERENCE BY DIVISION OF WAVEFRONT

The wave nature of light was demonstrated convincingly for the first time in 1801 by Thomas Young by a wonderfully simple experiment…He let a ray of sunlight into a dark room, placed a dark screen in front of it, pierced with two small pinholes, and beyond this, at some distance a white screen. He then saw two darkish lines at both sides of a bright line, which gave him sufficient encouragement to repeat the experiment, this time with spirit flame as light source, with a little salt in it, to produce the bright yellow sodium light. This time he saw a number of dark lines, regularly spaced; the first clear proof that light added to light can produce darkness. This phenomenon is called interference. Thomas Young had expected it because he believed in the wave theory of light.

—Dennis Gabor in his Nobel Lecture, December 11, 1971

Thomas Young had amazing broad interests and talents … From his discoveries in medicine and science, Helmholtz concluded: 'His was one of the most profound minds that the world has ever seen.'

—From the Internet

Learning Objectives

After reading this chapter, the reader should be able to:

LO 1: explain the interference pattern produced on water surface.
LO 2: understand the coherence phenomenon.
LO 3: discuss interference patterns produced by light waves.
LO 4: understand Young's interference pattern and obtain expression for fringe width.
LO 5: discuss intensity distribution in the interference pattern.
LO 6: illustrate Fresnel's two-mirror arrangement and biprism to produce interference pattern.
LO 7: describe interference pattern when the slit is illuminated by white light.
LO 8: discuss displacement of fringes in the interference pattern when a thin sheet is introduced in one of the beams.
LO 9: illustrate Lloyd's mirror arrangement for producing interference patterns.
LO 10: understand the phase change that occurs when a light beam undergoes reflection.

14.1 INTRODUCTION

In the previous chapter, we had considered the superposition of one-dimensional waves propagating on a string and had shown that there is a variation of energy density along the length of the string due to the interference of two waves (see Fig. 13.5). In general, whenever two waves superpose, one obtains an intensity distribution which is known as the interference pattern. In this chapter, we will consider the interference pattern produced by waves emanating from two point sources. It may be mentioned that with sound waves the interference pattern can be observed without much difficulty because the two interfering waves maintain a constant phase relationship; this is also the case for microwaves. However, for light waves, due to the very process of emission, one cannot observe interference between the waves from two independent sources,* although the interference does take place (see Sec. 14.4). Thus, one tries to derive interfering

* It is difficult to observe the interference pattern even with two laser beams unless they are phase locked.

waves from a single wave so that the phase relationship is maintained. The methods to achieve this can be classified under two broad categories. Under the first category, in a typical arrangement, a beam is allowed to fall on two closely spaced holes and the two beams emanating from the holes interfere. This method is known as **division of wavefront** and will be discussed in detail in this chapter. In the other method, known as **division of amplitude**, a beam is divided at two or more reflecting surfaces and the reflected beams interfere. This will be discussed in the next chapter. We must, however, emphasize that the present and the following chapters are based on one underlying principle, namely the superposition principle.

It is also possible to observe interference using multiple beams; this is known as **multiple beam interferometry** and will be discussed in Chapter 16. It will be shown that multiple beam interferometry offers some unique advantages over two beam interferometry.

14.2 INTERFERENCE PATTERN PRODUCED ON THE SURFACE OF WATER LO 1

We consider surface waves emanating from two point sources in a water tank. We may have, for example, two sharp needles vibrating up and down at the points S_1 and S_2 (see Fig. 14.1). Although water waves are not strictly transverse, we will, for the sake of simplicity, assume water waves to produce displacements which are transverse to the direction of propagation.

If there was only one needle (say at S_1) vibrating with a certain frequency v then circular ripples would have spread out from the point S_1. The wavelength would have been v/v and the crests and troughs would have moved outwards (see Fig. 16 in the prelim pages). Similarly for the vibrating needle at S_2. However, if both needles are vibrating, then waves emanating from S_1 will interfere with the waves emanating from S_2. We assume that the needle at S_2 vibrates in phase with the needle at S_1, i.e., S_1 and S_2 go up simultaneously, they also reach the lowest position at the same time. Thus, if at a certain instant, the disturbance emanating from the source S_1 produced a crest at a distance ρ from S_1 then the disturbance from S_2 would also produce a crest at a distance ρ from S_2, etc. This is explicitly shown in Fig. 14.1, where the solid curves represent (at a particular instant) the positions of the crests due to disturbances emanating from S_1 and S_2. Similarly, the dashed curves represent (at the same instant) the positions of the troughs. Notice that at all points on the perpendicular bisector OZ the disturbances reaching

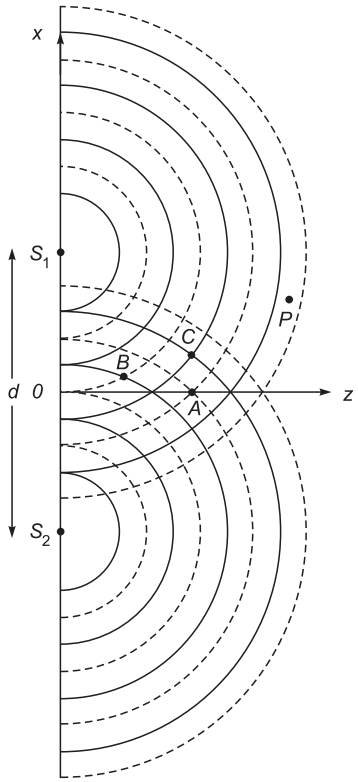

Fig. 14.1 Waves emanating from two pointsources S_1 and S_2 vibrating in phase. The solid and the dashed curves represent the positions of the crests and troughs, respectively.

from S_1 and from S_2 will always be in phase. Consequently, at an arbitrary point A (on the perpendicular bisector) we may write the resultant disturbance as

$$y = y_1 + y_2 = 2a \cos \omega t \qquad \textbf{(14.1)}$$

where y_1 (= $a \cos \omega t$) and y_2 (= $a \cos \omega t$) represent the displacements at the point A due to S_1 and S_2, respectively. We see that the amplitude at A is twice the amplitude produced by each one of the source. It should be noted that at $t = \frac{T}{4}\left(=\frac{1}{4v}=\frac{\pi}{2\omega}\right)$ the displacements produced at the point A by each of the source would be zero and the resultant will also be zero. This is also obvious from Eq. (14.1).

Next, let us consider a point B such that

$$S_2B - S_1B = \lambda/2 \qquad \textbf{(14.2)}$$

At such a point the disturbance reaching from the source S_1 will always be out of phase with the disturbance reaching from S_2. This follows from the fact that the disturbance reaching the point B from the source S_2 must have started half a period (= $T/2$) earlier than the disturbance reaching B

from S_1. Consequently, if the displacement at B due to S_1 is given by

$$y_1 = a \cos \omega t$$

then the displacement at B due to S_2 would be given by

$$y_2 = a \cos (\omega t - \pi) = -a \cos \omega t$$

and the resultant $y = y_1 + y_2$ is zero at all times. Such a point corresponds to destructive interference and is known as a node and corresponds to minimum intensity. It may be pointed out that the amplitudes of the two vibrations reaching the point B will not really be equal as it is at different distances from S_1 and S_2. However, if the distances involved are large (in comparison to the wavelength), the two amplitudes will be very nearly equal and the resultant intensity will be very nearly zero.

In a similar manner, we may consider a point C such that

$$S_2C - S_1C = \lambda$$

where the phase of the vibrations (reaching from S_1 and S_2) are exactly the same as at the point A. Consequently, we will again have constructive interference. In general, if a point P is such that

$$S_2P \sim S_1P = n\lambda \text{ (maxima)} \qquad \textbf{(14.3)}$$

$n = 0, 1, 2, \ldots$, then the disturbances reaching the point P from the two sources will be in phase, the interference will be constructive and the intensity will be maximum. On the other hand, if the point P is such that

$$S_2P \sim S_1P = \left(n + \frac{1}{2}\right)\lambda \text{ (minima)} \qquad \textbf{(14.4)}$$

then the disturbances reaching the point P from the two sources will be out of phase, the interference will be destructive and the intensity will be minimum. The actual interference pattern produced from two point sources vibrating in phase in a ripple tank is shown in Fig. 14.2.

Example 14.1 The intensity at the point which neither satisfies Eq. (14.3) nor Eq. (14.4) will neither be a maximum nor zero. Consider a point P such that $S_2P - S_1P = \lambda/3$. Find the ratio of the intensity at the point P to that at a maximum.

Solution: If the disturbance reaching the point P from S_1 is given by

$$y_1 = a \cos \omega t$$

then the disturbance from S_2 would be given by

$$y_2 = a \cos \left(\omega t - \frac{2\pi}{3}\right)$$

because a path difference of $\lambda/3$ corresponds to a phase difference of $\frac{2\pi}{3}$. Thus the resultant displacement would be

$$y = y_1 + y_2$$

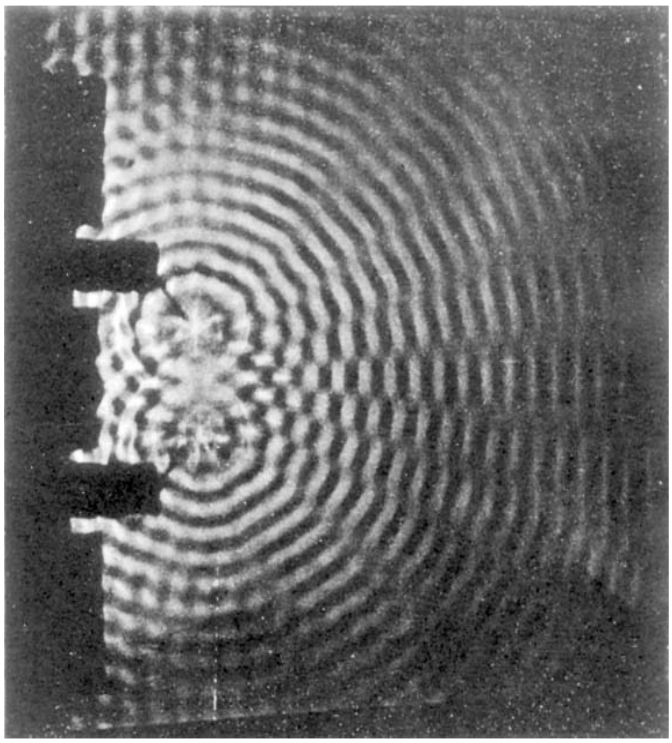

Fig. 14.2 The actual interference pattern produced from two point sources vibrating in phase in a ripple tank (After Ref. 14.9, used with permission).

$$= a\left[\cos \omega t + \cos\left(\omega t - \frac{2\pi}{3}\right)\right]$$

$$= 2a \cos\left(\omega t - \frac{\pi}{3}\right)\cos\frac{\pi}{3}$$

$$= a \cos\left(\omega t - \frac{\pi}{3}\right)$$

The intensity is therefore 1/4th of the intensity at the maxima. In a similar manner, one can calculate the intensity at any other point.

Example 14.2 The locus of points which correspond to minima are known as nodal lines. Show that the equation of a nodal line is a hyperbola. Also obtain the locus of points which correspond to maxima.

Solution: For the sake of generality, we find the locus of the point P which satisfies the following equation:

$$S_2P - S_1P = \Delta \tag{14.5}$$

Thus, if $\Delta = n\lambda$, we have a maximum and if $\Delta = \left(n + \frac{1}{2}\right)\lambda$ we have a minimum. We choose the midpoint of S_1S_2 as the origin, with the x-axis along S_1S_2 and the z-axis perpendicular to it (see Figs. 14.1 and 14.3). We will find the locus of the point P on the x-z plane. If the distance between S_1 and S_2 is d, then the coordinates of the points S_1 and S_2 are $\left(+\frac{d}{2}, 0\right)$ and $\left(-\frac{d}{2}, 0\right)$, respectively. Let the coordinates of the point P [see Fig. 14.3] be (x, z). Then,

$$S_1P = \left[\left(x - \frac{d}{2}\right)^2 + z^2\right]^{1/2}$$

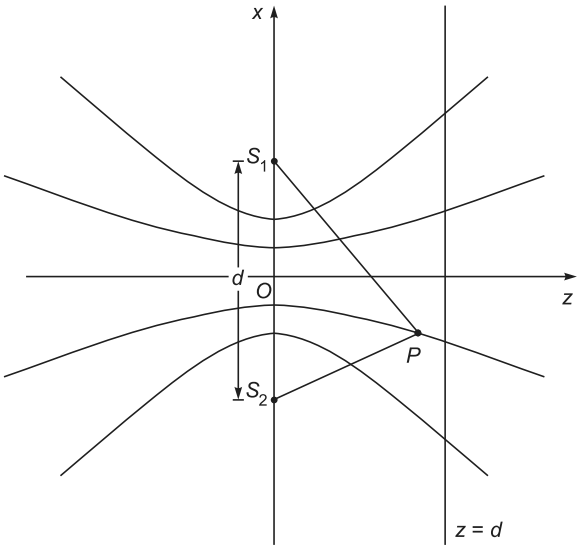

Fig. 14.3 The nodal curves.

and

$$S_2P = \left[\left(x + \frac{d}{2}\right)^2 + z^2\right]^{1/2}$$

Therefore,

$$S_2P - S_1P = \left[\left(x + \frac{d}{2}\right)^2 + z^2\right]^{1/2} - \left[\left(x - \frac{d}{2}\right)^2 + z^2\right]^{1/2} = \Delta$$

or

$$\left(x + \frac{d}{2}\right)^2 + z^2 = \left(x - \frac{d}{2}\right)^2 + z^2 + \Delta^2 + 2\Delta\left[\left(x - \frac{d}{2}\right)^2 + z^2\right]^{1/2}$$

or

$$2xd - \Delta^2 = 2\Delta\left[\left(x - \frac{d}{2}\right)^2 + z^2\right]^{1/2}$$

On squaring, and simplifying we obtain

$$\frac{x^2}{\frac{1}{4}\Delta^2} - \frac{z^2}{\frac{1}{4}(d^2 - \Delta^2)} = 1 \tag{14.6}$$

which is the equation of a hyperbola. When $\Delta = \left(n + \frac{1}{2}\right)\lambda$ the curve correspond to minima and when $\Delta = n\lambda$ the curve correspond to maxima. For large value of x and z the curve asymptotically tend to the straight lines

$$z = \pm\left(\frac{d^2 - \Delta^2}{\Delta^2}\right)^{1/2} x \tag{14.7}$$

It may be pointed out that $S_2P \sim S_1P$ equals d only on the x-axis and that there is no point P for which $S_2P \sim S_1P > d$. Now, it appears from Eq. (14.6) that when $\Delta > d$, the resulting equation is an ellipse which we know is impossible. The fallacy is a result of the fact that because of a few squaring operations, Eq. (14.6) also represents the locus of all those points for which $S_2P + S_1P = \Delta$ and obviously in this case Δ can exceed d.

Example 14.3 Consider a line parallel to the x-axis at a distance D from the origin (see Fig. 14.3). Show that for path differences small compared to d, the points on the line where minimum intensity occurs are equally spaced.

Solution: The equation of the line would be

$$z = D \tag{14.8}$$

Further at large distances from the origin the equation of the nodal lines would be

$$z = \pm\left(\frac{d^2 - \Delta_n^2}{\Delta_n^2}\right)^{1/2} x \qquad \textbf{(14.9)}$$

where $\Delta_n = \left(n + \frac{1}{2}\right)\lambda$; $n = 0, 1, 2,\ldots$Clearly the points at which minima will occur (on the line $z = D$) would be given by

$$x_n = \pm\left(\frac{\Delta_n^2}{d^2 - \Delta_n^2}\right)^{1/2} D$$

$$= \pm\frac{\Delta_n}{d}\left(1 - \frac{\Delta_n^2}{d^2}\right)^{-1/2} D$$

$$\approx \pm\left(n + \frac{1}{2}\right)\frac{\lambda D}{d} \qquad \textbf{(14.10)}$$

where we have assumed $\Delta_n << d$. Thus, the points corresponding to minima will be (approximately) equally spaced with a spacing of $\lambda D/d$.

Example 14.4 Till now we have assumed the needles at S_1 and S_2 (see Fig. 14.1) to vibrate in phase. Assume now that the needles vibrate with a phase difference of π and obtain the nodal lines. Generalize the result for an arbitrary phase difference between the vibrations of the two needles.

Solution: The two needles S_1 and S_2 vibrate out of phase. Thus if, at any instant, the needle at S_1 produces a crest at a distance R from it then the needle at S_2 would produce a trough at a distance R from S_2. Therefore, at all points on the perpendicular bisector OZ (see Fig. 14.4) the two vibrations will always be out of phase and we will have minimum. On the other hand, at the point B which satisfies the equation

$$S_2B - S_1B = \lambda/2$$

the two vibrations will be in phase and we will have maximum. Thus, because of the initial phase difference of π, the conditions for maxima and minima are reversed, i.e., when

$$S_2P \sim S_1P = \left(n + \frac{1}{2}\right)\lambda \text{ (maxima)}$$

the interference will be constructive and we will have maxima, and when

$$S_2P \sim S_1P = n\lambda \text{ (minima)}$$

the interference will be destructive and we will have minima. Notice that one again obtains a stationary interference pattern with nodal lines as hyperbolae.

The above analysis can easily be generalized for arbitrary phase difference between the two needles. Assume, for example, that there is a phase difference of $\pi/3$, i.e., if there is a crest at a distance R from S_1 then there is a crest at a distance $R - \lambda/6$ from S_2. Consequently, the condition

$$S_1P - S_2P = n\lambda + \frac{\lambda}{6};\ n = 0, \pm 1, \pm 2,\ldots$$

will correspond to maxima.

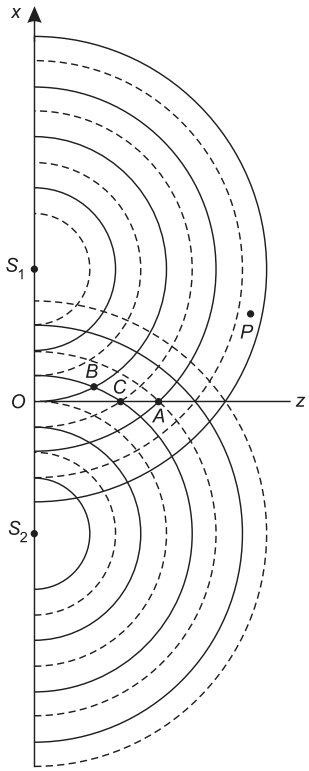

Fig. 14.4 Waves emanating from two point sources S_1 and S_2 vibrating out of phase.

14.3 COHERENCE LO 2

From the above examples, we find that whenever the two needles vibrate with a constant phase difference, a stationary interference pattern is produced. The positions of the maxima and minima will, however, depend on the phase difference in the vibration of the two needles. Two sources which vibrate with a fixed phase difference between them are said to be coherent.

We next assume that the two needles are sometimes vibrating in phase, sometimes vibrating out of phase, sometimes vibrating with a phase difference of $\pi/3$, etc., then the interference pattern will keep on changing. If the phase difference changes with such great rapidity that a stationary interference cannot be observed then the sources are said to be *incoherent*.

Let the displacement produced by the sources at S_1 and S_2 be given by

$$\left.\begin{aligned} y_1 &= a\cos\omega t \\ y_2 &= a\cos(\omega t + \phi) \end{aligned}\right\} \qquad \textbf{(14.11)}$$

then the resultant displacement would be

$$y = y_1 + y_2 = 2a\cos\phi/2\cos(\omega t + \phi/2) \qquad \textbf{(14.12)}$$

The intensity (I) which is proportional to the square of the amplitude can be written in the form

$$I = 4I_0 \cos^2\phi/2 \qquad \textbf{(14.13)}$$

where I_0 is the intensity produced by each one of the source individually. Clearly if $\phi = \pm\pi, \pm 3\pi, \ldots$, the resultant intensity will be zero and we will have minima. On the other hand, when $\phi = 0, \pm 2\pi, \pm 4\pi, \ldots$, the intensity will be maximum (= $4I_0$). However, if the phase difference between the sources S_1 and S_2 (i.e., ϕ) is changing with time, the observed intensity will be given by

$$I = 4I_0 \left\langle \cos^2 \frac{\phi}{2} \right\rangle \qquad \textbf{(14.14)}$$

where $\langle \ldots \rangle$ denotes the time average of the quantity inside the angular brackets; the time average of a time dependent function is defined by the following relation:

$$\langle f(t) \rangle = \frac{1}{\tau} \int_{-\tau/2}^{+\tau/2} f(t)\, dt \qquad \textbf{(14.15)}$$

where τ represents the time over which the averaging is carried out. For example, if the interference pattern is viewed by a normal eye, this averaging will be over about 1/10th of a second; for a camera with exposure time 0.001 sec, τ = 0.001 sec. etc. Clearly, if ϕ varies in a random manner in times which are small compared to τ, then $\cos^2 \phi/2$ will randomly vary between 0 and 1 and $< \cos^2 \phi/2 >$ would be 1/2 [see also Sec. 14.6]. For such a case,

$$I = 2I_0 \qquad \textbf{(14.16)}$$

which implies that if the sources are incoherent then the resultant intensity is the sum of the two intensities and there is no variation of intensity! Thus, if one (or both) of the two vibrating sources are turned on and off in a random manner (such that the phase difference between the vibrations of the two sources varies rapidly) then the interference phenomenon will not be observed. We will discuss this point again in Sec. 14.6 and also in Chapter 17.

14.4 INTERFERENCE OF LIGHT WAVES

LO 3

Till now we have considered interference of waves produced on the surface of water. We will now discuss the interference pattern produced by light waves; however, for light waves it is difficult to observe a stationary interference pattern. For example, if we use two conventional light sources (like two sodium lamps) illuminating two pinholes (see Fig. 14.5), we will not observe any interference pattern on the screen. This can be understood from the following reasoning: In a conventional light source, light comes from a large number of independent atoms; each atom emitting light for about 10^{-10} sec, i.e., light emitted by an atom is essentially a pulse lasting for only 10^{-10} seconds* Even if the atoms were emitting under similar conditions, waves from different atoms would differ in their initial phases.

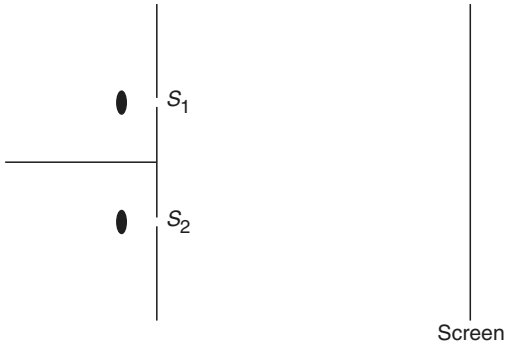

Fig. 14.5 If two sodium lamps illuminate two pinholes S_1 and S_2, no interference pattern will be observed on the screen.

Consequently, light coming out from the holes S_1 and S_2 will have a fixed phase relationship for a period of about 10^{-10} seconds, hence the interference pattern will keep on changing every billionth of a second. The eye can notice intensity changes which last at least for a tenth of a second and hence we will observe a uniform intensity over the screen. However, if we have a camera whose time of shutter opening can be made less than 10^{-10} seconds then the film will record an interference pattern**. We summarise the above results by noting that light beams from two independent sources do not have any fixed phase relationship***, as such they do not produce any stationary interference pattern.

Thomas Young in 1801 devised an ingenious but simple method to lock the phase relationship between the two sources. The trick lies in the division of a single wavefront into two; these two split wavefronts act as if they emanated from two sources having a fixed phase relationship and,

* Since the optical frequencies are of the order of 10^{15} s^{-1}, such a short pulse consists of about a million oscillations; thus it is almost monochromatic (see Chapter 17).

** This interference pattern will be a set of dark and bright bands only if the light waves have the same state of polarization. This can, however, be easily done by putting two polaroids in front of S_1 and S_2. We should mention here that by using two independent laser beams it has been possible to record the interference pattern (see Chapter 17).

*** Such sources are termed as incoherent sources.

therefore, when these two waves were allowed to interfere, a stationary interference pattern was obtained. In the actual experiment, a light source illuminates the pinhole S (see Fig. 14.6). Light diverging from this pinhole fell on a barrier which contained two pinholes S_1 and S_2 which were very close to one another and were located equidistant from S. Spherical waves emanating from S_1 and S_2 (see Fig. 14.7) were coherent and on the screen beautiful interference fringes were obtained. In order to show that this was indeed an interference effect, Young showed that the fringes on the screen disappear when S_1 (or S_2) is covered up. Young explained the interference pattern by considering the principle of superposition, and by measuring the distance between the fringes he calculated the wavelength. Figure 14.7 shows the section of the wavefront on the plane containing S, S_1 and S_2 (which is the $x - z$ plane).

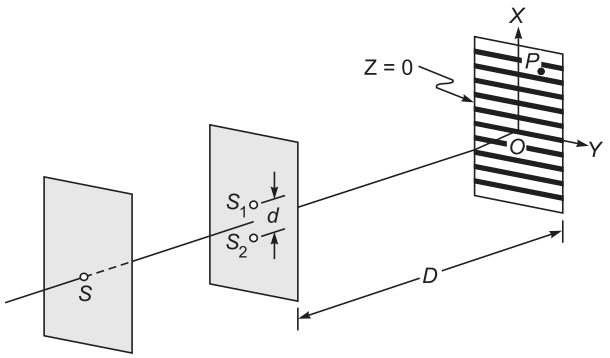

Fig. 14.6 Young's arrangement to produce interference pattern.

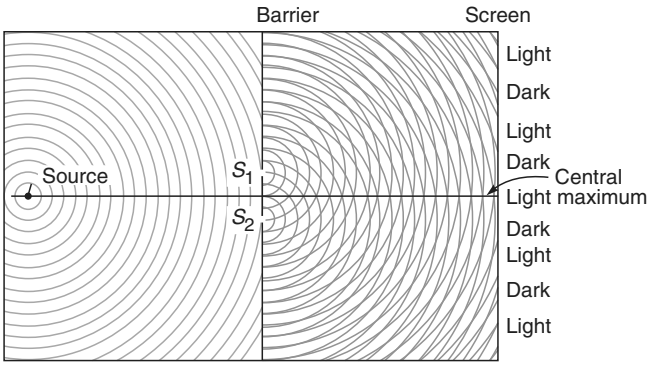

Fig. 14.7 Sections of the spherical wavefronts emanating from S, S_1 and S_2 (Adapted from Ref. 14.7; used with permission).

14.5 THE INTERFERENCE PATTERN

LO 4

In this section, we will first obtain an expression for the fringe width and then we will show that the fringes are strictly hyperbolic.

Let S_1 and S_2 represent the two pinholes of the Young's double hole arrangement. We would determine the positions of maxima and minima on the line LL' which is parallel to the x axis and lies in the plane containing S, S_1 and S_2 (see Fig. 14.8). We will show that the interference pattern (around the point O) consists approximately of a series of dark and bright lines perpendicular to the plane of Fig. 14.8; O being the foot of the perpendicular from the point S on the screen.

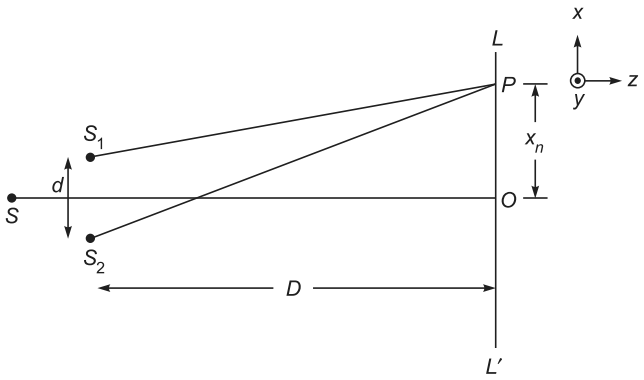

Fig. 14.8 Arrangement for producing Young's interference pattern.

For an arbitrary point P (on the line LL') to correspond to a maximum we must have

$$S_2P - S_1P = n\lambda;\ n = 0, 1, 2, 3,\ldots \qquad \textbf{(14.17)}$$

Now,

$$(S_2P)^2 - (S_1P)^2 = \left[D^2 + \left(x_n + \frac{d}{2}\right)^2\right] - \left[D^2 + \left(x_n - \frac{d}{2}\right)^2\right] \qquad \textbf{(14.18)}$$

$$= 2x_nd$$

where $S_1S_2 = d$ and $OP = x_n$. Thus,

$$S_2P - S_1P = \frac{2x_nd}{S_2P + S_1P} \approx \frac{x_nd}{D} \qquad \textbf{(14.19)}$$

where in the last step we have replaced $S_2P + S_1P$ by $2D$ which will be valid when $D >>> d, x_n$. For example, for $d =$ 0.02 cm, $D = 50$ cm, and $OP = 0.5$ cm (which corresponds to typical values in a light interference experiment) we will have

$$S_2P + S_1P = \sqrt{(50)^2 + (0.51)^2} + \sqrt{(50)^2 + (0.49)^2}$$

$$= 100.005 \text{ cm}$$

Thus, if we replace $S_2P + S_1P$ by $2D$, the error involved is about 0.005%. Using Eqs. (14.17) and (14.19) we obtain

$$x_n = \frac{n\lambda D}{d} \tag{14.20}$$

Thus, the bright (and dark) fringes are equally spaced and the distance between two consecutive bright (or dark) fringes is given by

$$\beta = x_{n+1} - x_n = \frac{\lambda D}{d} \tag{14.21}$$

which is the expression for the fringe width.

We will next determine the shape of the interference pattern on the screen LL' and show that the fringes are a set of hyperbolae. We assume the origin to be at the point O and the z-axis to be perpendicular to the plane of the screen LL' as shown in Fig. 14.8. The screen LL' corresponds to the plane $z = 0$; thus the coordinates of an arbitrary point P on the screen will be $(x, y, 0)$. The coordinates of the point S_1 and S_2 would be $\left(+\frac{d}{2}, 0, -D\right)$ and $\left(-\frac{d}{2}, 0, -D\right)$, respectively. Thus,

$$S_2P - S_1P = \left[\left(x + \frac{d}{2}\right)^2 + y^2 + D^2\right]^{1/2} - \left[\left(x - \frac{d}{2}\right)^2 + y^2 + D^2\right]^{1/2}$$

$$= \Delta \text{ (say)}$$

or

$$\left(x + \frac{d}{2}\right)^2 + y^2 + D^2 = \left\{\Delta + \left[\left(x - \frac{d}{2}\right)^2 + y^2 + D^2\right]^{1/2}\right\}^2$$

Simple manipulations will give us

$$(d^2 - \Delta^2)x^2 - \Delta^2 y^2 = \Delta^2\left[D^2 + \frac{1}{4}(d^2 - \Delta^2)\right]$$

For $\Delta = 0$ (zero path difference) we must have $x = 0$ which implies that the central (bright) fringe is along the y-axis; this is rigorously true. In general, the above equation can always be written in the form

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \tag{14.22}$$

where

$$a^2 = \frac{\Delta^2}{(d^2 - \Delta^2)}\left[D^2 + \frac{1}{4}(d^2 - \Delta^2)\right]$$

and

$$b^2 = D^2 + \frac{1}{4}(d^2 - \Delta^2) \tag{14.23}$$

Equation (14.22) represents a hyperbola. On rearranging, we get

$$x = \pm\left(\frac{\Delta^2}{d^2 - \Delta^2}\right)^{1/2}\left[y^2 + D^2 + \frac{1}{4}(d^2 - \Delta^2)\right]^{1/2} \tag{14.24}$$

Obviously for $y^2 << D^2$, we may neglect y^2 inside the square brackets and the loci are straight lines parallel to the y-axis. Thus, we obtain straight line fringes on the screen. We must remember that we had assumed point sources and we obtained straight line fringes. It is easy to see that if we had slits instead of point sources, each pair of points would have produced the same straight line fringes which would have overlapped with each other–thus we would again obtain straight line fringes. The fringes so produced are said to be non-localized; they can be photographed by just placing a film on the screen; they can also be seen through an eyepiece.

14.6 THE INTENSITY DISTRIBUTION

LO 5

Let $\mathbf{E}_1$ and $\mathbf{E}_2$ be the electric fields produced at the point P by S_1 and S_2 respectively (see Fig. 14.8). The electric fields $\mathbf{E}_1$ and $\mathbf{E}_2$ will, in general, have different directions and different magnitudes. However, if the distances S_1P and S_2P are very large in comparison to the distance S_1S_2, the two fields will almost be in the same direction. Thus, we may write

$$\left.\begin{aligned}\mathbf{E_1} &= \hat{\mathbf{i}}E_{01}\cos\left(\frac{2\pi}{\lambda}S_1P - \omega t\right)\\ \text{and}\quad \mathbf{E_2} &= \hat{\mathbf{i}}E_{02}\cos\left(\frac{2\pi}{\lambda}S_2P - \omega t\right)\end{aligned}\right\} \tag{14.25}$$

where $\hat{\mathbf{i}}$ represents the unit vector along the direction of either of the electric fields. The resultant field will be given by

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 = \hat{\mathbf{i}}\left[E_{01}\cos\left(\frac{2\pi}{\lambda}S_1P - \omega t\right) + E_{02}\cos\left(\frac{2\pi}{\lambda}S_2P - \omega t\right)\right] \tag{14.26}$$

The intensity (I) will be proportional to the square of the electric field and will be given by

$$I = KE^2 \tag{14.27}$$

or

$$I = K\left[E_{01}^2\cos^2\left(\frac{2\pi}{\lambda}S_1P - \omega t\right) +\right.$$

$$E_{02}^2 \cos^2\left(\frac{2\pi}{\lambda} S_2P - \omega t\right) +$$

$$E_{01}\, E_{02}\left\{\cos\left[\frac{2\pi}{\lambda}(S_2P - S_1P)\right] +\right.$$

$$\left.\cos\left[2\omega t - \frac{2\pi}{\lambda}(S_2P + S_1P)\right]\right\}\Bigg] \qquad \textbf{(14.28)}$$

where K is a proportionality constant.* For an optical beam the frequency is very large ($\omega \approx 10^{15}$ sec^{-1}) and all the terms depending on ωt will vary with extreme rapidity (10^{15} times in a second); consequently, any detector would record an average value of various quantities. Now,

$$\left\langle \cos^2(\omega t - \theta)\right\rangle = \frac{1}{2\tau}\int_{-\tau}^{+\tau} \frac{1+\cos[2(\omega t-\theta)]}{2}\, dt$$

$$= \frac{1}{2} + \frac{1}{16\pi}\frac{T}{\tau}\left\{\left[\sin 2(\omega t - \theta)\right]_{-\tau}^{+\tau}\right\}$$

where $T = \frac{2\pi}{\omega}$ ($\approx 2\pi \times 10^{-15}$ sec for an optical beam). For any practical detector** $\frac{T}{\tau} <<< 1$ and since the quantity between the curly brackets will always be between –2 and +2, we may write

$$\langle \cos^2(\omega t - \theta)\rangle \approx \frac{1}{2} \qquad \textbf{(14.29)}$$

The factor $\cos(2\omega t - \phi)$ will oscillate between +1 and –1 and its average will be zero as can indeed be shown mathematically. Thus the intensity, that a detector will record, will be given by

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta \qquad \textbf{(14.30)}$$

where

$$\delta = \frac{2\pi}{\lambda}(S_2P - S_1P) \qquad \textbf{(14.31)}$$

represents the phase difference between the displacements reaching the point P from S_1 and S_2. Further

$$I_1 = \frac{1}{2}KE_{01}^2$$

represents the intensity produced by the source S_1 if no light from S_2 is allowed to fall on the screen; similarly $I_2 = \frac{1}{2}KE_{02}^2$ represents the intensity produced by the source S_2 if no light from S_1 is allowed to fall on the screen. From Eq. (14.30) we may deduce the following:

(a) The maximum and minimum values of $\cos\delta$ are +1 and –1, respectively; as such the maximum and minimum values of I are given by

$$\left.\begin{aligned} I_{\max} &= (\sqrt{I_1} + \sqrt{I_2})^2 \\ I_{\min} &= (\sqrt{I_1} - \sqrt{I_2})^2 \end{aligned}\right\} \qquad \textbf{(14.32)}$$

and

The maximum intensity occurs when

$$\delta = 2n\pi, \quad n = 0, 1, 2, \ldots$$

or

$$S_2P \sim S_1P = n\lambda,$$

and the minimum intensity occurs when

$$\delta = (2n+1)\pi\,;\, n = 0, 1, 2, \ldots$$

or

$$S_2P \sim S_1P = \left(n + \frac{1}{2}\right)\lambda$$

Notice that when $I_1 = I_2$, the intensity minimum is zero. In general, $I_1 \neq I_2$ and the minimum intensity is not zero.

(b) If the holes S_1 and S_2 are illuminated by different light sources (see Fig. 14.5), then the phase difference δ will remain constant for about 10^{-10} sec (see discussion in Sec. 14.3) and thus δ would also vary with time*** in a random way. If we now carry out the averaging over time scales which are of the order of 10^{-8} sec, then

$$\langle \cos\delta\rangle = 0$$

and we obtain

$$I = I_1 + I_2$$

Thus, for two incoherent sources, the resultant intensity is the sum of the intensities produced by each one of the sources independently and no interference pattern is observed.

(c) In the arrangement shown in Fig. 14.6, if the distances S_1P and S_2P are large in comparison to d, then

$$I_1 \approx I_2 = I_0 \text{ (say)}$$

and

$$I = 2I_0 + 2I_0\cos\delta = 4I_0\cos^2\frac{\delta}{2} \qquad \textbf{(14.33)}$$

---

* Equation (14.27) will be derived in Sec. 23.5. In free space, the constant K will be shown to be equal to $\varepsilon_0 c^2$ where ε_0 (=8.854 × 10^{-12} Coul^2/N–m^2) represents the permittivity of free space and c, the speed of light in free space.

** For a normal eye, $\tau \approx 0.1$ s; thus $T/\tau \approx 6 \times 10^{-14}$; even for a detector having 1 nsec as the resolution time, $T/\tau \approx 6 \times 10^{-5}$.

*** Notice that this variation occurs in times of the order of 10^{-10} sec which is about a million times longer than the times for variation of the intensity due to the terms depending on ωt. Thus we are justified in first carrying out the averaging which leads to Eq. (14.30).

The intensity distribution (which is often termed as the $\cos^2$ pattern) is shown in Fig. 14.9. The actual fringe pattern (as it will appear on the screen) is shown in Fig. 14.10. Figures 14.10 (a) and (b) correspond to d = 0.005 mm ($\beta \approx 5$ mm) and d = 0.025 mm ($\beta \approx 1$ mm), respectively. Both figures correspond to D = 5 cm and $\lambda = 5 \times 10^{-5}$ cm. The values of the parameters are such that one can see the hyperbolic nature of the fringe pattern in Fig. 14.10(a).

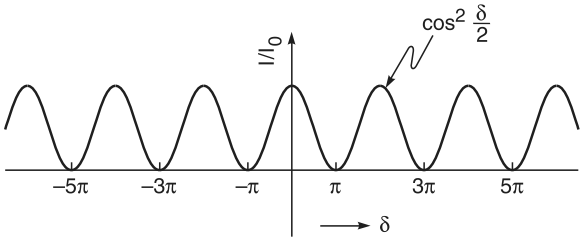

Fig. 14.9 The variation of intensity with δ.

Example 14.5 Instead of considering two point sources, we consider the superposition of two plane waves as shown in Fig. 14.11(a). The wave vectors for the two waves are given by

$$\mathbf{k}_1 = -\hat{\mathbf{x}}\ k \sin\theta_1 + \hat{\mathbf{z}}\ k \cos\theta_1$$

and

$$\mathbf{k}_2 = +\hat{\mathbf{x}}\ k \sin\theta_2 + \hat{\mathbf{z}}\ k \cos\theta_2$$

where $k = 2\pi/\lambda$ and θ_1 and θ_2 are defined in Fig. 14.11(a). Thus the electric fields of the two waves are described by the following two equations:

$$E_1 = E_{01} \cos [\mathbf{k_1} \cdot \mathbf{r} - \omega t]$$
$$= E_{01} \cos [-k x \sin\theta_1 + k z \cos\theta_1 - \omega t]$$
$$E_2 = E_{02} \cos [\mathbf{k_2} \cdot \mathbf{r} - \omega t]$$
$$= E_{02} \cos [k x \sin\theta_2 + k z \cos\theta_2 - \omega t]$$

where we have assumed both electric fields along the same direction (say along the y-axis); if we further assume $E_{01} = E_{02} = E_0$ and $\theta_1 = \theta_2 = \theta$ then the resultant field will be given by

$$E = 2E_0 \cos (k x \sin\theta) \cos (k z \cos\theta - \omega t)$$

Thus, the intensity distribution on the photograph plate LL' will be given by

$$I = 4 I_0 \cos^2(k x \sin\theta)$$

and the fringe pattern will be strictly straight lines (parallel to the y-axis) with fringe width given by

$$\beta = \frac{\lambda}{2 \sin\theta}$$

Figure 14.11(b) shows the computer generated interference pattern on the screen LL' for $\theta = \pi/6$ and λ = 5000 Å. Thus $\beta = \lambda$ = 0.0005 mm.

Example 14.6 In this example, we once again consider the interference pattern produced by 2 point sources S_1 and S_2 on a plane PP' which is perpendicular to the line joining S_1 and S_2 [see Fig. 14.12(a)]. Obviously, on the plane PP', the locus of the point P for which

$$S_1P - S_2P = \text{constant}$$

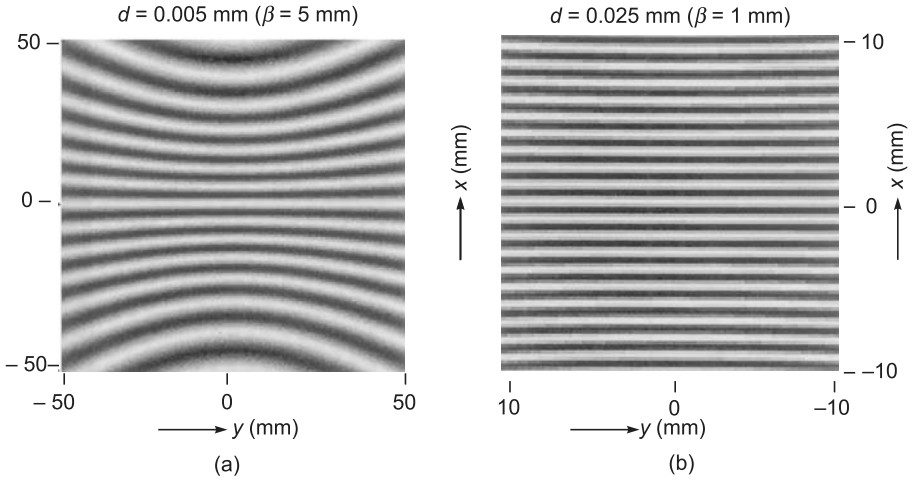

Fig. 14.10 Computer generated fringe pattern produced by two point sources S_1 and S_2 on the screen LL' (see Fig. 14.8); (a) and (b) correspond to d = 0.005 mm and 0.025 mm, respectively (both figures correspond to D = 5 cm and $\lambda = 5 \times 10^{-5}$ cm).

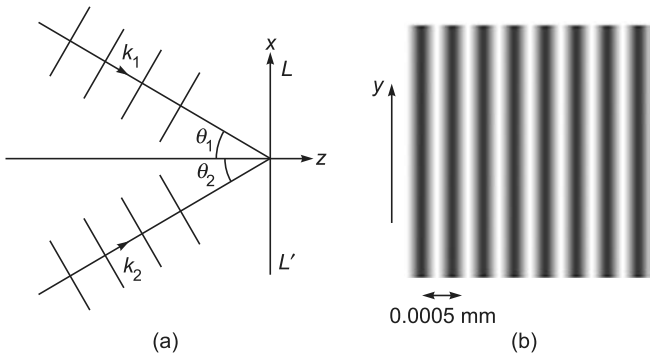

Fig. 14.11 (a) The superposition of two plane waves on LL'. (b) Computer generated interference pattern on the screen LL' for $\theta_1 = \theta_2 = \pi/6$ and $\lambda = 5000$ Å. The fringes are parallel to the y-axis.

will be a circle. Figures 14.12(b) and (c) show the fringe patterns for $D = 20$ cm and $D = 10$ cm; for both figures $S_1S_2 = d = 0.05$ mm and $\lambda = 5000$ Å. Obviously, if O represents the centre of the fringe pattern then

$$S_1O - S_2O = d = 100\ \lambda$$

Thus (for this value of d) the central spot will be bright for all values of D and will correspond to $n = 100$. The first and second bright circles will correspond to a path difference of 99 λ and 98 λ, respectively. Similarly, the first and second dark rings in the interference pattern will correspond to a path difference of 99.5 λ and 98.5 λ, respectively. The radii of the fringes can be calculated by using the formula given in Problem 14.10.

Example 14.7 We finally consider the interference pattern produced on PP' by the superposition of a plane wave incident normally and a spherical wave emanating from the point O (see Fig. 14.13). The plane wave will be given by

$$E_1 = E_0 \cos(kz - \omega t + \phi)$$

and the spherical wave will be given by

$$E_2 = \frac{A_0}{r} \cos(kr - \omega t)$$

where r is the distance measured from the point O which is assumed to be the origin. Now, on the plane PP' ($z = D$)

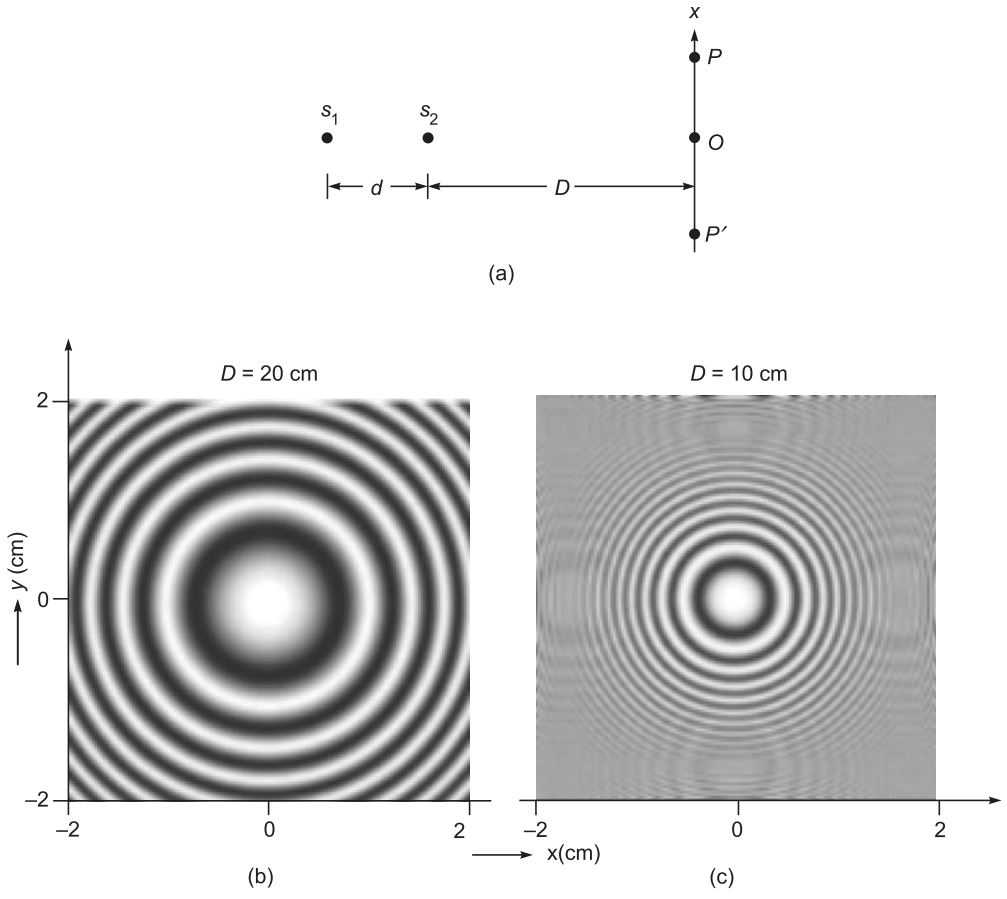

Fig. 14.12 (a) S_1 and S_2 represent two coherent sources, (b) and (c) show the interference fringes observed on the screen PP' when $D = 20$ cm and $D = 10$ cm, respectively.

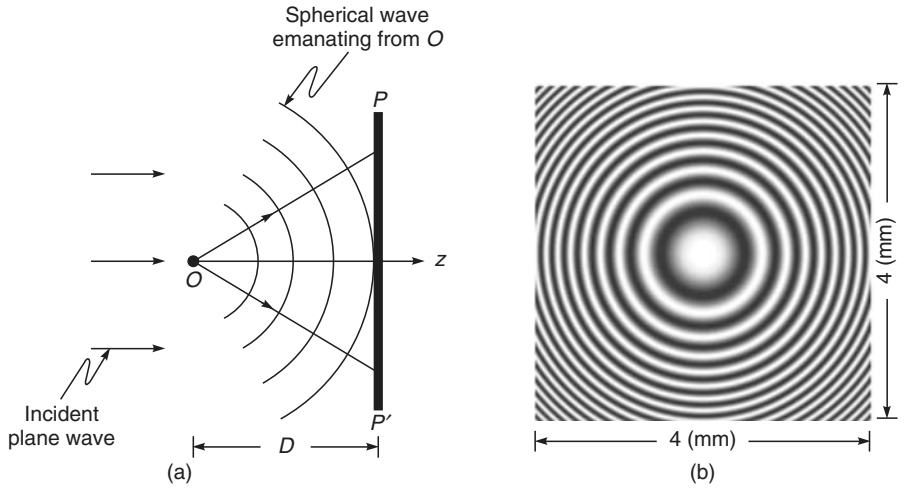

Fig. 14.13 (a) Superposition of a plane wave and a spherical wave emanating from the point O; (b) shows the interference fringes observed on the screen PP'.

$$r = (x^2 + y^2 + D^2)^{\frac{1}{2}} \approx D\left[1 + \frac{x^2 + y^2}{2D^2}\right]$$

$$\approx D + \frac{x^2 + y^2}{2D}$$

where we have assumed $x, y << D$. On the plane $z = D$, the resultant field will be given by

$$E = E_1 + E_2$$
$$\approx E_0 \cos(kD - \omega t + \phi) + \frac{A_0}{D}\cos\left[kD + \frac{k}{2D}(x^2 + y^2) - \omega t\right]$$

Thus,

$$\left\langle E^2 \right\rangle = \frac{1}{2}E_0^2 + \frac{1}{2}\left(\frac{A_0}{D}\right)^2 + E_0\frac{A_0}{D}\cos\left[\frac{k}{2D}(x^2 + y^2) - \phi\right]$$

If we assume that

$$\frac{A_0}{D} \approx E_0$$

i.e., the amplitude of the spherical wave (on the plane PP') is the same as the amplitude of the plane wave, then

$$\left\langle E^2 \right\rangle \approx 2E_0^2 \cos^2\left(\frac{k}{4D}(x^2 + y^2) - \frac{1}{2}\phi\right)$$

and we would obtain circular interference fringes as shown in Fig. 14.13(b). If r_m and r_{m+p} denote the radii of m^{th} and $(m + p)^{\text{th}}$ bright ring, then

$$r_{m+p}^2 - r_m^2 = 2p\lambda D$$

14.6.1 Moire Fringes

We may mention here that Moire fringes can be very effectively used to study the formation of fringe patterns. In Fig. 14.14, we have shown the overlapping of two simple patterns from which one can understand the formation of bright and dark fringes when two plane waves propagate in slightly different directions. In a classroom, it can be easily demonstrated by having a periodic pattern on a transparency and overlapping it with its own photocopy at different angles. Similarly, if one overlaps a circular pattern (on a transparency) with its own copy, one obtains the hyperbolic fringes as shown in Fig. 14.15. (To get a clearer fringe pattern, you may have to view the patterns from a greater distance.) In Sec. 17.5, we have shown how the beat phenomenon can be understood by observing the Moire fringes obtained by the overlapping of two patterns of slightly different periods (see Fig. 17.13).

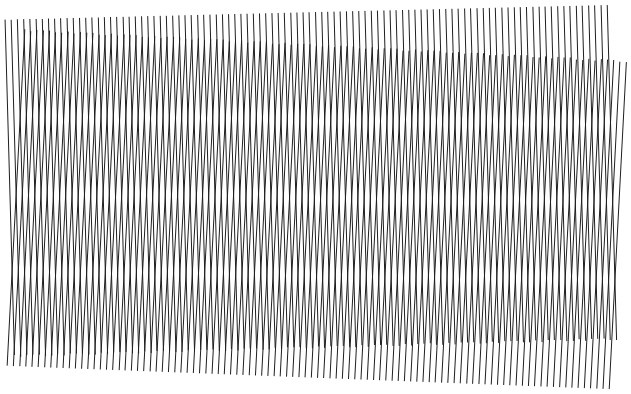

Fig. 14.14 The Moire Pattern produced by two overlapping straight line patterns.

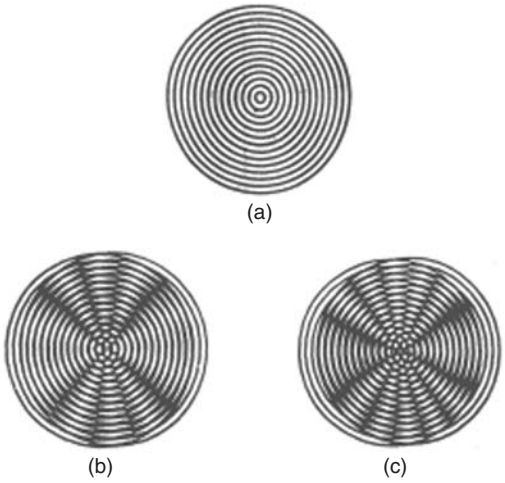

Fig. 14.15 The Moire pattern produced by two overlapping circular patterns. You will see clear hyperbolic fringes if you put the pattern at a greater distance from the eye. The circular pattern was provided by Dr. R. E. Bailey.

Example 14.8 Consider a parallel beam of light (from a distant source S' like a star incident at an angle θ) on two slits S_1 and S_2 as shown in Fig. 14.16. Obviously, the path difference between the waves emanating from the slits S_1 and S_2 will be given by

$$X S_2 = d \sin \theta$$

Therefore, the intensity distribution on the screen due to S' will be given by

$$I = I_0 \cos^2 \frac{\delta}{2}$$

where

$$\delta = \frac{2\pi}{\lambda}[XS_2 + S_2P - S_1P] = \frac{2\pi}{\lambda}[S_2P - S_1P) + d \sin \theta]$$

$$= \frac{2\pi}{\lambda}\left[\frac{xd}{D} + d \sin \theta\right]$$

Thus, the intensity distribution (due to light coming from the distant source S') will be given by

$$I' = I_0 \cos^2\left(\frac{\pi}{\lambda}\left[\frac{xd}{D} + d \sin \phi\right]\right)$$

Similarly, if there is light incident from another distant source S'' (at an angle ϕ) then the corresponding intensity distribution on the screen will be given by

$$I'' = I_0 \cos^2\left(\frac{\pi}{\lambda}\left[\frac{xd}{D} - d \sin \phi\right]\right)$$

The resultant intensity distribution will be given by

$$I = I' + I''$$

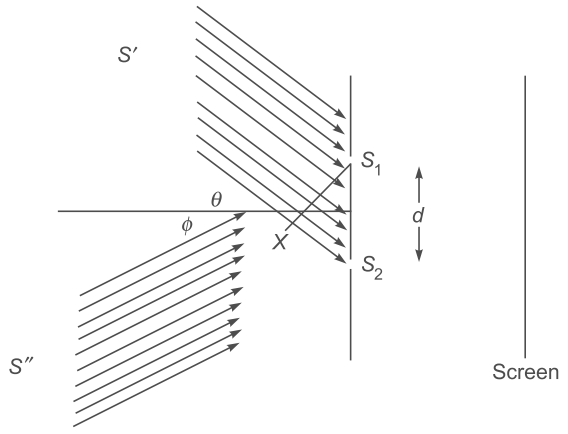

Fig. 14.16 Two distant incoherent sources S' and S'' illuminate the slits S_1 and S_2.

Example 14.9 This example presupposes the knowledge of half-wave plates (see Sec. 22.6) and therefore the reader may skip this example until he has gone through Chapter 22.

Consider a y-polarized beam light beam incident on a double hole system as shown in Fig. 14.17. Behind the hole S_1 we have put a half-wave plate H_1 whose optic axis is along the y direction and behind the hole S_2 we have put a half-wave plate H_2 whose optic axis is along the x direction. Thus, as discussed in Sec. 22.6, in H_1, a y-polarized beam will propagate with velocity c/n_e and in H_2, a y-polarized beam will propagate with velocity c/n_o. In calcite $n_e < n_o$; and, in a half-wave plate a phase change of π is introduced between the o-wave and the e-wave. Thus, the whole fringe pattern will shift by $\beta/2$ where β is the fringe width.

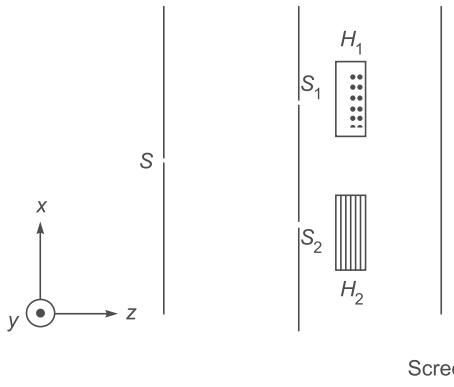

Fig. 14.17 H_1 and H_2 are half-wave plates placed in front of the slits S_1 and S_2. The optic axis of H_1 and H_2 are along y and x directions, respectively.

14.7 FRESNEL'S TWO-MIRROR ARRANGEMENT LO 6

After Young's double hole interference experiment, Fresnel devised a series of arrangements to produce the interference pattern. One of the experimental arrangements, known as the **Fresnel two-mirror arrangement**, is shown in Fig. 14.18; it consists of two plane mirrors which are inclined to each other at a small angle θ and touching at the point M. S represents a narrow slit placed perpendicular to the plane of the paper.

A portion of the wavefront from S gets reflected from M_1M and illuminates the region AD of the screen. Another portion of the wavefront gets reflected from the mirror MM_2 and illuminates the region BC of the screen. Since these two wavefronts are derived from the same source they are coherent. Thus in the region BC, one observes interference fringes. The formation of the fringes can also be understood as being due to the interference of the wavefronts from the virtual sources S_1 and S_2 of S formed by the mirrors M_1 and M_2, respectively. From simple geometric considerations, it can be shown that the points S, S_1 and S_2 lie on a circle whose centre is at the point M. Further, if the angle between the mirrors is θ, then the angle S_1SS_2 is also θ and the angle S_1MS_2 is 2θ. Thus, S_1S_2 is $2R\theta$, where R is the radius of the circle.

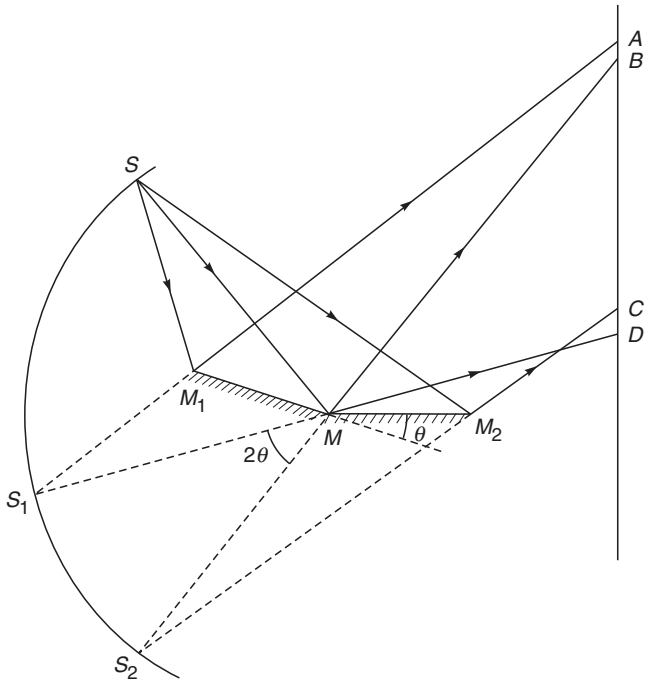

Fig. 14.18 Fresnel's two-mirror arrangement.

14.8 FRESNEL BIPRISM LO 6

Fresnel devised yet another simple arrangement for the production of interference pattern. He used a biprism, which was actually a simple prism, the base angles of which are extremely small ($\sim 20'$). The base of the prism is shown in Fig. 14.19 and the prism is assumed to stand perpendicular to the plane of the paper. S represents the slit which is also placed perpendicular to the plane of the paper. Light from the slit S gets refracted by the prism and produces two virtual images S_1 and S_2. These images act as coherent sources and produce interference fringes on the right of the biprism. The fringes can be viewed through an eyepiece. If n represents the refractive index of the material of the biprism and α the base angle, then $(n-1)\alpha$ is approximately the angular deviation produced by the prism and, therefore, the distance S_1S_2 is $2a(n-1)\alpha$, where a represents the distance from S to the base of the prism. Thus, for $n = 1.5$, $\alpha \approx (20') \approx 5.8 \times 10^{-3}$ radians, $a \approx 2$ cm, one gets $d = 0.012$ cm.

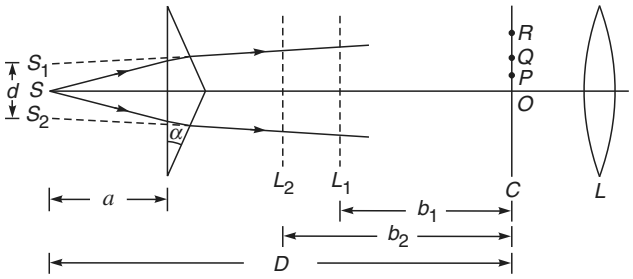

Fig. 14.19 Fresnel's biprism arrangement. C and L represent the positions of the crosswires and the eyepiece, respectively. In order to determine d one introduces a lens between the biprism and the crosswires; L_1 and L_2 represent the two positions of the lens where the slits are clearly seen.

The biprism arrangement can be used for the determination of wavelength of an almost monochromatic light like the one coming from a sodium lamp. Light from the sodium lamp illuminates the slit S and interference fringes can be easily viewed through the eyepiece. The fringe width (β) can be determined by means of a micrometer attached to the eyepiece. Once β is known, λ can be determined by using the following relation:

$$\lambda = \frac{d\beta}{D} \qquad \textbf{(14.34)}$$

It may be mentioned that in order to determine d, one need not measure the value of α. In fact the distances d and D can easily be determined by placing a convex lens between the

biprism and the eyepiece. For a fixed position of the eyepiece there will be two positions of the lens (shown as L_1 and L_2 in Fig. 14.19) where the images of S_1 and S_2 can be seen at the eyepiece.* Let d_1 be the distance between the two images when the lens is at the position L_1 (at a distance b_1 from the eyepiece). Let d_2 and b_2 be the corresponding distances when the lens is at L_2. Then, it can easily be shown that

$$d = \sqrt{d_1 d_2}$$

and

$$D = b_1 + b_2$$

Typically for $d \approx 0.01$ cm, $\lambda \approx 6 \times 10^{-5}$ cm, $D \approx 50$ cm, $\beta \approx 0.3$ cm.

In the above arrangement we have considered a slit instead of a point source. Since each pair of points S_1 and S_2 produce (approximately) straight line fringes, the slit will also produce straight line fringes of increased intensity.

14.9 INTERFERENCE WITH WHITE LIGHT

LO 7

We will now discuss the interference pattern when the slit is illuminated by white light. The wavelengths corresponding to the violet and red end of the spectrum are about 4×10^{-5} cm and 7×10^{-5} cm, respectively. Clearly, the central fringe produced at the point O (Fig. 14.19) will be white because all wavelengths will constructively interfere here. Now, slightly below (or above) the point O the fringes will become colored. For example, if the point P is such that

$$S_2P \sim S_1P = 2 \times 10^{-5} \text{ cm} \left(= \frac{\lambda_{\text{violet}}}{2} \right)$$

then complete destructive interference will occur only for the violet color. Partial destructive interference will occur for other wavelengths. Consequently, we will have a line devoid of the violet color and will appear reddish. The point Q which satisfies

$$S_2Q \sim S_1Q = 3.5 \times 10^{-5} \text{ cm} \left(= \frac{\lambda_{\text{red}}}{2} \right)$$

will be devoid of the red color. It will correspond to almost constructive interference for the violet color. No other wavelength (in the visible region) will neither constructively nor destructively interfere. Thus following the white central fringe we will have colored fringes; when the path difference is about 2×10^{-5} cm the fringe will be red in color, then the color will gradually change to violet. The colored fringes will soon disappear because at points far away from O there will be so many wavelengths (in the visible region) which will constructively interfere that we will observe uniform white illumination. For example, at a point R, such that $S_2R \sim S_1R = 30 \times 10^{-5}$ cm, wavelengths corresponding to $30 \times 10^{-5}/n$ (n = 1, 2,…) will constructively interfere. In the visible region, these wavelengths will be 7.5×10^{-5} cm (red), 6×10^{-5} cm (yellow), 5×10^{-5} cm (greenish yellow) and 4.3×10^{-5} cm (violet). Further, wavelengths corresponding to $30 \times 10^{-5}/\left(n + \frac{1}{2}\right)$ will destructively interfere; thus, in the visible region, the wavelengths 6.67×10^{-5} cm (orange), 5.5×10^{-5} cm (yellow), 4.6×10^{-5} cm (indigo) and 4.0×10^{-5} cm (violet) will be absent. The color of such light, as seen by the unaided eye, will be white. Thus, with white light one gets a white central fringe at the point of zero path difference along with a few colored fringes on both the sides, the color soon fading off to white. While using a white light source, if we put a red (or green) filter in front of our eye, we will see the interference pattern corresponding to the red (or green) light.

In the usual interference pattern with a nearly monochromatic source (like a sodium lamp) a large number of interference fringes are obtained and it is extremely difficult to determine the position of the central fringe. In many interference experiments, it is necessary to determine the position of the central fringe and, as has been discussed above, this can easily be done by using white light as a source.

As discussed above, when we observe interference pattern using a white light source, we will see only few colored fringes. However, if we put a red filter in front of our eye, the fringe pattern (corresponding to the red color) will suddenly appear. If we replace the red filter by a green filter in front of our eye, the fringe pattern corresponding to the green color will appear.

14.10 DISPLACEMENT OF FRINGES

LO 8

We will now discuss the change in the interference pattern produced by introducing a thin transparent plate in the path of one of the two interference beams as shown in Fig. 14.20. Let t be the thickness of the plate and let n be its refractive

* This method is similar to the displacement method for the determination of the focal length of a convex lens.

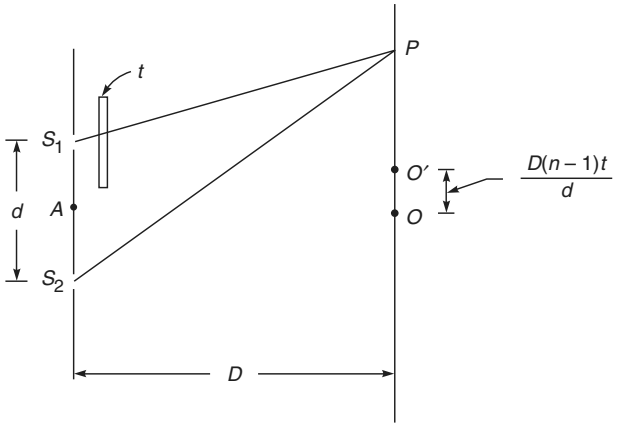

Fig. 14.20 If a thin transparent sheet (of thickness t) is introduced in one of the beams, the fringe pattern gets shifted by a distance $(n - 1)\, t\, D/d$.

index. It is easily seen from the figure that light reaching the point P from S_1 has to traverse a distance t in the plate and a distance $S_1P - t$ in air. Thus the time required for the light to reach from S_1 to the point P is given by

$$\frac{S_1P - t}{c} + \frac{t}{v} = \frac{1}{c}[S_1P - t + nt]$$

$$= \frac{1}{c}\,[S_1P + (n-1)t] \qquad \textbf{(14.35)}$$

where $v\ \left(=\frac{c}{n}\right)$ represents the speed of light in the plate. Equation (14.35) shows that by introducing the thin plate the effective optical path increases by $(n - 1)t$. Thus, when the thin plate is introduced, the central fringe (which corresponds to equal optical path from S_1 and S_2) is formed at the point O' where

$$S_1O' + (n-1)t = S_2O'$$

Since [see Eq. (14.19)]

$$S_2O' - S_1O' \approx \frac{d}{D}\ OO'$$

therefore,

$$(n-1)t = \frac{d}{D}\ OO' \qquad \textbf{(14.36)}$$

Thus the fringe pattern gets shifted by a distance Δ which is given by the following equation:

$$\Delta = \frac{D(n-1)\,t}{d} \qquad \textbf{(14.37)}$$

The above principle enables us to determine the thickness of extremely thin transparent sheets (like that of mica) by measuring the displacement of the central fringe. Further, if white light is used as a source, the displacement of the central fringe is easy to measure.

Example 14.10 In a double-slit interference arrangement, one of the slits is covered by a thin mica sheet whose refractive index is 1.58. The distances S_1S_2 and AO (see Fig. 14.20) are 0.1 cm and 50 cm, respectively. Due to the introduction of the mica sheet the central fringe gets shifted by 0.2 cm. Determine the thickness of the mica sheet.

Solution: $\Delta = 0.2$ cm; $d = 0.1$ cm; $D = 50$ cm

Hence,

$$t = \frac{d\Delta}{D(n-1)} = \frac{0.1 \times 0.2}{50 \times 0.58}$$

$$\approx 6.7 \times 10^{-4} \text{ cm}$$

Example 14.11 In an experimental arrangement similar to the one discussed in the above example, one finds that by introducing the mica sheet the central fringe occupies the position that was originally occupied by the eleventh bright fringe. If the source of light is a sodium lamp ($\lambda = 5893$ Å) determine the thickness of the mica sheet.

Solution: The point O' (see Fig. 14.20) corresponds to the eleventh bright fringe, thus,

$$S_2O' - S_1O' = 11\lambda = (n-1)t = 0.58t$$

14.11 THE LLOYD'S MIRROR ARRANGEMENT LO 9

In this arrangement, light from a slit S_1 is allowed to fall on a plane mirror at grazing incidence (see Fig. 14.21). The light directly coming from the slit S_1 interferes with the light reflected from the mirror forming an interference pattern in the region BC of the screen. One may thus consider the slit S_1 and its virtual image S_2 to form two coherent sources which produce the interference pattern. It should be noted that at grazing incidence one really need not have a mirror; even a dielectric surface has very high reflectivity (see Chapter 23).

As can be seen from Fig. 14.21, the central fringe cannot be observed on the screen unless the latter is moved to the position $L'_1L'_2$, where it touches the end of the reflector. Alternatively, one may introduce a thin mica sheet in the path of the direct beam so that the central fringe appears in the region BC. (This is discussed in detail in Problem 14.2) Indeed, if the central fringe is observed with white light, it is found to be dark. This implies that the reflected beam under-

goes a sudden phase change of π on reflection. Consequently, when the point P on the screen is such that

$$S_2P - S_1P = n\lambda, n = 0, 1, 2, 3,\ldots$$

we will get minima (i.e., destructive interference). On the other hand, if

$$S_2P - S_1P = \left(n + \frac{1}{2}\right)\lambda$$

we will get maxima.

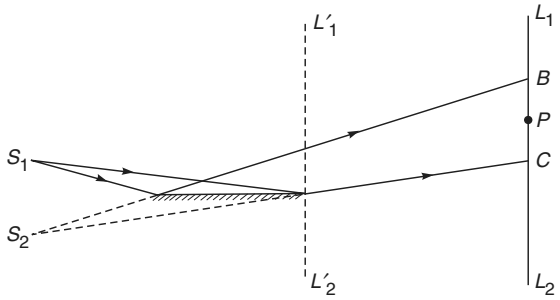

Fig. 14.21 The Lloyd's mirror arrangement.

In the next section, using the principle of optical reversibility, we will show that if there is an abrupt phase change of π when light gets reflected by a denser medium, then no such abrupt phase change occurs when reflection takes place at a rarer medium.

14.12 PHASE CHANGE ON REFLECTION

We will now investigate the reflection of light at an interface between two media using the principle of optical reversibility. According to this principle, in the absence of any absorption, a light ray that is reflected or refracted will retrace its original path if its direction is reversed.*

Consider a light ray incident on an interface of two media of refractive indices n_1 and n_2 as shown in Fig. 14.22(a). Let the amplitude reflection and transmission coefficients be r_1 and t_1, respectively. Thus, if the amplitude of the incident ray is a, then the amplitudes of the reflected and refracted rays would be ar_1 and at_1, respectively.

We now reverse the rays and we consider a ray of amplitude at_1 incident on medium 1 and a ray of amplitude ar_1 incident on medium 2 as shown in Fig. 14.22(b). The ray of amplitude at_1 will give rise to a reflected ray of amplitude at_1r_2 and a transmitted ray of amplitude at_1t_2 where r_2 and t_2 are the amplitude reflection and transmission coefficients when a ray is incident from medium 2 on medium 1. Similarly, the ray of amplitude ar_1 will give rise to a ray of amplitude ar_1^2 and a refracted ray of amplitude ar_1t_1. According to the principle of optical reversibility the two rays of amplitudes ar_1^2 and at_1t_2 must combine to give the incident ray of Fig. 14.22(a); thus,

$$ar_1^2 + at_1t_2 = a$$

or

$$t_1t_2 = 1 - r_1^2 \qquad \textbf{(14.38)}$$

Further, the two rays of amplitudes at_1r_2 and ar_1t_1 must cancel each other, i.e.,

$$at_1r_2 + ar_1t_1 = 0$$

or

$$r_2 = -r_1 \qquad \textbf{(14.39)}$$

Since we know from the Lloyd's mirror experiment that an abrupt phase change of π occurs when light gets reflected by a denser medium, we may infer from Eq. (14.39) that no such abrupt phase change occurs when light gets reflected by a rarer medium. This is indeed borne out by experiments. Equations (14.38) and (14.39) are known as **Stokes' relations**.

In Chapter 24, we will calculate the amplitude reflection and transmission coefficients for plane waves incident on a dielectric and also on a conductor. It will be shown that the coefficients satisfy Stokes' relations; the phase change on reflection will also be discussed there.

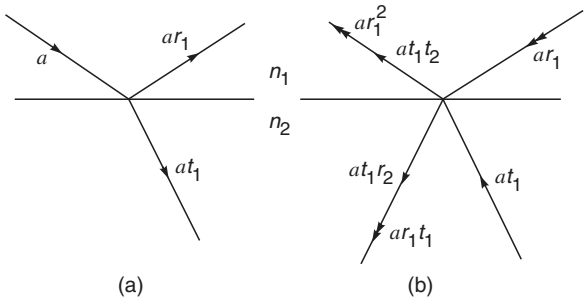

Fig. 14.22 (a) A ray traveling in a medium of refractive index n_1 incident on a medium of refractive index n_2. (b) Rays of amplitude ar_1 and at_1 incident on a medium of refractive index n_1.

* This principle is consequence of time reversal invariance according to which processes can run either way in time; for more details see Refs. 14.3 and 14.8.

Summary

- In 1801, Thomas Young devised an ingenious but simple method to lock the phase relationship between two sources of light. The trick lies in the division of a single wavefront into two; these two split wavefronts act as if they emanated from two sources having a fixed phase relationship and, therefore, when these two waves were allowed to interfere, a stationary interference pattern is obtained.
- For two coherent point sources, almost straight-line interference fringes are formed on some planes and by measuring the fringe width (which represents the distance between two consecutive fringes) one can calculate the wavelength.
- On a plane which is normal to the line joining the two coherent point sources, the fringe pattern is circular.
- In the Young's double-slit interference pattern, if we use a white light source, one gets a white central fringe at the point of zero path difference along with a few colored fringes on both the sides, the color soon fading off to white. If we now introduce a very thin slice of transparent material (like mica) in the path of one of the interfering beams, the fringes get displaced and by measuring the displacement of fringes, one can calculate the thickness of the mica sheet.

Problems

14.1 In the Young's double-hole experiment (see Fig. 14.6), the distance between the two holes is 0.5 mm, $\lambda = 5 \times 10^{-5}$ cm and D = 50 cm. What will be the fringe width?

14.2 Figure 14.23 represents the layout of Lloyd's mirror experiment. S is a point source emitting waves of frequency 6×10^{14} s^{-1}. A and B represent the two ends of a mirror placed horizontally and LOM represents the screen. The distances SP, PA, AB and BO are 1 mm, 5 cm, 5 cm and 190 cm, respectively. (a) Determine the position of the region where the fringes will be visible and calculate the number of fringes. (b) Calculate the thickness of a mica sheet (n = 1.5) which should be introduced in the path of the direct ray so that the lowest fringe becomes the central fringe. The velocity of light is 3×10^{10} cm/s.

[**Ans:** (a) 2 cm, 40 fringes, (b) 38 μm]

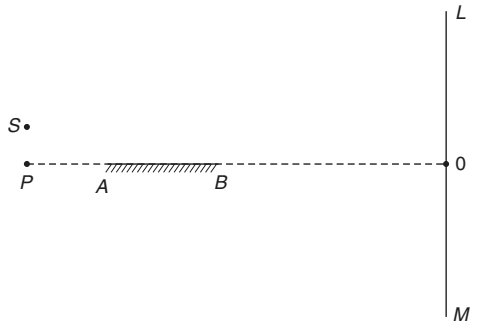

Fig. 14.23 Figure for Problem 14.2.

14.3 **(a)** In the Fresnel's biprism arrangement, show that $d = 2(n - 1)\, a\alpha$ where a represents the distance from the source to the base of the prism (see Fig. 14.19), α is the angle of the biprism and n the refractive index of the material of the biprism.

(b) In a typical biprism arrangement b/a = 20, and for sodium light ($\lambda \simeq$ 5893 Å) one obtains a fringe width of 0.1 cm; here b is the distance between the biprism and the screen. Assuming n = 1.5, calculate the angle α.

[**Ans:** $\simeq$ 0.71°]

14.4 In the Young's double-hole experiment a thin mica sheet (n = 1.5) is introduced in the path of one of the beams. If the central fringe gets shifted by 0.2 cm, calculate the thickness of the mica sheet. Assume d = 0.1 cm, and D = 50 cm.

14.5 In order to determine the distance between the slits in the Fresnel biprism experiment, one puts a convex lens in between the biprism and the eyepiece. Show that if $D > 4f$ one will obtain two positions of the lens where the image of the slits will be formed at the eyepiece; here f is the focal length of the convex lens and D is the distance between the slit and the eyepiece. If d_1 and d_2 are the distances between the images (of the slits) as measured by the eyepiece, then show that $d = \sqrt{d_1 d_2}$. What would happen if $D < 4f$?

14.6 In the Young's double hole experiment, interference fringes are formed using sodium light which predominantly comprises of two wavelengths (5890 Å and 5896 Å). Obtain the regions on the screen where the fringe pattern will disappear. You may assume d = 0.5 mm and D = 100 cm.

14.7 If one carries out the Young's double-hole interference experiment using microwaves of wavelength 3 cm, discuss the nature of the fringe pattern if d = 0.1 cm, 1 cm, and 4 cm. You may assume D = 100 cm. Can you use Eq. (14.21) for the fringe width?

14.8 In the Fresnel's two-mirror arrangement (see Fig. 14.18) show that the points S, S_1, and S_2 lie on a circle and $S_1S_2 = 2b\theta$ where $b = MS$ and θ is the angle between the mirrors.

14.9 In the double-hole experiment using white light, consider two points on the screen, one corresponding to a path difference of 5000 Å and the other corresponding to a path difference of 40000 Å. Find the wavelengths (in the visible region) which correspond to constructive and destructive interference. What will be the color of these points?

14.10 **(a)** Consider a plane which is normal to the line joining two point coherent sources S_1 and S_2 as shown in Fig. 14.14. If $S_1P - S_2P = \Delta$, then show that

$$y = \frac{1}{2\Delta}(d^2 - \Delta^2)^{\frac{1}{2}}[4D^2 + 4Dd + (d^2 - \Delta^2)]^{\frac{1}{2}}$$

$$\approx \frac{D}{\Delta}\sqrt{(d - \Delta)(d + \Delta)}$$

where the last expression is valid for $D >> d$.

(b) For λ = 0.5 μm, d = 0.4 mm and D = 20 cm; $S_1O - S_2O = 800\,\lambda$. Calculate the value of $S_1P - S_2P$ for the point P to be first dark ring and first bright ring.

[**Ans:** 0.39975 mm, 0.3995 mm]

14.11 In continuation of the above problem calculate the radii of the first two dark rings for

(a) D = 20 cm and **(b)** D = 10 cm.

[**Ans:** (a) ≈ 0.71 cm and 1.22 cm]

14.12 In continuation of Problem 14.10 assume that d = 0.5 mm, $\lambda = 5 \times 10^{-5}$ cm and D = 100 cm. Thus the central (bright) spot will correspond to n = 1000. Calculate the radii of the first, second and third bright rings which will correspond to n = 999, 998 and n = 997, respectively.

14.13 Using the expressions for the amplitude reflection and transmission coefficients (derived in Chapter 24), show that they satisfy Stokes' relations.

14.14 Assume a plane wave incident normally on a plane containing two holes separated by a distance d. If we place a convex lens behind the slits, show that the fringe width, as observed on the focal plane of the lens, will be $f\lambda/d$ where f is the focal length of the lens.

14.15 In Problem 14.14, show that if the plane (containing the holes) lies in the front focal plane of the lens, then the interference pattern will consist of exactly parallel straight lines. However, if the plane does not lie on the front focal plane, the fringe pattern will be hyperbolae.

14.16 In the Young's double-hole experiment calculate I/I_{max} where I represents the intensity at a point where the path difference is $\lambda/5$.

15.1 INTRODUCTION

In the previous chapter, we discussed the interference pattern produced by division of a wavefront; for example, light coming out of a pinhole was allowed to fall on two holes, and spherical waves emanating from these two holes produced the interference pattern. In this chapter, we will consider the formation of interference pattern by division of amplitude; for example, if a plane wave falls on a thin film then the wave reflected from the upper surface interferes with the wave reflected from the lower surface. Such studies have many practical applications and also explain phenomena like the formation of beautiful colors produced by a soap film illuminated by white light.

15.2 INTERFERENCE BY A PLANE PARALLEL FILM WHEN ILLUMINATED BY A PLANE WAVE LO 1

If a plane wave is incident normally on a thin* film of uniform thickness d (see Fig. 15.1) then the waves reflected from the upper surface interfere with the waves reflected from the lower surface. In this section, we will study this interference pattern. In order to observe the interference pattern without obstructing the incident beam, we use a partially reflecting plate G as shown in Fig. 15.1. Such an arrangement also enables us to eliminate the direct beam from reaching the photographic plate P (or the eye). The plane wave may be produced by placing an illuminated pinhole at the focal point of a corrected lens; alternatively, it may just be a beam coming out of a laser.

Let the solid and the dashed lines in Fig. 15.2 represent the positions of the crests** (at any particular instant of time) corresponding to the waves reflected from the upper and lower surfaces of the film, respectively.*** Clearly, the wave reflected from the lower surface of the film traverses an additional optical path of $2nd$, where n represents the refractive index of the material of the film. Further, if the film is placed in air, then the wave reflected from the upper surface of the film will undergo a sudden change in phase of π (see Sec. 14.12) and as such the conditions for destructive or constructive interference will be given by

$$2nd = m\lambda \quad \text{destructive interference} \qquad \textbf{(15.1a)}$$

$$= \left(m + \frac{1}{2}\right)\lambda \quad \text{constructive interference} \qquad \textbf{(15.1b)}$$

where m = 0, 1, 2,…and λ represents the free space wavelength.

Thus, if we place a photographic plate at P (see Fig. 15.1), then the plate will receive uniform illumination; it will be dark when $2nd = m\lambda$ and bright when $2nd = \left(m + \frac{1}{2}\right)\lambda$; m = 0, 1, 2,… Instead of placing the photographic plate, if we try

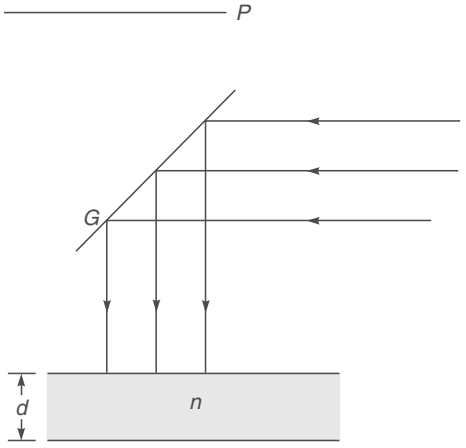

Fig. 15.1 The normal incidence of a parallel beam of light on a thin film of refractive index n and thickness d. G denotes a partially reflecting plate and P represents a photographic plate.

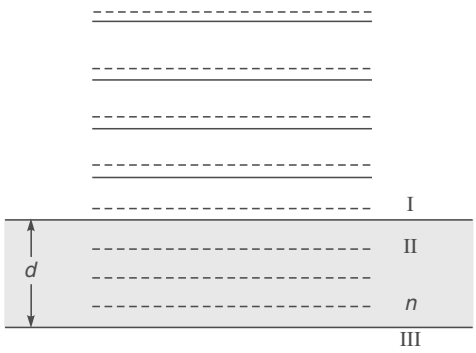

Fig. 15.2 The solid and the dashed lines represent the crests of the waves reflected from the upper surface and from the lower surface of the thin film. Notice that the distance between the consecutive crests inside the film is less than the corresponding distance in medium I.

* Why the film should be thin is explained in Sec. 15.7.

** Notice that the distance between consecutive crests in the film is less than the corresponding distance in air. This is because of the fact that the effective wavelength in a medium of refractive index n is λ/n.

*** In general, the wave reflected from the lower surface of the film will suffer multiple reflections. The effect of such multiple reflections is neglected (see Chapter 16).

to view the film (from the top) with naked eye, then the film will appear to be uniformly illuminated.

It may be noted that the amplitudes of the waves reflected from the upper and lower surfaces will, in general, be slightly different; and as such the interference will not be completely destructive. However, with appropriate choice of the refractive indices of media II and III, the two amplitudes can be made very nearly equal (see Example 15.1).

For an air film between two glass plates (see Fig. 15.3) no phase change will occur on reflection at the glass-air interface, but a phase change of π will occur on reflection at the air-glass interface and the conditions for maxima and minima will remain the same. On the other hand, if the I medium is crown glass (n = 1.52), the II medium is an oil of refractive index 1.60 and the III medium is flint glass (n = 1.66) then a phase change of π will occur at both the reflections and the conditions for maxima and minima would be

$$2nd = \left(m + \frac{1}{2}\right)\lambda \qquad \text{minima} \qquad \textbf{(15.2a)}$$

$$= m\lambda \qquad \text{maxima} \qquad \textbf{(15.2b)}$$

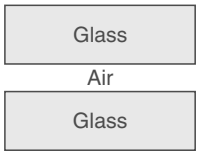

Fig. 15.3 Thin film of air formed between two glass plates.

In general, whenever the refractive index of the II medium lies in between the refractive indices of the I and the III media, then the conditions of maxima and minima would be given by Eqs. (15.2a) and (15.2b).

We next consider the oblique incidence of the plane wave on the thin film (see Fig. 15.4). Once again, the wave reflected from the upper surface of the film interferes with the wave reflected from the lower surface of the film. The latter traverses an additional optical path Δ, which is given by (see Fig. 15.5):

$$\Delta = n_2(BD + DF) - n_1 BC \qquad \textbf{(15.3)}$$

where C is the foot of the perpendicular from the point F on BG. We will show in the next section that

$$\Delta = 2n_2 d \cos\theta' \qquad \textbf{(15.4)}$$

where θ' is the angle of refraction.

For a film placed in air, a phase change of π will occur when reflection takes place at the point B and as such, the conditions of destructive and constructive interference would be given by

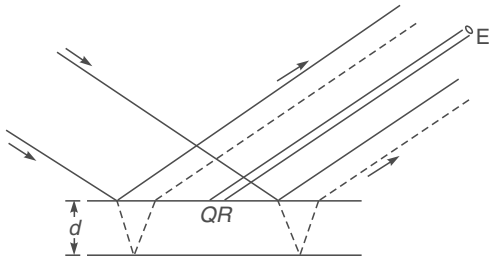

Fig. 15.4 The oblique incidence of a plane wave on a thin film. The solid and dashed lines denote the boundary of the wave reflected from the upper surface and from the lower surface of the film. The eye E receives the light reflected from the region QR.

$$\Delta = 2n_2 d \cos\theta' = m\lambda \qquad \text{minima} \qquad \textbf{(15.5a)}$$

$$= \left(m + \frac{1}{2}\right)\lambda \qquad \text{maxima} \qquad \textbf{(15.5b)}$$

If we place a photographic plate at P (see Fig. 15.5) it will receive uniform illumination; if we try to view the film with naked eye (at the position E — see Fig. 15.4) then only light rays reflected from a small position QR of the film will reach the eye. The image formed at the retina will be dark or bright depending on the value of Δ (see Eq. 15.5).

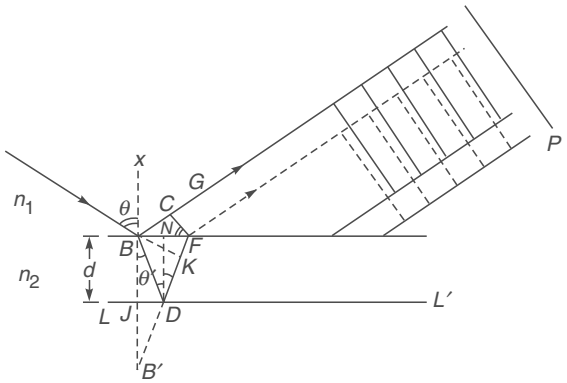

Fig. 15.5 Calculation of the optical path difference between the waves reflected from the upper surface of the film and from the lower surface of the film. The solid and the dashed lines represent the corresponding positions of the crests. P denotes a photographic plate.

15.3 THE COSINE LAW LO 2

In this section, we will show that the wave reflected from the lower surface of the film traverses an additional optical path which is given by the following expression:

$$\Delta\,[= n_2(BD + DF) - n_1 BC] = 2n_2 d \cos\theta' \qquad \textbf{(15.6)}$$

Let θ and θ' denote the angles of incidence and refraction respectively. We drop a perpendicular BJ from the point B on the lower surface LL' and extend BJ and FD to the point B' where they meet (see Fig. 15.5). Clearly,

$$\angle JBD = \angle BDN = \angle NDF = \theta'$$

where N is the foot of the perpendicular drawn from the point D on BF. Now,

$$\angle BDJ = \frac{\pi}{2} - \theta'$$

and $$\angle B'DJ = \pi - \left[\left(\frac{\pi}{2} - \theta'\right) + \theta' + \theta'\right] = \frac{\pi}{2} - \theta'$$

Thus, $BD = BD'$ and $BJ = JB' = d$

or $BD + DF = B'D + DF = B'F$

Hence, $$\Delta = n_2 B'F - n_1 BC \qquad \textbf{(15.7)}$$

Now, $\angle CFB = \angle CBX = \theta$

$$BC = BF \sin\theta = \frac{KF}{\sin\theta'} \sin\theta = \frac{n_2}{n_1} KF \qquad \textbf{(15.8)}$$

where K is the foot of the perpendicular from B on $B'F$. Substituting the above expression for BC in Eq. (15.7), we get

$$\Delta = n_2 B'F - n_2 KF = n_2 B'K$$

or $$\Delta = 2n_2 d \cos\theta' \qquad \textbf{(15.9)}$$

which is known as the **cosine law**.

15.4 NON-REFLECTING FILMS

LO 3

One of the important applications of the thin film interference phenomenon discussed in Sec. 15.2 lies in reducing the reflectivity of lens surfaces; this we plan to discuss in this section. However, for a quantitative understanding of the phenomenon, we will have to assume that when a light beam (propagating in a medium of refractive index n_1) is incident normally on a dielectric of refractive index n_2 then the amplitudes of the reflected and the transmitted beams are related to that of the incident beam through the following relations* [see Fig. 15.6(a)]:

$$a_r = \frac{n_1 - n_2}{n_1 + n_2} a_i \qquad \textbf{(15.10a)}$$

$$a_t = \frac{2n_1}{n_1 + n_2} a_i \qquad \textbf{(15.10b)}$$

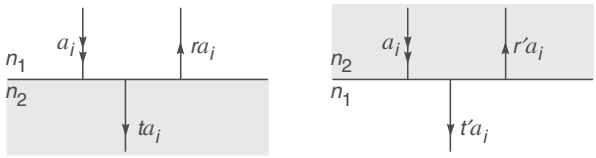

Fig. 15.6 (a) If a plane wave of amplitude a_i, propagating in a medium of refractive index n_1, is incident normally on a medium of refractive index n_2, then the amplitudes of the reflected and the transmitted beams are a_r and a_t, respectively. Similarly, (b) corresponds to the case when the beam (propagating in a medium of refractive index n_2) is incident on a medium of refractive index n_1.

where a_i, a_r and a_t are the amplitudes of the incident beam, reflected beam and the transmitted beam respectively. Notice that when $n_2 > n_1$, a_r is negative showing that when a reflection occurs at a denser medium a phase change of π occurs. The amplitude reflection and transmission coefficients r and t are, therefore, given by [see Sec. 24.2]:

$$r = \frac{n_1 - n_2}{n_1 + n_2} \qquad \textbf{(15.11a)}$$

$$t = \frac{2n_1}{n_1 + n_2} \qquad \textbf{(15.11b)}$$

If r' and t' are the reflection and transmission coefficients where light propagating in a medium of refractive index n_2 is incident on a medium of refractive index n_1 [see Fig. 15.6(b)], then

$$r' = \frac{n_2 - n_1}{n_2 + n_1} = -r \qquad \textbf{(15.12)}$$

$$t' = \frac{2n_2}{n_1 + n_2} \qquad \textbf{(15.13)}$$

and

$$1 - tt' = 1 - \frac{4n_1 n_2}{(n_1 + n_2)^2} = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 = r^2 \qquad \textbf{(15.14)}$$

Equations (15.13) and (15.14) represent the Stokes' relations (see Sec. 14.12).

We will now discuss the application of the thin film interference phenomenon in reducing the reflectivity of lens surfaces. We all know that in many optical instruments (like a telescope) there are many interfaces and the loss of intensity due to reflections can be severe. For example, for near normal incidence,** the reflectivity of crown glass surface in air is

* These relations can be derived from electromagnetic theory; see Eqs. (24.67)–(24.72) (with $\theta_1 = \theta_2 = 0$).

** In all what follows in this section, we will assume near normal incidence.

$$\left(\frac{n-1}{n+1}\right)^2 = \left(\frac{1.5-1}{1.5+1}\right)^2 \simeq 0.04,$$

i.e., 4% of the incident light is reflected. For a dense flint glass $n \simeq 1.67$ and about 6% of light is reflected. Thus, if we have a large number of surfaces, the losses at the interfaces can be considerable. In order to reduce these losses, lens surfaces are often coated with a $\lambda/4n$ thick 'non-reflecting film'; the refractive index of the film being less than that of the lens. For example, glass (n = 1.5) may be coated with an MgF_2 film (see Fig. 15.7) and the film thickness d should be such that*

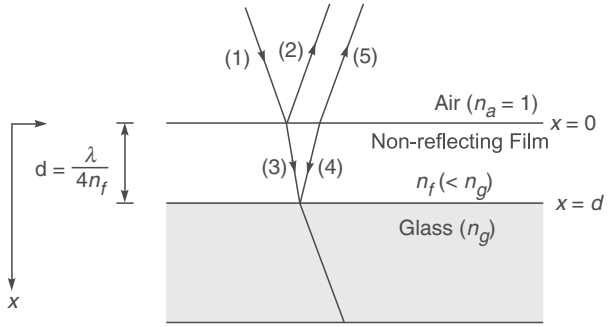

Fig. 15.7 If a film (having a thickness of $\lambda/4n_f$ and having refractive index less than that of the glass) is coated on the glass, then waves reflected from the upper surface of the film destructively interfere with the waves reflected from the lower surface of the film. Such a film is known as a non-reflecting film.

$$2n_f d = \frac{1}{2}\lambda$$

or
$$d = \frac{\lambda}{4n_f} \qquad \textbf{(15.15)}$$

where we have assumed near normal incidence [i.e., $\cos\theta' \cong 1$; see Eq. (15.9)] and n_f represents the refractive index of the film; for MgF_2, n_f= 1.38. Thus, if we assume λ to be 5.0×10^{-5} cm (which roughly corresponds to the center of the visible spectrum), we will have

$$d = \frac{5.0\times10^{-5}\text{ cm}}{4\times1.38} \approx 0.9\times10^{-5}\text{ cm}$$

Figure 15.8 shows a comparison between a grasses lens without anti-reflective coating (top) and a lens with anti-reflective coating (bottom). Note the reflection of the photographer in the top lens and the tinted reflection in the bottom. We would like to emphasize the following points:

(a) Let n_a, n_f and n_g be the refractive indices of air, non-reflecting film and glass, respectively. If a is the amplitude of the incident wave then the amplitudes of the reflected and refracted waves (the corresponding rays shown as (2) and (3) in Fig. 15.7) would be

$$-\frac{n_f - n_a}{n_f + n_a}\,a \text{ and } \frac{2n_a}{n_f + n_a}\,a$$

respectively (we have assumed near normal incidence). The amplitudes of the waves corresponding to rays (4) and (5) would be

$$-\frac{2n_a}{n_f + n_a}\frac{n_g - n_f}{n_g + n_f}\,a$$

and
$$-\frac{2n_a}{n_f + n_a}\frac{n_g - n_f}{n_g + n_f}\frac{2n_f}{n_f + n_a}\,a$$

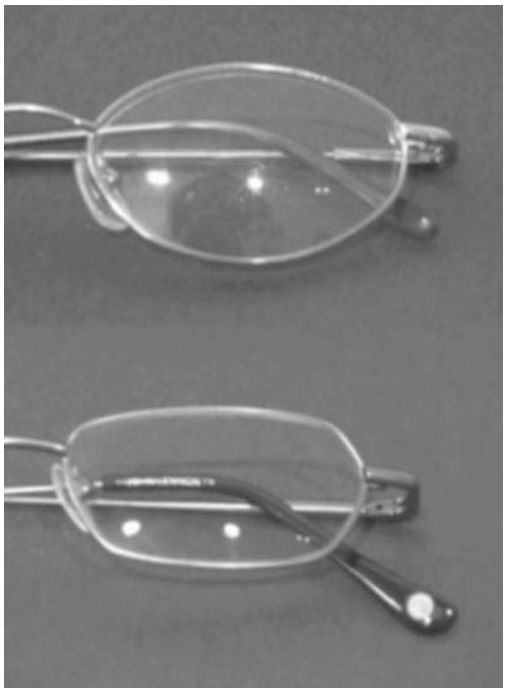

Fig. 15.8 Comparison between a glasses lens without anti-reflective coating (top) and a lens with anti-reflective coating (bottom). Note the reflection of the photographer in the top lens and the tinted reflection in the bottom. [*Figure adapted from http://en.wikipedia.org/wiki/Optical_coating (Photograph taken by Justin Lebar); a color photograph appears as Fig. 18 in the prelim pages–used with permission of Mr. J. Lebar*].

* Since the refractive index of the non-reflecting film is greater than that of air and less than that of the glass, abrupt phase change of π occurs at both the reflections. Consequently, when $2nd\cos\theta' = m\lambda$ there would be constructive interference and when $2nd\cos\theta' = (m+\frac{1}{2})\lambda$ there would be destructive interference.

respectively. Now, for complete destructive interference, the waves corresponding to rays (2) and (5) should have the same amplitude, i.e.,

$$-\frac{n_f - n_a}{n_f + n_a} a = -\frac{2n_a}{n_f + n_a}\frac{n_g - n_f}{n_g + n_f}\frac{2n_f}{n_f + n_a} a \quad \textbf{(15.16)}$$

$$\text{or} \quad \frac{n_f - n_a}{n_f + n_a} = \frac{n_g - n_f}{n_g + n_f} \quad \textbf{(15.17)}$$

where we have used the fact that $\frac{4n_a n_f}{(n_f + n_a)^2}$ is very nearly equal to unity; for $n_a = 1$ and $n_f = 1.4$,

$$\frac{4n_a n_f}{(n_f + n_a)^2} \approx 0.97$$

On simplification we obtain

$$n_f = \sqrt{n_a n_g} \quad \textbf{(15.18)}$$

If the first medium is air then $n_a = 1$ and with $n_g = 1.66$ (dense flint glass) n_f should be 1.29 and when $n_g = 1.5$ (light crown glass) n_f should be 1.22. We note that the refractive indices of magnesium fluoride and cryolite are 1.38 and 1.36, respectively. Now for a $\frac{\lambda}{4n}$ thick film, the reflectivity will be about

$$\left[\frac{n_f - n_a}{n_f + n_a} - \frac{n_g - n_f}{n_g + n_f}\right]^2 \quad \textbf{(15.19)}$$

Thus, for $n_a = 1$, $n_f = 1.38$ and $n_g = 1.5$ the reflectivity will be about 1.3%. In the absence of the film, the reflectivity would have been about 4%. The reduction of reflectivity is much more pronounced for the dense flint glass. This technique of reducing the reflectivity is known as **blooming**.

(b) The film is non-reflecting only for a particular value of λ; in Eq. (15.15) λ was assumed to be 5000 Å. For a polychromatic light, the film's non-reflecting property will be falling off when λ is greater or less than the above value. However, the effect is not serious. For example, for the MgF_2 film on crown glass at 5000 Å, the reflectivity rises by about 0.5% as one goes either to the red or the violet end of the visible spectrum. In Sec. 15.4.2, we will discuss why we should use a $\frac{\lambda}{4n}$ thick film and not $\frac{3\lambda}{4n}$ or $\frac{5\lambda}{4n}$ thick film, although the latter will also give destructive interference for the chosen wavelength.

(c) As in the case of Young's double slit experiment there is no loss of energy; there is merely a redistribution of energy. The energy appears mostly in the transmitted beam.

15.4.1 Mathematical Expressions for the Reflected Waves

It may be worthwhile to carry out a bit of mathematical analysis for the anti-reflecting film shown in Fig. 15.7. We assume $n_g > n_f > n_a$ and that the x-axis is pointing downwards with $x = 0$ at the upper surface of the film. The displacement associated with the incident wave (propagating in the $+x$ direction) is given by

$$y_1 = a \cos(\omega t - k_a x); \quad k_a = \frac{\omega}{c} n_a \quad \textbf{(15.20)}$$

Thus at $x = 0$, $y_1 = a \cos \omega t$. The reflected wave (shown as 2) would therefore be

$$y_2 = -a|r_1| \cos(\omega t + k_a x); \quad \textbf{(15.21)}$$

where

$$|r_1| = \left|\frac{n_f - n_a}{n_f + n_a}\right| \quad \textbf{(15.22)}$$

is a positive quantity. The minus sign in Eq. (15.21) represents the sudden phase change of π at $x = 0$. The transmitted wave (shown as 3) would be given by

$$y_3 = at_1 \cos(\omega t - k_f x); \quad k_f = \frac{\omega}{c} n_f \quad \textbf{(15.23)}$$

where

$$t_1 = \frac{2n_a}{n_f + n_a} \quad \textbf{(15.24)}$$

Thus the displacement at $x = d$ [associated with wave (3)] is

$$y_3 = at_1 \cos(\omega t - k_f d) \quad \textbf{(15.25)}$$

Therefore, the wave reflected from the lower surface [wave (4), which would be propagating in the negative x-direction] is given by

$$y_4 = -at_1|r_2| \cos[\omega t + k_f(x - 2d)]$$

$$|r_2| = \left|\frac{n_g - n_f}{n_g + n_f}\right| \quad \textbf{(15.26)}$$

where the phase factor is adjusted such that at $x = +d$ we obtain the phase given by Eq. (15.25). The wave (5) would therefore be given by

$$y_5 = -at_1|r_2|t_2 \cos[\omega t + k_a x - 2k_f d] \quad \textbf{(15.27)}$$

Assuming the amplitudes of y_2 and y_5 to be approximately the same, destructive interference (between y_2 and y_5) would occur if

$$2k_f d = \pi, 3\pi, \ldots \quad \textbf{(15.28)}$$

or

$$d = \frac{\lambda_f}{4}, \frac{3\lambda_f}{4}, \frac{5\lambda_f}{4}, \ldots \quad \lambda_f = \frac{\lambda}{n_f} \quad \textbf{(15.29)}$$

15.4.2 Rigorous Expressions for Reflectivity

In the above section, we have considered two-beam interference and have neglected multiple reflections at the lower and upper surfaces. The effect of multiple reflections will be discussed in Sec. 16.2; however, such an effect is automatically taken into account when we solve Maxwell's equations incorporating the appropriate boundary conditions. In Sec. 24.4, we will carry out such an analysis and will show that the reflectivity (at normal incidence) of a dielectric film of the type shown in Fig. 15.7 is given by [see Sec. 24.10 and Eq. (24.94)]:*

$$R = \frac{r_1^2 + r_2^2 + 2r_1 r_2 \cos 2\delta}{1 + r_1^2 r_2^2 + 2r_1 r_2 \cos 2\delta} \qquad \textbf{(15.30)}$$

where

$$r_1 = \frac{n_a - n_f}{n_a + n_f} \quad \text{and} \quad r_2 = \frac{n_f - n_g}{n_f + n_g} \qquad \textbf{(15.31)}$$

represent the Fresnel reflection coefficients at the first and second interface, respectively, and

$$\delta = \frac{2\pi}{\lambda} n_f d \qquad \textbf{(15.32)}$$

d being the thickness of the film and, as before, λ represents the free space wavelength. Elementary differentiation shows us that $dR/d\delta = 0$ when $\sin 2\delta = 0$. Indeed for $r_1 r_2 > 0$,

$$\cos 2\delta = -1 \text{ (minima)} \qquad \textbf{(15.33)}$$

represents the condition for minimum reflectivity and when this condition is satisfied, the reflectivity is given by

$$R = \left(\frac{r_1 - r_2}{1 - r_1 r_2}\right)^2 = \left(\frac{n_a n_g - n_f^2}{n_a n_g + n_f^2}\right)^2 \qquad \textbf{(15.34)}$$

where we have used Eq. (15.18). Thus the film is non-reflecting when

$$n_f = \sqrt{n_a n_g}$$

consistent with Eq. (15.18). Now, the condition $\cos 2\delta = -1$ implies

$$2\delta = \frac{4\pi}{\lambda} n_f d = (2m+1)\pi;\ m = 0, 1, 2, \ldots \qquad \textbf{(15.35)}$$

or

$$d = \frac{\lambda}{4n_f}, \frac{3\lambda}{4n_f}, \frac{5\lambda}{4n_f}, \ldots \qquad \textbf{(15.36)}$$

In Fig. 15.9, we have plotted the reflectivity as a function of δ for

$$n_a = 1 \qquad n_g = 1.5 \qquad \textbf{(15.37)}$$

and

$$n_f = \sqrt{n_a n_g} \simeq 1.225$$

As expected, R is maximum ($\approx 4\%$) when $\delta = 0, \pi, 2\pi, \ldots$ and the film is anti-reflecting ($R = 0$) when $\delta = \pi/2, 3\pi/2, \ldots$ implying $d = \lambda/4n_f, 3\lambda/4n_f, \ldots$ As an example, let us suppose that we wish to make the film antireflecting at $\lambda = 6000$ Å; then from Eq. (15.26), the thickness of the film could be

$$1224.5 \text{ Å} \quad \text{or} \quad 3673.5 \text{ Å} \quad \text{or} \quad 6122.5 \text{ Å}, \ldots$$

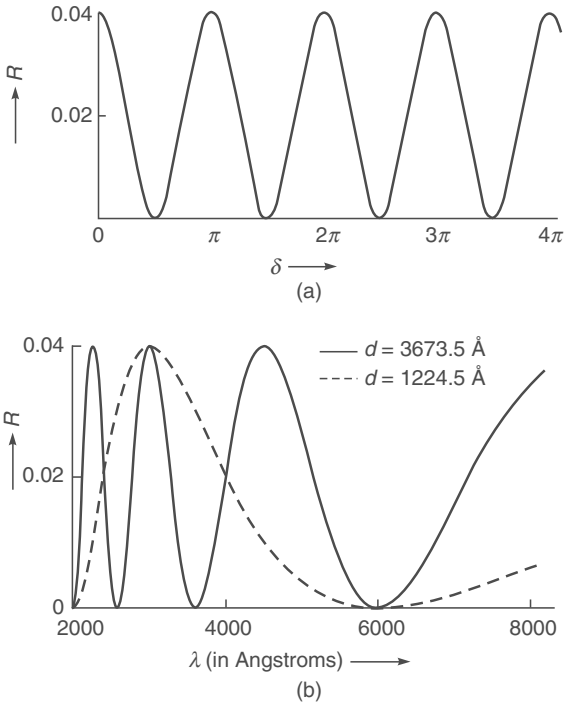

Fig. 15.9 (a) Variation of the reflectivity of a film as a function of δ ($= 2\pi n_f d/\lambda$) for $n_a = 1$, $n_g = 1.5$ and $n_f = \sqrt{n_a n_g} \simeq 1.225$. Notice that the reflectivity is zero for $\delta = \pi/2, 3\pi/2, 5\pi/2, \ldots$ (b) Wavelength variation of the reflectivity for a film of thickness 1224.5 Å (dashed curve) and of thickness 3673.5 Å (solid curve) with $n_a = 1$, $n_g = 1.5$ and $n_f = \sqrt{n_a n_g} \approx 1.225$. Notice that both films are non-reflecting at 6000 Å.

* Equation (15.30) is actually valid even for oblique incidence with r_1, r_2 and δ defined appropriately (see Sec. 24.10).

of the film will reach the eye. If the eye is focussed at infinity then parallel light coming in a particular direction reaching the eye would have originated from nearby points of the extended source and the intensity produced on the retina would depend on the value of $2nd \cos \theta'$ which is the same for all parallel rays like S_1Q, S_2Q', etc. (see Fig. 15.22). Rays emanating in a different direction (like S_1R, S_2R', etc.) would correspond to a different value of θ' and would focus at a different point on the retina. Since θ' is constant over the circumference of a cone (whose axis is normal to the film and whose vertex is at the eye), the eye will see dark and bright concentric rings, with the center lying along the direction $\theta' = 0$. Such fringes, produced by a film of uniform thickness, are known as **Haidinger fringes**. They are also known as **fringes of equal inclination** because the changes in the optical path are due to the changes in the direction of incidence and hence in the value of θ'. In Sec. 15.10 we will discuss the Michelson interferometer where such fringes are observed.

15.8 INTERFERENCE BY A FILM WITH TWO NON-PARALLEL REFLECTING SURFACES

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Till now we have assumed the film to be of uniform thickness. We will now discuss the interference pattern produced by a film of varying thickness. Such a film may be produced by a wedge which consists of two non-parallel plane surfaces [see Fig. 15.23(a)].

We first consider a parallel beam of light incident normally on the upper surface of the film [see Fig. 15.23(a)]. In Fig. 15.23(b) the successive positions of the crests (at a particular instant of time) reflected from the upper surface and from the lower surface of the film are shown by solid and dashed lines, respectively. Obviously, a photographic plate P will record straight line interference fringes which will be parallel to the edge of the wedge (the edge is the line passing through the point O and perpendicular to the plane of the paper). The dots in the figure indicate the positions of maxima. In order to find the distance between two consecutive fringes on the film we note that for the point A to be bright*

$$n(LM + MA) = \left(m + \frac{1}{2}\right)\lambda; \quad m = 0, 1, 2, \ldots \qquad \textbf{(15.56)}$$

[see Fig. 15.23(a)]. However, when the wedge angle ϕ is very small (which is indeed the case for practical systems)

$$LM + MA \approx 2AA'$$

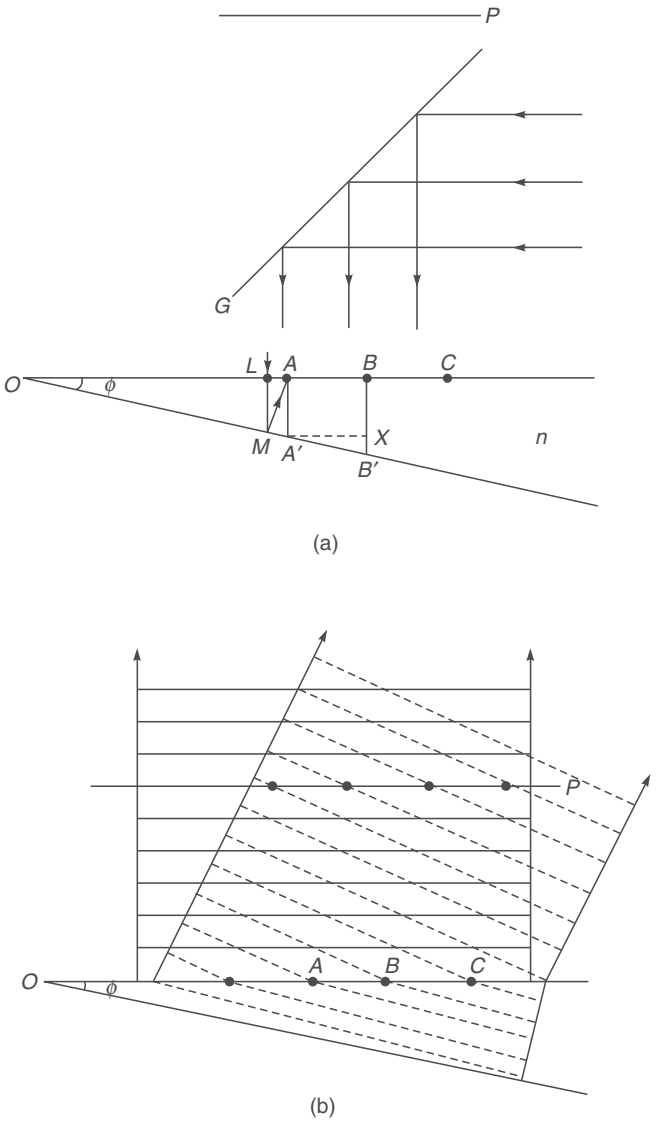

Fig. 15.23 (a) A parallel beam of light incident on a wedge. (b) The solid and the dashed lines represent the positions of the crests (at a particular instant of time) corresponding to the waves reflected from the upper surface and from the lower surface respectively. The maxima will correspond to the intersection of the solid and dashed lines. The fringes will be perpendicular to the plane of the paper.

where AA' represents the thickness of the film at A. Thus, the condition for the point A to be bright is

$$2nAA' \approx \left(m + \frac{1}{2}\right)\lambda \qquad \textbf{(15.57)}$$

* We are assuming here that the beam undergoes a sudden phase change of π when it gets reflected by the upper surface. The expression for the fringe width (Eq. 15.60) is, however, independent of this condition.

Similarly, the next bright fringe will occur at the point B where

$$2nBB' \approx \left(m+\frac{3}{2}\right)\lambda \qquad \textbf{(15.58)}$$

Thus, $2n(BB'-AA') \approx \lambda$

or $$XB' \approx \lambda/2n \qquad \textbf{(15.59)}$$

But $XB' = (A'X)\tan\phi$

or $$A'X = \beta \approx \frac{\lambda}{2n\phi} \qquad \textbf{(15.60)}$$

where β represents the fringe width and we have assumed ϕ to be small. Such fringes are commonly referred to as **fringes of equal thickness**.

On the other hand, for a point source, the fringe pattern will be similar to the parallel film case; i.e., for near normal incidence, the pattern will be very nearly the same as produced by two sources S' and S'' (Fig. 15.24). (Notice that the point S'' is not vertically below S'; this is a consequence of the fact that the two surfaces of the film are not parallel.) The intensity of an arbitrary point Q will be determined by the following equations:

$$[SA + n(AB + BC) + CQ] - [SD + DQ] = \left(m+\frac{1}{2}\right)\lambda \quad \text{maxima} \qquad \textbf{(15.61)}$$

$$= m\lambda \quad \text{minima}$$

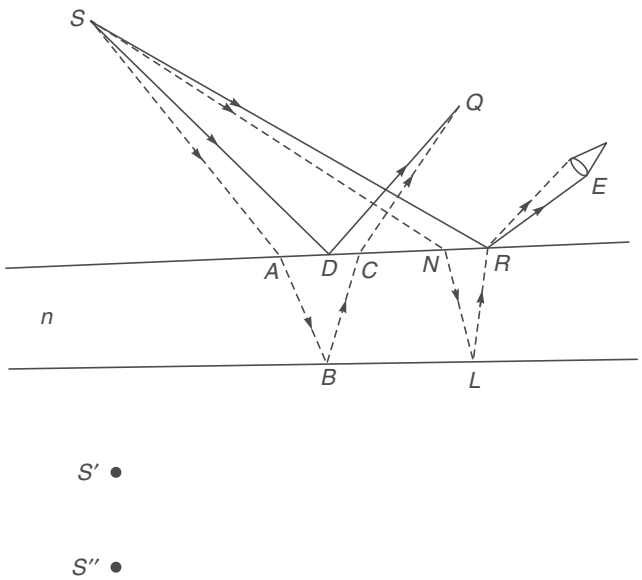

Fig. 15.24 Light from a point source illuminating a wedge. E represents the lens of the eye.

If we view the film with naked eye (say at the position E — see Fig. 15.24) then only a small portion of the film (around the point R) would be visible and the point R will be bright or dark as the optical path difference $[\{SN + n(NL + LR)\} - SR]$ is $\left(m+\frac{1}{2}\right)\lambda$ or $m\lambda$, respectively. One can similarly discuss the case when the eye is focussed for infinity.

We next consider the illumination by an extended source S as shown in Fig. 15.25. Since the extended source can be assumed to consist of a large number of independent point sources, each point source will produce its own pattern on a photographic plate P. Consequently, no definite fringe pattern will be observed.* However, if we view the film with a camera (or with a naked eye) and if the camera is focussed on the upper surface of the film then a particular point on the film will appear dark or bright depending on the fact that whether $2nd$ is $m\lambda$ or $\left(m+\frac{1}{2}\right)\lambda$ (see Fig. 15.25) — we are assuming near normal incidence. It may be seen in the figure that interference at the point Q may occur due to light coming from different points on the extended source, but if the incidence is near normal then the intensity at the point Q will be determined entirely by the thickness of the film at that

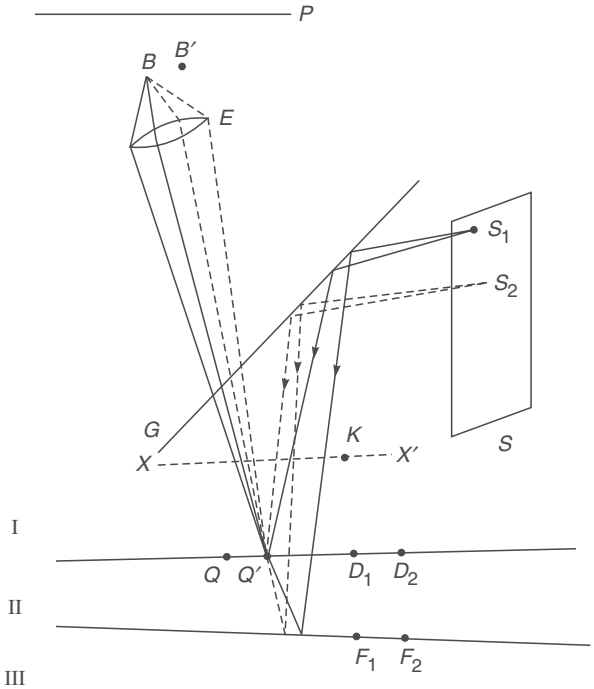

Fig. 15.25 Localized interference fringes produced by an extended source S. Fringes will be seen only when the eye is focussed on the upper surface of the film.

* There is, however, one exception to this, when the extended source is taken to a very large distance, then the light rays reaching the plate G will be approximately parallel and an interference pattern (of low contrast) will be formed on the plate P. The same phenomenon will also occur if instead of moving the extended source we take the plate P far away from the wedge.

place. Similarly, the intensity at the point Q' will be determined by the thickness of the film at Q'; however, the point Q' will be focussed at a different point B' on the retina of the eye. The fringes will be straight lines parallel to the edge of the film OO' (Fig. 15.26). It should be emphasized that all along we are assuming near normal incidence and the fact that the wedge angle is extremely small. These assumptions are indeed valid for practical systems.

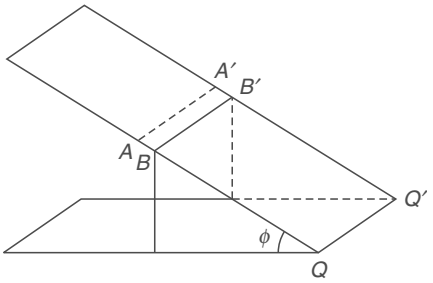

Fig. 15.26 The fringes formed by a wedge will be parallel to the edge OO'.

It is of interest to mention that if we focus the camera on a plane XX', which is slightly above the film, then no definite interference pattern will be observed. This follows from the fact that the light waves reaching the point K from S_2 undergo reflection at the points D_2 and F_2 and the light waves reaching K from S_1 undergo reflection at the points D_1 and F_1. Since the thickness of the film is not uniform, the waves reaching K from S_1 may produce brightness, whereas the waves reaching from S_2 may produce darkness. Thus, in order to view the fringes, one must focus the camera on the upper surface of the film, and in this sense, the fringes are said to be localized. It is left as an exercise for the reader to verify that if the camera is focussed for infinity, no definite interference pattern will be recorded.

Till now we have assumed the film to be 'thin'; the question now arises as to how thin the film should be. In order to obtain an interference pattern, there should be definite phase relationship between the waves reflected from the upper surface of the film and from the lower surface of the film. Thus the path difference Δ (= $2nd \cos \theta'$) should be small compared to the coherence length.* For example, if we are using the D_1 line of an ordinary sodium lamp ($\lambda = 5.890 \times 10^{-5}$ cm), the coherence length is of the order of 1 cm and for fringes to be visible Δ should be much less than 1 cm. It should be pointed out that there is no particular value of Δ for which the fringes disappear; but as the value of Δ increases, the contrast of the fringes becomes poorer. A laser beam has a very high coherence length and fringes can be visible even for path differences much greater than 1 m. On the other hand, if we use a white light source no fringes will be visible for $\Delta \gtrsim 2 \times 10^{-4}$ cm (see Sec. 14.9).

It should be pointed out that interference also occurs in region III (see Fig. 15.27) between the directly transmitted beam and the beam which comes out of the film after suffering two reflections, first from the lower surface and then from the upper surface of the film. However, the two amplitudes will be very different and the fringes will have very poor contrast (see Example 15.1).

Example 15.1 Consider a film of refractive index 1.36 in air. Assuming near normal incidence ($\theta \approx 0$), show that whereas the amplitudes of the reflected rays (1) and (5) (Fig. 15.27) are nearly equal, the amplitudes of the transmitted rays (4) and (7) are quite different. (This is the reason why the fringes observed in transmission have very poor contrast.)

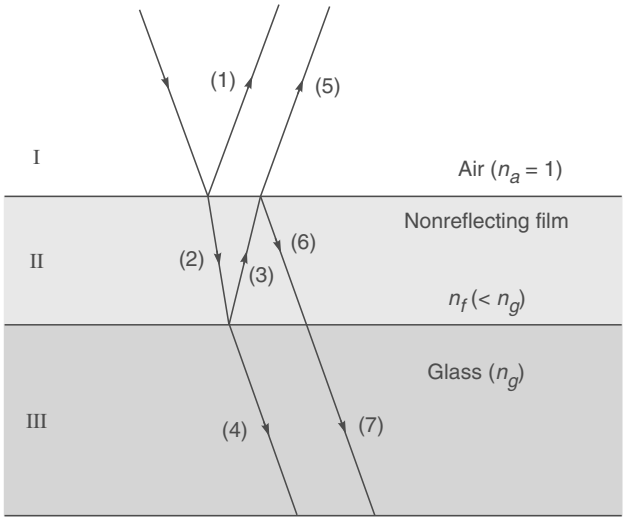

Fig. 15.27 In general, whereas the amplitude of (1) and (5) are nearly the same, the amplitudes of (2) and (6) are quite different.

Solution: Let the amplitude of the incident ray be a and let the amplitudes of the rays (1), (2), (3),…be denoted by a_1, a_2,… etc. Using Eqs. (15.10a) and (15.10b), we get

$$a_1 = \frac{1-n}{1+n}\, a = -\frac{0.36}{2.36}\, a \approx -0.153a$$

$$a_2 = \frac{2}{1+n}\, a = \frac{2}{2.36}\, a \approx 0.847a$$

* Coherence length is defined in Sec. 17.1. If a source remains coherent for a time τ, then the coherence length (L) will be about $c\tau_c$ where c is the speed of light in free space. Thus for $\tau_c \sim 10^{-10}$ sec, $L \sim 3$ cm.

$$a_3 = \frac{n-1}{n+1} a_2 = \frac{0.36}{2.36} \times 0.874a \approx 0.129a$$

$$a_5 = \frac{2n}{n+1} a_3 = \frac{2 \times 1.36}{2.36} \times 0.129a \approx 0.149a$$

$$a_4 = \frac{2n}{1+n} a_2 = \frac{2 \times 1.36}{2.36} \times 0.847a \approx 0.977a$$

$$a_7 = \frac{2n}{n+1} a_6 = \frac{2n}{n+1} \cdot \frac{n-1}{n+1} a_3 = \frac{2 \times 1.36 \times .36}{(2.36)^2} a_3$$

$$\approx 0.023a$$

We first note that the sign of a_5 is opposite to that of a_1 which is a consequence of the fact that a sudden phase change of π occurs when the ray gets reflected at the point B. Further the magnitude of a_5 is nearly equal to that of a_1. On the other hand $|a_7| << |a_4|$. This is the reason why the interference fringes formed in transmission have poor contrast.

15.9 COLORS OF THIN FILMS

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We have seen in the previous section that if light from an extended monochromatic source (like a sodium lamp) is incident normally on a wedge, then equally spaced dark and bright fringes will be observed. The distance between two consecutive bright (or dark) fringes is determined by the wedge angle, the wavelength of light and by the refractive index of the film. If we use a polychromatic source (like an incandescent lamp) we will observe colored fringes. Further, if instead of a wedge we have a film of arbitrarily varying thickness we will again observe fringes, each fringe representing the locus of constant film thickness (see Fig. 15.28). This is indeed what we see when sunlight falls on a soap bubble or on a thin film of oil on water; see Fig. 13 in the prelim pages. It should be mentioned that if the optical path difference between the waves reflected from the upper surface of the film and from the lower surface of the film exceeds a few wavelengths, the interference pattern will be washed out due to the overlapping of interference patterns of many colors and no fringes will be seen (see Sec. 14.9). Thus, in order to see the fringes with white light, the film should not be more than few wavelength thick.

Fig. 15.28 A typical fringe pattern produced by an airfilm formed between two glass surfaces (which are not optically flat) and placed in contact with each other. Whenever the thickness of the airfilm is $m\lambda/2$, we obtain a dark fringe and when the thickness is $\left(m+\frac{1}{2}\right)\lambda/2$, we obtain a bright fringe. Each fringe describes a focus of equal thickness of the film. [*Photograph courtesy: Prof. R.S. Sirohi*].

15.10 NEWTON'S RINGS

If we place a plano-convex lens on a plane glass surface, a thin film of air is formed between the curved surface of the lens (AOB) and the plane glass plate (POQ)—see Fig. 15.29.

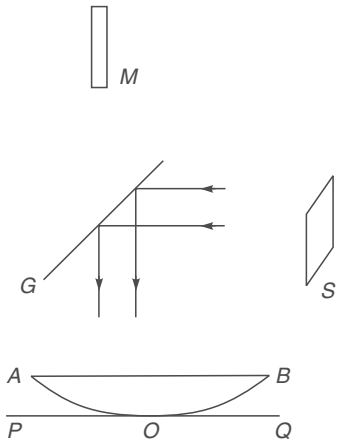

Fig. 15.29 An arrangement for observing Newton's rings. Light from an extended source S is allowed to fall on a thin film formed between the plano-convex lens AOB and the plane glass plate POQ. M represents a traveling microscope.

The thickness of the air film is zero at the point of contact O and increases as one moves away from the point of contact. If we allow monochromatic light (such as from a sodium lamp) to fall on the surface of the lens, then the light reflected from the surface AOB interferes with the light reflected from the surface POQ. For near normal incidence (and considering points very close to the point of contact) the optical path difference between the two waves is very nearly equal to $2nt$, where n is the refractive index of the film and t the thickness of the film. Thus, whenever the thickness of the air film satisfies the condition

$$2nt = \left(m + \frac{1}{2}\right)\lambda;\ m = 0, 1, 2,\ldots \qquad \textbf{(15.62)}$$

we will have maxima. Similarly, the condition

$$2nt = m\lambda \qquad \textbf{(15.63)}$$

will correspond to minima. Since the convex side of the lens is a spherical surface, the thickness of the air film will be constant over a circle (whose center will be at O) and we will obtain concentric dark and bright rings. These rings are known as **Newton's rings**.* It should be pointed out that in order to observe the fringes, the microscope (or the eye) has to be focussed on the upper surface of the film (see the discussion in Sec. 15.7).

The radii of various rings can easily be calculated. As mentioned earlier, the thickness of the air film will be constant over a circle whose center is at the point of contact O. Let the radius of the mth dark ring be r_m and if t is the thickness of the air film where the mth dark ring appears to be formed, then

$$r_m^2 = t(2R - t) \qquad \textbf{(15.64)}$$

where R represents the radius of curvature of the convex surface of the lens (see Fig. 15.30). Now $R \approx 100$ cm and $t \lesssim 10^{-3}$ cm, thus we may neglect t in comparison to $2R$ to obtain

$$r_m^2 \approx 2Rt$$

or

$$2t \approx \frac{r_m^2}{R} \qquad \textbf{(15.65)}$$

Substituting this in Eq. (15.63), we get

$$r_m^2 \approx m\lambda R\ ;\ m = 0, 1, 2,\ldots \qquad \textbf{(15.66)}$$

which implies that the radii of the rings vary as square root of natural numbers. Thus the rings will become close to each other as the radius increases (see Fig. 15.31). Between the two dark rings there will be a bright ring whose radius will be $\sqrt{m + \frac{1}{2}}\ \lambda R$.

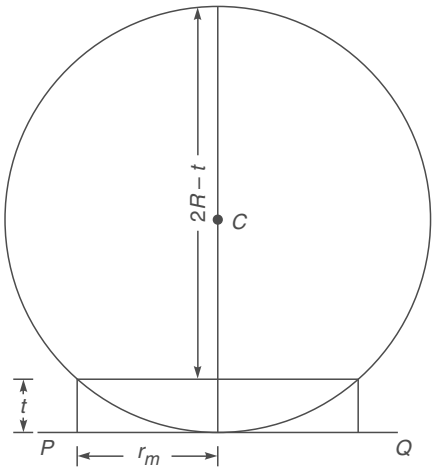

Fig. 15.30 r_m represents the radius of the mth dark ring; the thickness of the air film (where the mth dark ring is formed) is t.

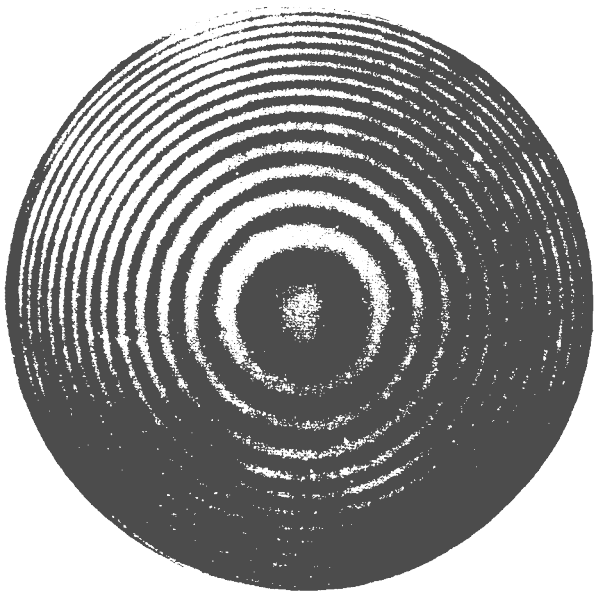

Fig. 15.31 Newton's rings as observed in reflection. The rings observed with transmitted light are of much poorer contrast. [*Photograph courtesy: Dr. G. Bose*].

Newton's rings can easily be observed in the laboratory by using an apparatus as shown in Fig. 15.29. Light from an extended source (emitting almost monochromatic light, like a sodium lamp) is allowed to fall on a glass plate which partially reflects the beam. This reflected beam falls on the plano-convex lens–glass plate arrangement and Newton's rings can easily be observed by viewing directly or through

* Boyle and Hooke had independently observed the fringes earlier but Newton was the first to measure their radii and make an analysis. The proper explanation was given by Thomas Young. Also see 'Milestones' in the beginning of this chapter.

a travelling microscope M. Actually, one really need not have plano-convex lens; the rings would be visible even when a biconvex lens is used.

Typically for $\lambda = 6 \times 10^{-5}$ cm, and $R = 100$ cm

$$r_m = 0.0774 \sqrt{m} \text{ cm} \qquad \textbf{(15.67)}$$

Thus the radii of the first, second and third dark rings would be approximately 0.0774 cm, 0.110 cm and 0.134 cm, respectively. Notice that the spacing between the second and third dark rings is smaller than the spacing between the first and second dark rings.

Equation (15.63) predicts that the central spot should be dark. Normally, with the presence of minute dust particles the point of contact is really not perfect and the central spot may not be perfectly dark. Thus, while carrying out the experiment one should measure the radii of the mth and the $(m+p)$th ring ($p \approx 10$) and take the difference in the squares of the radii ($r^2_{m+p} - r^2_m = p\lambda R$), which is indeed independent of m. Usually, the diameter can be more accurately measured and in terms of the diameters the wavelength is given by the following expression:

$$\lambda = \frac{D^2_{m+p} - D^2_m}{4pR} \qquad \textbf{(15.68)}$$

The radius of curvature can be accurately measured with the help of a spherometer and therefore by carefully measuring the diameters of dark (or bright) rings one can experimentally determine the wavelength.

It may be mentioned that if a liquid of refractive index n is introduced between the lens and the glass plate, the radii of the dark rings would be given by

$$r_m = (m\lambda R/n)^{1/2} \qquad \textbf{(15.69)}$$

Equation (15.69) may be compared with Eq. (15.66). Further, if the refractive indices of the material of the lens and of the glass plate are different and if the refractive index of the liquid lies in between the two values, the central spot will be bright as in Fig. 15.31 and Eq. (15.69) would give the radii of the bright rings.

An important practical application of the principle involved in the Newton's rings experiment lies in the determination of the optical flatness of a glass plate. Consider a glass surface placed on another surface whose flatness is known. If a monochromatic light beam is allowed to fall on this combination and the reflected light is viewed by a microscope, then, in general, dark and bright patches will be seen (Fig. 15.28). The space between the two glass surfaces forms an air film of varying thickness and whenever this thickness becomes $m\lambda/2$, we see a dark spot and when this thickness becomes $\left(m + \frac{1}{2}\right) \lambda/2$ we see a bright spot. Two consecutive dark fringes will be separated by the air film whose thickness will differ by $\lambda/2$. Consequently, by measuring the distance between consecutive dark and bright fringes one can calculate the optical flatness of a glass plate.

When we observe Newton's rings using a white light source, we will have situation similar to that discussed in Sec. 14.9, i.e., we will see only few colored fringes. However, if we put a red filter in front of our eye, the fringe pattern (corresponding to the red color) will suddenly appear. If we replace the red filter by a green filter in front of our eye, the fringe pattern corresponding to the green color will appear; this is similar to the discussion we had in Sec. 14.9.

Example 15.2 Consider the formation of Newton's rings by monochromatic light of $\lambda = 6.4 \times 10^{-5}$ cm. Assume the point of contact to be perfect. Now slowly raise the lens vertically above the plate. As the lens moves gradually away from the plate, discuss the ring pattern as seen through the microscope. Assume the radius of the convex surface to be 100 cm.

Solution: Since the point of contact is perfect, the central spot will be dark, the first dark ring will form at P where $PA = \lambda/2$, and the radius of this ring, OA, will be $\sqrt{\lambda R}$ (= 0.080 cm) –see Fig. 15.32(a). Similarly, the radius of the second dark ring will be $OB = \sqrt{2\lambda R}$ (= 0.113 cm). If we now raise the lens by $\frac{\lambda}{4}$(=1.6×10^{-5} cm) then $2t$ corresponding to the central spot would be $\lambda/2$ and instead of the dark spot at the centre we will now have a bright spot. The radii of the first and the second dark rings will be

$$OA_1 = \left(\frac{1}{2}\lambda R\right)^{1/2} = 0.0566 \text{ cm}$$

and

$$OB_1 = \left(\frac{3}{2}\lambda R\right)^{1/2} = 0.098 \text{ cm}$$

respectively [see Fig. 15.32(b)]. If the lens is further moved by $\lambda/4$ (see Fig. 15.32(c)], then the first dark ring collapses to the center and the central spot will be dark. The ring which was originally at Q now shifts to Q_2; similarly the ring at R [Fig. 15.32(a)] collapses to R_2 [Fig. 15.32(c)].

Thus, as the lens is moved upward the rings collapse to the centre. Hence if we can measure the distance by which the lens is moved upward and also count the number of dark spots that have collapsed to the center, we can determine the wavelength. For example, in the present case, if the lens is moved by 6.4×10^{-3} cm, 200 rings will collapse to the center. If one carries out this experiment it will be observed that the 200th dark ring will slowly converge to the center and when the lens has moved exactly by 6.4×10^{-3} cm it has exactly come to the center.

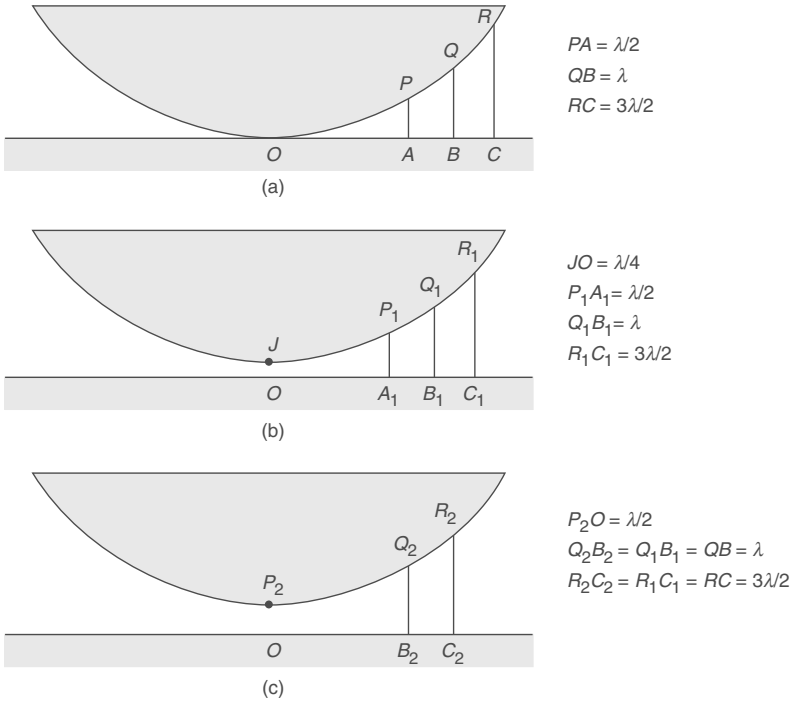

Fig. 15.32 The rings collapse to the center as the lens is moved away from the plate.

Example 15.3 Consider the formation of Newton's rings when two closely spaced wavelengths are present; for example, the D_1 and D_2 lines of sodium (λ_1 = 5890 Å and λ_2 = 5896 Å). What will be the effect of the presence of these two wavelengths as the lens is gradually moved away from the plate? What will happen if the sodium lamp is replaced by a white light source?

Solution: We will first assume that the lens is in contact with the plane glass plate [see Fig. 15.32(a)]. Since the two wavelengths are very close, the bright and dark rings of λ_1 superpose on the bright and dark rings of λ_2, respectively. This can easily be seen by calculating the radii of the ninth dark and bright ring for each wavelength.

For $\lambda = 5.890 \times 10^{-5}$ cm,

$$\text{radius of the ninth bright ring} = \sqrt{\left(9+\frac{1}{2}\right)\lambda R}$$

$$= \sqrt{9.5 \times 5.890 \times 10^{-5} \times 100}$$

$$= 0.236548 \text{ cm}$$

$$\text{radius of the ninth dark ring} = \sqrt{9\lambda R}$$

$$= 0.230239 \text{ cm}$$

Similarly, for $\lambda = 5.896 \times 10^{-5}$ cm,

$$\text{radius of the ninth bright ring} = \sqrt{9.5 \times 5.896 \times 10^{-5} \times 100}$$

$$= 0.236669 \text{ cm}$$

and

$$\text{radius of the ninth dark ring} = \sqrt{9 \times 5.896 \times 10^{-3}}$$

$$= 0.230356 \text{ cm}$$

Thus the rings almost exactly superpose on each other. However, for large values of m, the two ring patterns may produce uniform illumination. To be more specific, when the air film thickness t is such that

$$2t = m\lambda_1 = \left(m+\frac{1}{2}\right)\lambda_2$$

or
$$\frac{2t}{\lambda_2} - \frac{2t}{\lambda_1} = \frac{1}{2} \tag{15.70}$$

then around that point the fringe system will completely disappear; i.e., the bright ring for the wavelength λ_1 will fall on the dark ring for the wavelength λ_2 and conversely. Thus the contrast will be zero and no fringe pattern will be visible. Rewriting Eq. (15.70) we get

$$2t\,\frac{\lambda_1 - \lambda_2}{\lambda_1 \lambda_2} = \frac{1}{2}$$

or
$$2t = \frac{1}{2}\frac{\lambda_1\lambda_2}{\Delta\lambda} \approx \frac{1}{2}\frac{(5.893 \times 10^{-5})^2}{6 \times 10^{-8}}$$

$$\approx 3 \times 10^{-2} \text{ cm}$$

This will correspond to $m \approx 500$.

We shall see the effect of the same phenomenon if we slowly raise the convex lens in the upward direction as we had considered in Example 15.2. Let t_0 be the vertical distance through which the lens has been raised (see Fig. 15.33) and let t_0 be such that it satisfies the following equation:

$$\frac{2t_0}{\lambda_2} - \frac{2t_0}{\lambda_1} = \frac{1}{2}$$

or

$$t_0 = \frac{\lambda_1\lambda_2}{4(\lambda_1 - \lambda_2)}$$

Thus, if the point J (see Fig. 15.33) corresponds to a dark spot for λ_1 then it will correspond to a bright spot for λ_2 and conversely. Further, the nearby dark rings for λ_1, will almost fall at the same place as the bright rings for λ_2 and the interference pattern will be washed out. Thus viewing from a microscope we will not be able to see any ring pattern. Now, if the lens is further moved upwards by a distance t_0, then we will have

$$\frac{2t_1}{\lambda_2} - \frac{2t_1}{\lambda_1} = 1 \qquad \textbf{(15.71)}$$

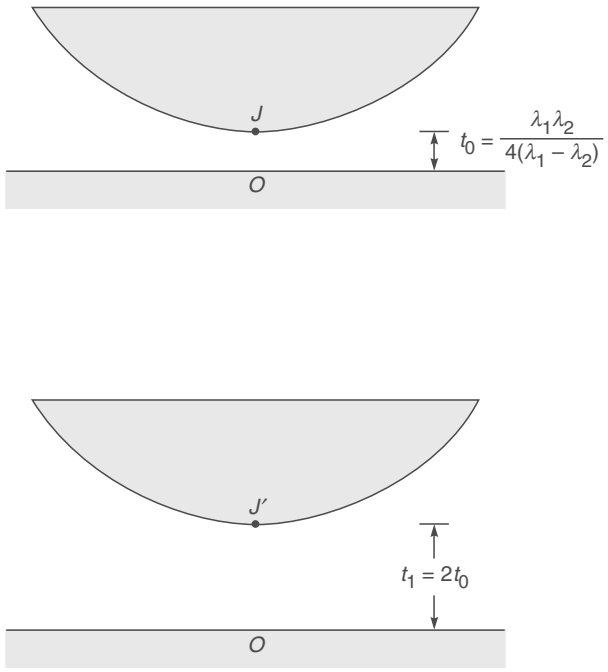

Fig. 15.33 In the Newton's rings experiment, if the light consists of two closely spaced wavelengths λ_1 and λ_2 (like the D_1 and D_2 lines of sodium) then, if the lens is separated by a distance t_0 $\left(= \frac{\lambda_1\lambda_2}{4(\lambda_1 - \lambda_2)}\right)$ interference fringes will be washed out. The fringes will reappear when the distance is $2t_0$.

where $t_1 = 2t_0$. Consequently, if the point J' corresponds to a dark spot for λ_1 then it will also correspond to a dark spot for λ_2. The fringe pattern will reappear but now with a slightly weaker contrast (see also Chapter 17).

In this way if we continue to move the lens upwards the fringe system will reappear every time the lens is moved up by a distance $2t_0$ $\left(\approx \frac{1}{2}\frac{\lambda_1\lambda_2}{\Delta\lambda}\right)$. This principle is used in a Michelson interferometer to measure the small wavelength difference $\Delta\lambda$, between two closely spaced lines (like the D_1 and D_2 lines of sodium).

It should be pointed out that for complete disappearance of the fringe pattern the intensities of the two lines λ_1 and λ_2 should be the same.

Another corollary of the above experiment consists in finding the change in the interference pattern (as we move up the convex lens) when we consider a single line of wavelength λ, but which has a width of $\Delta\lambda$. Thus we should assume all wavelengths between λ and $\lambda + \Delta\lambda$ to exist. By finding the approximate height at which the fringes disappear one can calculate $\Delta\lambda$. The coherence length (L) is related to $\Delta\lambda$ through the following relation (see Sec. 17.2):

$$L \sim \frac{\lambda^2}{\Delta\lambda} \qquad \textbf{(15.72)}$$

15.11 THE MICHELSON INTERFEROMETER* LO 10

A schematic diagram of the Michelson interferometer is shown in Fig. 15.34. S represents a light source (which may be a sodium lamp) and L represents a ground glass plate so that an extended source of almost uniform intensity is formed. G_1 is a beam splitter; i.e., a beam incident on G_1 gets partially reflected and partially transmitted. M_1 and M_2 are good quality plane mirrors having very high reflectivity. One of the mirrors (usually M_2) is fixed and the other (usually M_1) is capable of moving away or towards the glass plate G_1 along an accurately machined track by means of a screw. In the normal adjustment of the interferometer, the mirrors M_1 and M_2 are perpendicular to each other and G_1 is at 45° to the mirror.

Waves emanating from a point P get partially reflected and partially transmitted by the beam splitter G_1, and the two resulting beams are made to interfere in the following manner: The reflected wave [shown as (1) in Fig. 15.34] undergoes a further reflection at M_1 and this reflected wave gets (partially) transmitted through G_1; this is shown as (5) in the figure. The transmitted wave [shown as (2) in Fig. 15.34] gets reflected by M_2 and gets (partially) reflected by G_1 and results in the wave shown as (6) in the figure.

* The Michelson – Morley experiments using the Michelson interferometer to detect the motion of the earth with respect to the "ether" is discussed in Chapter 31.

Waves (5) and (6) interfere in a manner exactly similar to that shown in Fig. 15.22. This can easily be seen from the fact that if x_1 and x_2 are the distances of the mirrors M_1 and M_2 from the plate G_1, then to the eye the waves emanating from the point P will appear to get reflected by two parallel mirrors [M_1 and M_2' – see Fig. 15.34] separated by a distance $(x_1 \sim x_2)$. As discussed in Sec. 15.7, if we use an extended source, then no definite interference pattern will be obtained on a photographic plate placed at the position of the eye. Instead, if we have a camera focused for infinity, then on the focal plane we will obtain circular fringes, each circle corresponding to a definite value of θ (see Figs. 15.22 and 15.35); the circular fringes will look like the ones shown in Fig. 15.36. Now, if the beam splitter is just a simple glass plate, the beam reflected from the mirror M_2 will undergo an abrupt phase change of π (when getting reflected by the beam splitter) and since the extra path that one of the beams will traverse will be $2(x_1 \sim x_2)$, the condition for destructive interference will be

$$2\, d \cos\theta = m\, \lambda$$

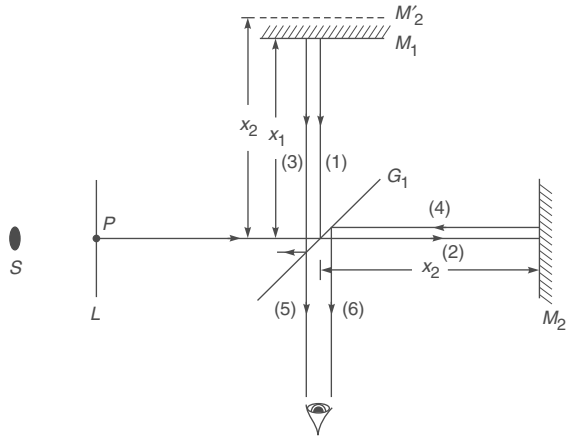

Fig. 15.34 Schematic of the Michelson interferometer.

where $m = 0, 1, 2, 3, \ldots\ldots$ and

$$d = x_1 \sim x_2$$

and the angle θ represents the angle that the rays make with the axis (which is normal to the mirrors as shown in Fig. 15.35). Similarly, the condition for a bright ring would be

$$2\, d \cos\theta = \left(m + \frac{1}{2}\right)\lambda$$

For example, for $\lambda = 6 \times 10^{-5}$ cm if $d = 0.3$ mm, the angles at which the dark rings will occur will be

$$\theta = \cos^{-1}\left(\frac{m}{1000}\right)$$

$$= 0^\circ, 2.56^\circ, 3.62^\circ, 4.44^\circ, 5.13^\circ, 5.73^\circ, 6.28^\circ, \ldots$$

corresponding to $m = 1000, 999, 998, 997, 996, 995, \ldots$ Thus the central dark ring in Fig. 15.36(a) corresponds to $m = 1000$, the first dark ring corresponds to $m = 999$, etc. If we now reduce the separation between the two mirrors so that $d = 0.15$ mm, the angles at which the dark rings will occur will be [see Fig.15.36(b)]

$$\theta = \cos^{-1}\left(\frac{m}{500}\right) = 0^\circ, 3.62^\circ, 5.13^\circ, 6.28^\circ, 7.25^\circ, \ldots$$

where the angles now correspond to $m = 500, 499, 498, 497, 496, 495, \ldots$Thus as we start reducing the value of d, the fringes will appear to collapse at the centre and the fringes become less closely placed. It may be noted that if d is now slightly decreased, say from 0.15 mm to 0.14985 mm,

$$2\, d = 499.5\, \lambda$$

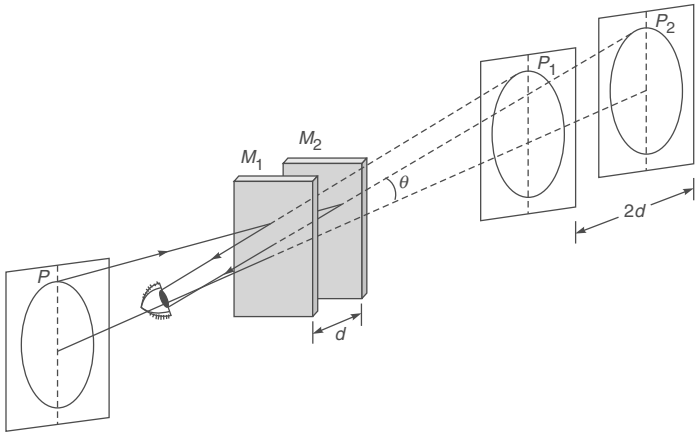

Fig. 15.35 A schematic of the formation of circular fringes [Adapted from Ref. 15.7].

the dark central spot in Fig.15.36(b) (corresponding to m = 500) would disappear and the central fringe will become bright. Thus, as d decreases, the fringe pattern tends to collapse towards the centre. (Conversely, if d is increased, the fringe pattern will expand.) Indeed, if N fringes collapse to the centre as the mirror M_1 moves by a distance d_0, then we must have

$$2\,d = m\,\lambda$$
$$2\,(d - d_0) = (m - N)\,\lambda$$

where we have put $\theta' = 0$ because we are looking at the central fringe. Thus,

$$\lambda = \frac{2d_0}{N} \qquad \textbf{(15.73)}$$

This provides us with a method for the measurement of the wavelength. For example, in a typical experiment, if one finds 1000 fringes collapse to the center as the mirror is moved through a distance of 2.90×10^{-2} cm, then

$$\lambda = 5800 \text{ Å}$$

The above method was used by Michelson for the standardization of the meter. He had found that the red cadmium line (λ = 6438.4696 Å) is one of the ideal monochromatic sources and as such this wavelength was used as a reference for the standardization of the meter. In fact, he defined the meter by the following relation:

1 meter = 1553164.13 red cadmium wavelengths,

the accuracy is almost one part in 10^9.

In an actual Michelson interferometer, the beam splitter G_1 consists of a plate (which may be about 1/2 cm thick), the back surface of which is partially silvered and the reflections occur at the back surface as shown in Fig. 15.37. It is immediately obvious that the beam (5) traverses the glass plate thrice and in order to compensate for this additional path, one introduces a 'compensating plate' G_2 which is exactly of the same thickness as G_1. The compensating plate is not really necessary for a monochromatic source because the additional path $2(n - 1)t$ introduced by G_1 can be compensated by moving the mirror M_1 by a distance $(n - 1)t$ where n is the refractive index of the material of the glass plate G_1.

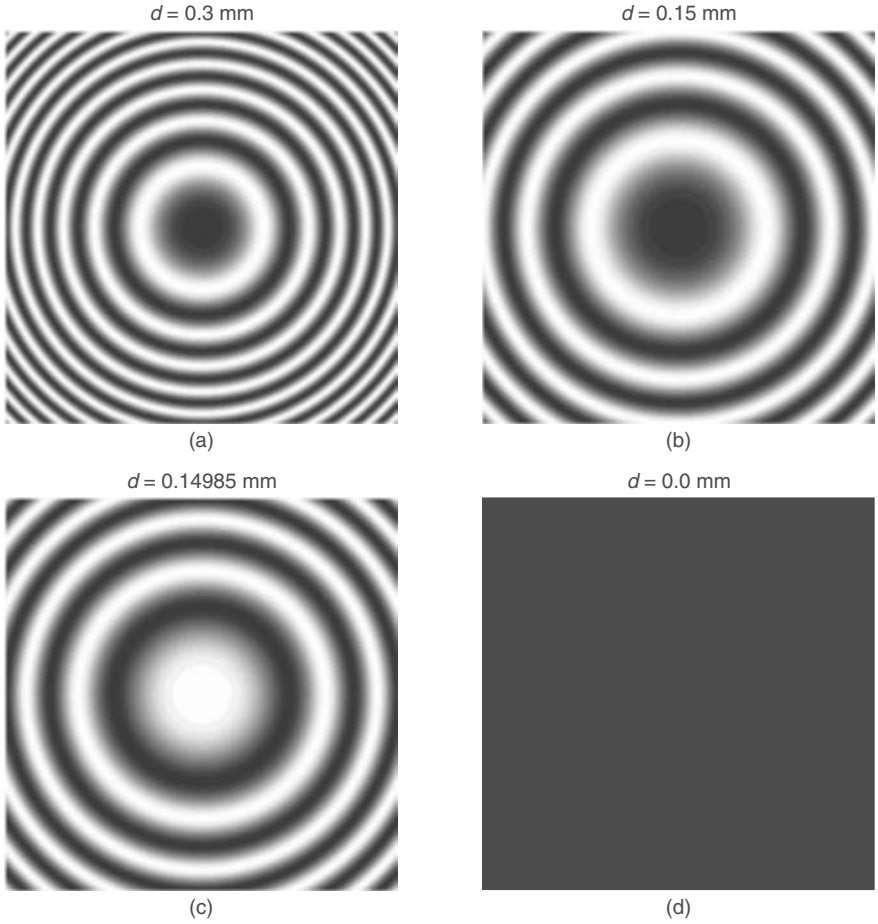

Fig. 15.36 Computer generated interference pattern produced by a Michelson interferometer.

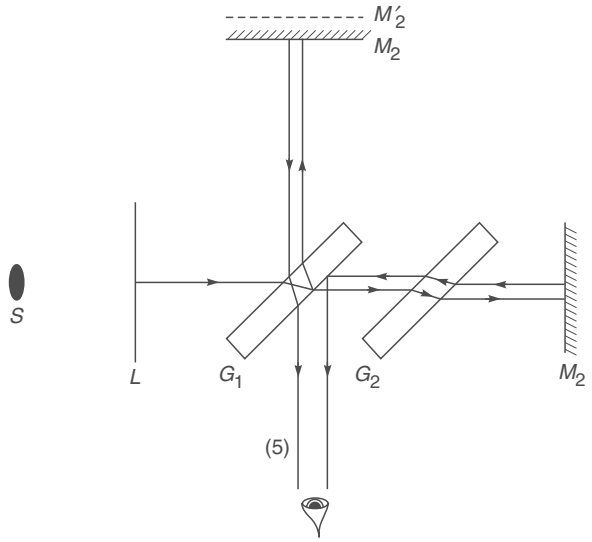

Fig. 15.37 In an actual interferometer there is also a compensating plate G_2.

However, for a white light source it is not possible to simultaneously satisfy the zero path-difference condition for all wavelengths, since the refractive index depends on wavelength. For example, for $\lambda = 6560$ Å and 4861 Å, the refractive index of crown glass is 1.5244 and 1.5330 respectively. If we are using a 0.5 cm thick crown glass plate as G_1, then M_1 should be moved by 0.2622 cm for λ = 6560 Å and by 0.2665 cm for λ = 4861 Å, the difference between the two positions corresponding to over hundred wavelengths! Thus, if we have a continuous range of wavelengths from 4861 Å to 6560 Å, the path difference between any pair of interfering rays (see Fig. 15.34) will vary so rapidly with wavelength that we would observe only a uniform white light illumination. However, in the presence of the compensating plate G_2, one would observe a few colored fringes around the point corresponding to zero path difference (see Sec. 14.9).

Michelson interferometer can also be used in the measurement of two closely spaced wavelengths. Let us assume that we have a sodium lamp which emits predominantly two closely spaced wavelengths 5890 Å and 5896 Å. The interferometer is first set corresponding to the zero path difference.* Near $d = 0$, both the fringe patterns will overlap. If the mirror M_1 is moved away (or towards) the plate G_1 through a distance d, then the maxima corresponding to the wavelength λ_1 will not, in general, occur at the same angle as λ_2. Indeed, if the distance d is such that

$$\frac{2d}{\lambda_1} - \frac{2d}{\lambda_2} = \frac{1}{2} \qquad \textbf{(15.74)}$$

and if $2d \cos \theta' = m\lambda_1$, then $2d \cos \theta' = \left(m + \frac{1}{2}\right)\lambda_2$. Thus, the maxima of λ_1 will fall on the minima of λ_2 and conversely, and the fringe system will disappear. It can easily be seen that if

$$\frac{2d}{\lambda_1} - \frac{2d}{\lambda_2} = 1 \qquad \textbf{(15.75)}$$

then interference pattern will again reappear. In general, if

$$\frac{2d}{\lambda_1} - \frac{2d}{\lambda_2}$$

is 1/2, 3/2, 5/2,…we will have disappearance of the fringe pattern and if it is equal to 1, 2, 3,…then the interference pattern will appear.

Instead of two discrete wavelengths, if the source consists of all wavelengths, lying between λ and $\lambda + \Delta\lambda$, then no interference pattern will be observed if

$$\frac{2d}{\lambda} - \frac{2d}{\lambda + \frac{\Delta\lambda}{2}} \gtrsim \frac{1}{2}$$

or

$$2d \gtrsim \frac{\lambda^2}{\Delta\lambda} \qquad \textbf{(15.76)}$$

In this case the fringes will not reappear because we have a continuous range of wavelengths rather than two discrete wavelengths (see Sec. 17.2).

Example 15.4 For a sodium lamp, the distance traversed by the mirror between two successive disappearances is 0.289 mm. Calculate the difference in the wavelengths of the D_1 and the D_2 lines. Assume λ = 5890 Å.

Solution: When the mirror moves through a distance 0.289 mm, the additional path introduced is 0.578 mm. Thus,

$$\frac{0.578}{\lambda} - \frac{0.578}{\lambda + \Delta\lambda} = 1$$

or

$$\Delta\lambda \simeq \frac{\lambda^2}{0.578} = \frac{(5890 \times 10^{-7})^2}{0.578} \text{ mm}$$
$$\simeq 6 \text{ Å}$$

Summary

- If a plane wave is incident normally on a thin film of uniform thickness d then the waves reflected from the upper surface interfere with the waves reflected from the lower surface. Indeed, for a film of thickness $\lambda/4n_f$ [where λ is the free space wavelength and n_f is the film refractive index which lies between the refractive indices of the two surrounding media], the wave reflected from the upper surface interferes destruc-

* The zero path difference is easily obtained by using white light where only a few coloured fringes, around $d = 0$, will be visible.

tively with the wave reflected from the lower surface and therefore the film acts like an anti-reflection layer.

- A medium consisting of a large number of alternate layers of high and low refractive indices of $n_0 + \Delta n$ and $n_0 - \Delta n$ of equal thickness d is called a periodic medium and the spatial period of the refractive index variation is denoted by Λ ($= 2d$). For $\Delta n << n_0$, if $d \approx \frac{\lambda}{4n_0}$ (where λ is the free space wavelength), the reflections arising out of the individual reflections from the various interfaces would all be in phase and would result in a strong reflection. Thus for strong reflection at a chosen (free space) wavelength λ_B, the period of the refractive index variation should be

$$\Lambda = 2d = \frac{\lambda_B}{2n_0}$$

This is referred to as the Bragg condition. This is the principle of operation of Fiber Bragg gratings.

- If we place a plano-convex lens on a plane glass surface, a thin film of air is formed between the curved surface of the lens and the plane glass plate. If we allow monochromatic light (such as from a sodium lamp) to fall (almost normally) on the surface of the lens, then the light reflected from the curved surface interferes with the light reflected from the plane surface. Since the convex side of the lens is a spherical surface, the thickness of the air film will be constant over a circle and we will see concentric dark and bright rings. These rings are known as Newton's rings. The radii of the concentric rings are such that the difference between the square of the radii of successive fringes is very nearly a constant.
- The Michelson interferometer was used by Michelson for the standardization of the meter. He had found that the red cadmium line (λ = 6438.4696 Å) is one of the ideal monochromatic sources and as such this wavelength was used as a reference for the standardization of the meter. In fact, he defined the meter by the following relation:

1 meter = 1553164.13 red cadmium wavelengths,

the accuracy is almost one part in 10^9.

- Michelson interferometer can also be used in the measurement of two closely spaced wavelengths.

Problems

15.1 A glass plate of refractive index 1.6 is in contact with another glass plate of refractive index 1.8 along a line such that a wedge of 0.5° is formed. Light of wavelength 5000 Å is incident vertically on the wedge and the film is viewed from the top. Calculate the fringe spacing. The whole apparatus is immersed in an oil of refractive index 1.7. What will be the qualitative difference in the fringe pattern and what will be the new fringe width?

15.2 Two plane glass plates are placed on top of one another and on one side a cardboard is introduced to form a thin wedge of air. Assuming that a beam of wavelength 6000 Å is incident normally, and that there are 100 interference fringes per centimeter, calculate the wedge angle.

15.3 Consider a non-reflecting film of refractive index 1.38. Assume that its thickness is 9×10^{-6} cm. Calculate the wavelengths (in the visible region) for which the film will be non-reflecting. Repeat the calculations for the thickness of the film to be 45×10^{-6} cm. Show that both the films will be non-reflecting for a particular wavelength but only the former one will be suitable. Why?

15.4 In the Newton's rings arrangement, the radius of curvature of the curved side of the plano-convex lens is 100 cm. For $\lambda = 6 \times 10^{-5}$ cm what will be the radii of the 9th and 10th bright rings?

15.5 In the Newton's rings arrangement, the radius of curvature of the curved surface is 50 cm. The radii of the 9th and 16th dark rings are 0.18 cm and 0.2235 cm. Calculate the wavelength.

(**Hint:** The use of Eq. (15.66) will give wrong results, why?) (**Ans.** 5015 Å)

15.6 In the Newton's rings arrangement if the incident light consists of two wavelengths 4000 Å and 4002 Å calculate the distance (from the point of contact) at which the rings will disappear. Assume that the radius of curvature of the curved surface is 400 cm.

(**Ans.** 4 cm)

15.7 In Problem 15.6, if the lens is slowly moved upward, calculate the height of the lens at which the fringe system (around the center) will disappear.

(**Ans.** 0.2 mm)

15.8 An equiconvex lens is placed on another equiconvex lens. The radii of curvatures of the two surfaces of the upper lens are 50 cm and those of the lower lens are 100 cm. The waves reflected from the upper and lower surface of the air film (formed between the two lenses) interfere to produce Newton's rings. Calculate the radii of the dark rings. Assume λ = 6000 Å.

[**Ans.** $0.0447\sqrt{m}$ cm]

15.9 In the Michelson interferometer arrangement, if one of the mirrors is moved by a distance 0.08 mm, 250 fringes cross the field of view. Calculate the wavelength.

[**Ans.** 6400 Å]

15.10 The Michelson interferometer experiment is performed with a source which consists of two wavelengths 4882 Å and 4886 Å. Through what distance does the mirror have to be moved between two positions of the disappearance of the fringes?

[**Ans.** 0.298 mm]

15.11 In the Michelson interferometer experiment, calculate the various values of θ' (corresponding to bright rings) for $d = 5 \times 10^{-3}$ cm. Show that if d is decreased to 4.997×10^{-3} cm the fringe corresponding to m = 200 disappears. What will be the corresponding values of θ'? Assume $\lambda = 5 \times 10^{-5}$ cm.

Chapter Eighteen

FRAUNHOFER DIFFRACTION I

No one has ever been able to define the difference between interference and diffraction satisfactorily. It is just a question of usage, and there is no specific, important physical difference between them. The best we can do is, roughly speaking, is to say that when there are only a few sources, say two, interfering, then the result is usually called interference, but if there is a large number of them, it seems that the word diffraction is more often used.

—Richard Feynman in *Feynman Lectures on Physics*, Vol. 1

Learning Objectives

After reading this chapter, the reader should be able to:

LO 1: describe diffraction phenomenon.
LO 2: derive the intensity distribution associated with the single-slit diffraction pattern.
LO 3: analyse diffraction by circular aperture.
LO 4: discuss the directionality of laser beams.
LO 5: discuss the limit of resolution.
LO 6: derive the intensity distributions associated with the 2 slit and N-slit Fraunhofer diffraction patterns.
LO 7: discuss the grating spectrun.
LO 8: discuss the diffraction of a plane wave incident obliquely on a grating.
LO 9: analyse X-ray diffraction and its experimental methods.
LO 10: discuss self-focussing or laser beams.

Important Milestones

1819 *Joseph Fraunhofer demonstrated the diffraction of light by gratings which were initially made by winding fine wires around parallel screws.*
1823 *Fraunhofer published his theory of diffraction.*
1835 *George Airy calculated the (Fraunhofer) diffraction pattern produced by a circular aperture.*

18.1 INTRODUCTION LO 1

Consider a plane wave incident on a long narrow slit of width b (see Fig. 18.1). According to geometrical optics one expects the region AB of the screen SS' to be illuminated and the remaining portion (known as the *geometrical shadow*) to be absolutely dark. However, if the observations are made carefully then one finds that if the width of the slit is not very large compared to the wavelength, then the light intensity in the region AB is not uniform and there is also some intensity inside the geometrical shadow. Further, if the width of the slit is made smaller, larger amounts of energy reach the geometrical shadow. This spreadingout of a wave when it passes through a narrow opening is usually referred to as diffraction and the intensity distribution on the screen is known as the **diffraction pattern**. We will discuss the

phenomenon of diffraction in this chapter and will show that the spreading out decreases with decrease in wavelength. Indeed, since the light wavelengths are very small ($\lambda \sim 5 \times 10^{-5}$ cm), the effects due to diffraction are not readily observed.

Fig. 18.1 If a plane wave is incident on an aperture then according to geometrical optics a sharp shadow will be cast in the region AB of the screen.

We should point out that there is not much of a difference between the phenomena of interference and diffraction, indeed, interference corresponds to the situation when we consider the superposition of waves coming out from a number of point sources and diffraction corresponds to the situation when we consider waves coming out from an area source like a circular or rectangular aperture or even a large number of rectangular apertures (like the diffraction grating).

The diffraction phenomena are usually divided into two categories: (i) Fresnel diffraction and (ii) Fraunhofer diffraction.

In the Fresnel class of diffraction, the source of light and the screen are, in general, at a finite distance from the diffracting aperture [see Fig. 18.2(a)]. In the Fraunhofer class of diffraction, the source and the screen are at infinite distances from the aperture; this is easily achieved by placing the source on the focal plane of a convex lens and placing the screen on the focal plane of another convex lens [see Fig. 18.2(b)]. The two lenses effectively moved the source and the screen to infinity because the first lens makes the light beam parallel and the second lens effectively makes the screen receive a parallel beam of light. It turns out that it is much easier to calculate the intensity distribution of a Fraunhofer diffraction pattern which we plan to do in this chapter. Further, the Fraunhofer diffraction pattern is not difficult to observe; all that one needs is an ordinary laboratory spectrometer; the collimator renders a parallel beam of light and the telescope receives parallel beams of light on its focal plane. The diffracting aperture is placed on the prism table. In Chapter 20, we will study the Fresnel class of diffraction and will discuss the transition from the Fresnel region to the Fraunhofer region.

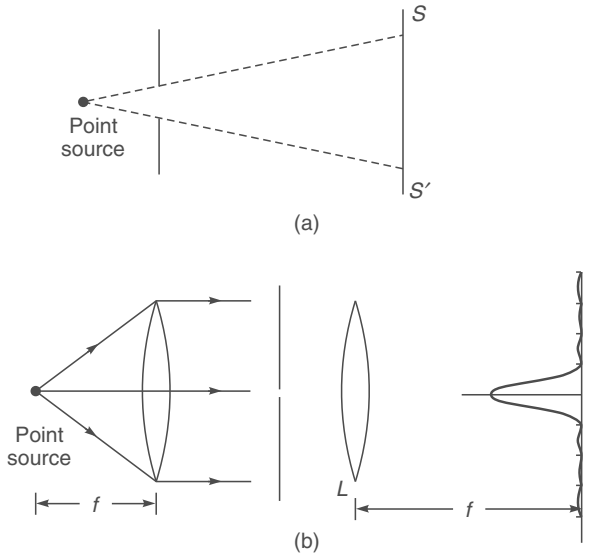

Fig. 18.2 (a) When either the source or the screen (or both) are at finite distances from the aperture, the diffraction pattern corresponds to the Fresnel class. (b) In the Fraunhofer class both the source and the screen are at infinity.

18.2 SINGLE-SLIT DIFFRACTION PATTERN

LO 2

We will first study the Fraunhofer diffraction pattern produced by an infinitely long slit of width b. A plane wave is assumed to fall normally on the slit and we wish to calculate the intensity distribution on the focal plane of the lens L [see Fig. 18.3(a)]. We assume that the slit consists of a large number of equally spaced point sources and that each point on the slit is a source of Huygens' secondary wavelets which interfere with the wavelets emanating from other points. Let the point sources be at $A_1, A_2, A_3, \ldots$ and let the distance between two consecutive points be Δ [see Fig. 18.3(b)]. Thus, if the number of point sources be n, then

$$b = (n-1)\Delta \qquad \textbf{(18.1)}$$

We will now calculate the resultant field produced by these n sources at the point P, P being an arbitrary point (on the focal plane of the lens) receiving parallel rays making an angle θ with the normal to the slit [see Fig. 18.3(b)]. Since the slit actually consists of a continuous distribution of sources, we will, in the final expression, let n go to infinity and Δ go to zero such that $n\Delta$ tends to b.

Now, at the point P, the amplitudes of the disturbances reaching from $A_1, A_2, \ldots$ will be very nearly the same because the point P is at a distance which is very large in comparison to b [see Fig. 18.3(b)]. However, because of even slightly different path lengths to the point P, the field produced by A_1 will differ in phase from the field produced by A_2.

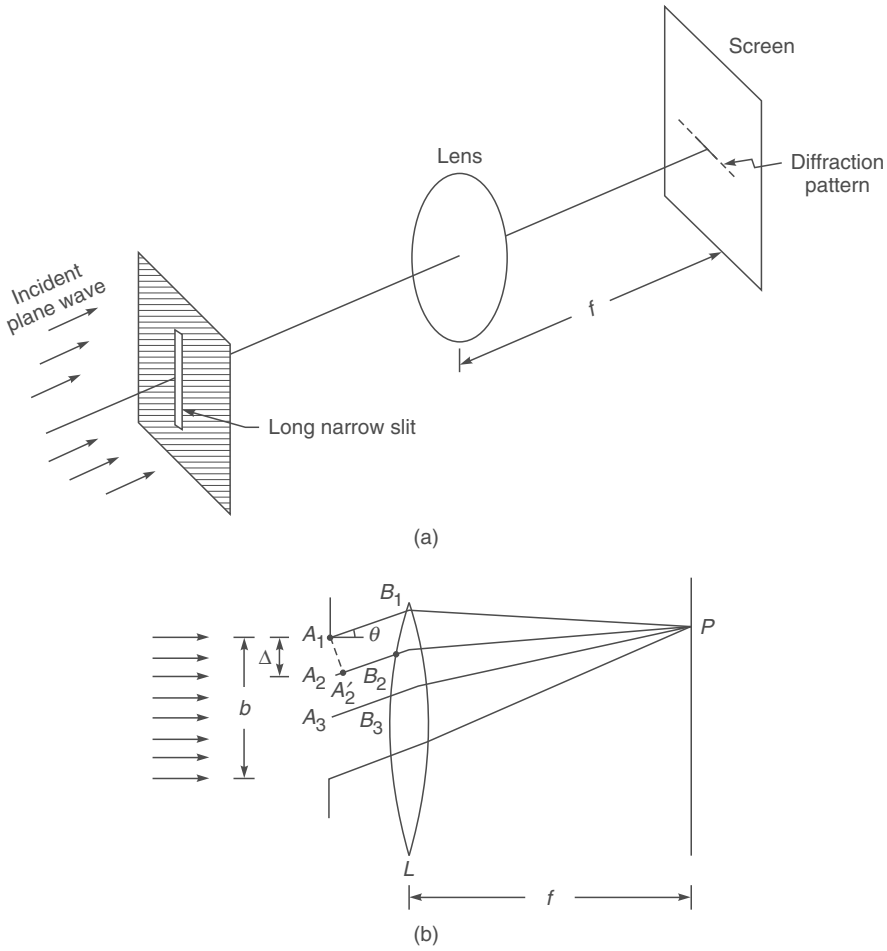

Fig. 18.3 (a) Diffraction of a plane wave incident normally on a long narrow slit of width b. Notice that the spreading occurs along the width of the slit. (b) In order to calculate the diffraction pattern, the slit is assumed to consist of a large number of equally spaced points.

For an incident plane wave, the points A_1, A_2,... are in phase and, therefore, the additional path traversed by the disturbance emanating from the point A_2 will be A_2A_2' where A_2' is the foot of the perpendicular drawn from A_1 on A_2B_2. This follows from the fact that the optical paths A_1B_1P and $A_2'B_2P$ are the same. If the diffracted rays make an angle θ with the normal to the slit then the path difference would be

$$A_2A_2' = \Delta \sin \theta$$

The corresponding phase difference, ϕ, would be given by

$$\phi = \frac{2\pi}{\lambda} \Delta \sin \theta \qquad \textbf{(18.2)}$$

Thus, if the field at the point P due to the disturbance emanating from the point A_1 is $a \cos \omega t$ then the field due to the disturbance emanating from A_2 would be $a \cos (\omega t - \phi)$. Now the difference in the phases of the disturbance reaching from the points A_2 and A_3 will also be ϕ and thus the resultant field at the point P would be given by

$$E = a[\cos \omega t + \cos (\omega t - \phi) + \ldots + \cos (\omega t - (n-1)\phi)] \qquad \textbf{(18.3)}$$

where

$$\phi = \frac{2\pi}{\lambda} \Delta \sin \theta$$

Now, we had shown in Sec. 11.7 that

$$\cos \omega t + \cos (\omega t - \phi) + \ldots + \cos [\omega t - (n-1)\phi]$$
$$= \frac{\sin n\phi/2}{\sin \phi/2} \cos\left[\omega t - \frac{1}{2}(n-1)\phi\right] \qquad \textbf{(18.4)}$$

Thus,

$$E = E_0 \cos\left[\omega t - \frac{1}{2}(n-1)\phi\right] \qquad \textbf{(18.5)}$$

where the amplitude E_θ of the resultant field would be given by*

$$E_\theta = a\frac{\sin(n\phi/2)}{\sin\phi/2} \tag{18.6}$$

In the limit of $n \to \infty$ and $\Delta \to 0$ in such a way that $n\Delta \to b$, we have

$$\frac{n\phi}{2} = \frac{\pi}{\lambda} n\Delta \sin\theta \to \frac{\pi}{\lambda} b \sin\theta$$

Further,

$$\phi = \frac{2\pi}{\lambda}\Delta \sin\theta = \frac{2\pi}{\lambda}\frac{b\sin\theta}{n}$$

would tend to zero and we may, therefore, write

$$E_\theta \approx \frac{a\sin\left(\frac{n\phi}{2}\right)}{\frac{\phi}{2}} = na\,\frac{\sin\frac{\pi b\sin\theta}{\lambda}}{\frac{\pi b\sin\theta}{\lambda}}$$

$$= A\,\frac{\sin\beta}{\beta} \tag{18.7}$$

where**

$$A = na$$

and

$$\beta = \frac{\pi b\sin\theta}{\lambda} \tag{18.8}$$

Thus,

$$E = A\,\frac{\sin\beta}{\beta}\cos(\omega t - \beta) \tag{18.9}$$

The corresponding intensity distribution is given by

$$I = I_0\,\frac{\sin^2\beta}{\beta^2} \tag{18.10}$$

where I_0 represents the intensity at $\theta = 0$

18.2.1 Positions of Maxima and Minima

The variation of the intensity with β is shown in Fig. 18.4(a). It is obvious from Eq. (18.10) that the intensity is zero when

$$\beta = m\pi,\ m \neq 0 \tag{18.11}$$

[When $\beta = 0$, $\frac{\sin\beta}{\beta} = 1$ and $I = I_0$ which corresponds to the maximum of the intensity.] Substituting the value of β one obtains

$$b\sin\theta = m\lambda;\ m = \pm 1, \pm 2, \pm 3, \ldots \text{ (minima)} \tag{18.12}$$

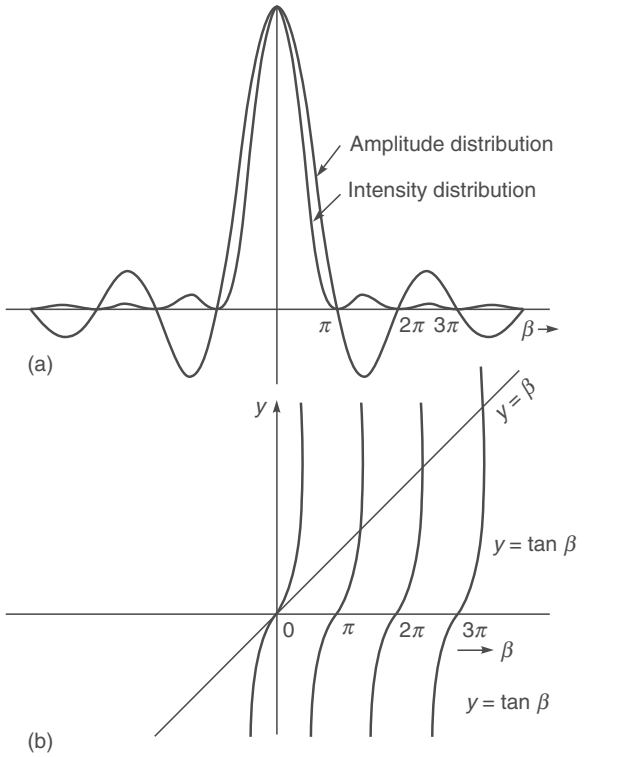

Fig. 18.4 (a) The intensity distribution corresponding to the single slit Fraunhofer diffraction pattern. (b) Graphical method for determining the roots of the equation $\tan\beta = \beta$.

as the conditions for minima. The first minimum occurs at $\theta = \pm\sin^{-1}\left(\frac{\lambda}{b}\right)$; the second minimum at $\theta = \pm\sin^{-1}\left(\frac{2\lambda}{b}\right)$, etc. Since $\sin\theta$ cannot exceed unity, the maximum value of m is the integer which is less than (and closest to) $\frac{b}{\lambda}$.

The positions of minima can directly be obtained by simple qualitative arguments. Let us consider the case $m = 1$. The angle θ satisfies the equation

$$b\sin\theta = \lambda \tag{18.13}$$

We divide the slit into two halves as shown in Fig. 18.5. Consider two points A and A' separated by a distance $b/2$. Clearly the path difference between the disturbances (reaching the point P) emanating from A and A' is $\frac{b}{2}\sin\theta$ which in this case is $\frac{\lambda}{2}$. The corresponding phase difference will be π and the resultant disturbance will be zero. Similarly, the disturbance from the point B will be cancelled by the disturbance reaching from the point B'. Thus, the resultant

* Equation (18.6) represents the amplitude distribution due to the interference of n point sources. Thus, for $n = 2$, the amplitude E_θ becomes $\cos\phi/2$ giving rise to $\cos^2\phi/2$ intensity distribution [cf. Eq. (14.13) of Chapter 14]. Notice that if we have a large number of equidistant sources oscillating in phase, then the propagation is only in cetain directions where the displacements add up in phase.

** We may mention here that in the limit $n \to \infty$ and $a \to 0$ the product na tends to a finite limit.

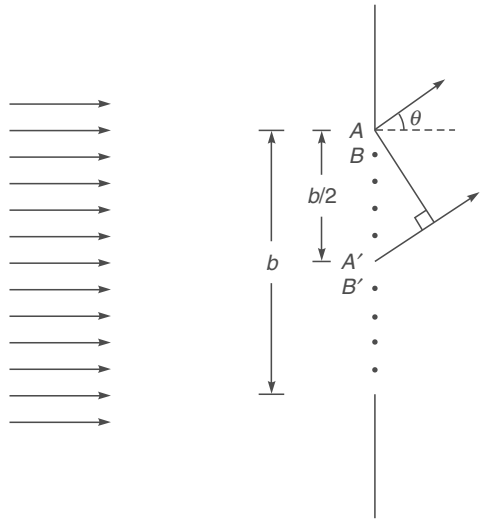

Fig. 18.5 The slit is divided into two halves for deriving the condition for the first minimum.

disturbance due to the upper half of the slit will be canceled by the disturbances reaching from the lower half and the resultant intensity will be zero. In a similar manner when

$$b \sin \theta = 2\lambda \qquad \textbf{(18.14)}$$

we divide the slit into four parts; the first and second quarters cancelling each other and the third and fourth quarters cancelling each other. Similarly when $m = 3$, the slit is divided into six parts and so on.

In order to determine the positions of maxima, we differentiate Eq. (18.10) with respect to β and set it equal to zero. Thus

$$\frac{dI}{d\beta} = I_0 \left[\frac{2 \sin \beta \cos \beta}{\beta^2} - \frac{2 \sin^2 \beta}{\beta^3} \right] = 0$$

or

$$\sin \beta [\beta - \tan \beta] = 0 \qquad \textbf{(18.15)}$$

The condition $\sin \beta = 0$, or $\beta = m\pi$ ($m \neq 0$) correspond to minima. The conditions for maxima are roots of the following transcendental equation

$$\tan \beta = \beta \quad \text{(maxima)} \qquad \textbf{(18.16)}$$

The root $\beta = 0$ corresponds to the central maximum. The other roots can be found by determining the points of intersections of the curves $y = \beta$ and $y = \tan \beta$ [see Fig. 18.4(b)]. The intersections occur at $\beta = 1.43\pi$, $\beta = 2.46\pi$, etc., and are known as the first maximum, the second maximum, etc. Since

$$\left[\frac{\sin (1.43\pi)}{1.43\pi} \right]^2$$

is about 0.0496, the intensity of the first maximum is about 4.96% of the central maximum. Similarly, the intensities of the second and third maxima are about 1.68% and 0.83% of the central maximum, respectively.

Example 18.1 A parallel beam of light is incident normally on a narrow slit of width 0.2 mm. The Fraunhofer diffraction pattern is observed on a screen which is placed at the focal plane of a convex lens whose focal length is 20 cm. Calculate the distance between the first two minima and the first two maxima on the screen. Assume $\lambda = 5 \times 10^{-5}$ cm and that the lens is placed very close to the slit.

Solution:

$$\frac{\lambda}{b} = \frac{5 \times 10^{-5}}{2 \times 10^{-2}} = 2.5 \times 10^{-3}$$

Now, the conditions for diffraction minima are given by $\sin \theta = m\lambda/b$. We assume θ to be small (measured in radians) so that we may write $\sin \theta \approx \theta$ (an assumption which will be justified by subsequent calculations); thus, on substituting the value of λ/b, we get

$$\theta \simeq 2.5 \times 10^{-3} \text{ and } 5 \times 10^{-3} \text{ radians}$$

as the angles of diffraction corresponding to the first and second minima, respectively. Notice that since

$$\sin (2.5 \times 10^{-3}) = 2.4999973 \times 10^{-3}$$

the error in the approximations $\sin \theta \simeq \theta$ is about 1 part in a million! These minima will be separated by a distance $(5 \times 10^{-3} - 2.5 \times 10^{-3}) \times 20 = 0.05$ cm on the focal plane of the lens. Similarly, the first and second maxima occur at

$$\beta = 1.43\pi \quad \text{and} \quad 2.46\pi$$

respectively. Thus,

$$b \sin \theta = 1.43\lambda \quad \text{and} \quad 2.46\lambda$$

or

$$\sin \theta = 1.43 \times 2.5 \times 10^{-3} \quad \text{and} \quad 2.46 \times 2.5 \times 10^{-3}$$

Consequently, the maxima will be separated by the distance given by

$$(2.46 - 1.43) \times 2.5 \times 10^{-3} \times 20 \simeq 0.05 \text{ cm}$$

Example 18.2 Consider, once again, a parallel beam of light ($\lambda = 5 \times 10^{-5}$ cm) to be incident normally on a long narrow slit of width 0.2 mm. A screen is placed at a distance of 3 m from the slit. Assuming that the screen is so far away that the diffraction is essentially of the Fraunhofer type, calculate total width of the central maximum.

Solution: As in Example 18.1, the first minimum occurs at $\theta \simeq 2.5 \times 10^{-3}$ radians; thus the total width of the central maximum is approximately given by

It is of interest to note that if we assume that the angular resolution of the human eye is primarily due to diffraction effects then it will be given by

$$\Delta\theta \sim \frac{\lambda}{D} \approx \frac{6\times10^{-5}}{2\times10^{-1}} = 3\times10^{-4}\ \text{rad.} \qquad \textbf{(18.40)}$$

where we have assumed the pupil diameter to be 2 mm. Thus, at a distance of 20 m, the eye should be able to resolve two points which are separated by a distance

$$3\times10^{-4}\times20 = 6\times10^{-3}\ \text{m} = 6\ \text{mm}$$

One can indeed verify that this result is qualitatively valid by finding the distance at which the millimetre scale will become blurred.

In the above discussion, we have assumed that the two object points produce identical (but displaced) Airy patterns. If that is not the case then the two central maxima will have different intensities; accordingly one has to set up a modified criterion for the limit of resolution such that the two maxima stand out.

18.5.1 Resolving Power of a Microscope

We next consider the resolving power of a microscope objective of diameter D as shown in Fig. 18.28. Let P and Q represent two closely spaced self-luminous point objects which are to be viewed through the microscope. Assuming the absence of any geometrical aberrations, rays emanating from the points P and Q will produce spherical wavefronts (after refraction through the lens) which will form Airy patterns around their paraxial image points P' and Q'. For the points P and Q to be just resolved, the point Q' should lie on the first dark ring surrounding the point P' and, therefore, we must have

$$\sin\alpha' \approx \frac{1.22\,\lambda}{D} = \frac{1.22\,\lambda_0}{n'D} \qquad \textbf{(18.41)}$$

where n and n' represent the refractive indices of the object and image spaces, λ_0 and $\lambda\,(=\lambda_0/n')$ represent the wavelength of light in free space and in the medium of refractive index n' respectively. The angle α' is defined in Fig. 18.28 and we have

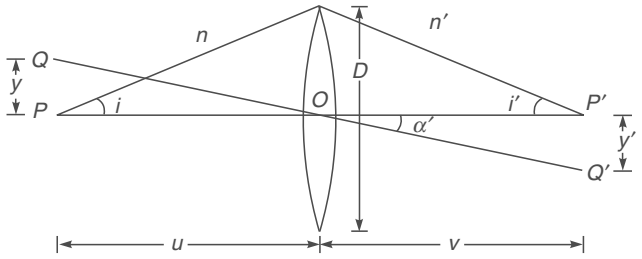

Fig. 18.28 The resolving power of a microscope objective.

$$\sin\alpha' \approx \frac{y'}{OP'} = \frac{y'\tan i'}{D/2} \approx \frac{y'\sin i'}{D/2} \qquad \textbf{(18.42)}$$

where we have assumed $\sin i' \approx \tan i'$, this is justified since the image distance (OP') is large compared to D. Using Eqs. (18.41) and (18.42), we get

$$y' \approx \frac{0.61\,\lambda_0}{n'\sin i'}$$

If we now use the sine law $n'y'\sin i' = ny\sin i$ [see Eq. (4.39) of Chapter 4], we get

$$y \approx \frac{0.61\,\lambda_0}{n\sin i} \qquad \textbf{(18.43)}$$

which represents the smallest distance that the microscope can resolve. The quantity $n\sin i$ is the numerical aperture of the optical system and the resolving power increases with increase in the numerical aperture. It is for this reason that in some microscopes the space between the object and the objective is filled with an oil—and they are referred to as 'oil immersion objectives'. Equation (18.43) also tells us that the resolving power increases with decrease in λ. As such, one often uses blue light (or even ultraviolet light) for the illumination of the object. For example, in an electron microscope the de Broglie wavelength of electrons accelerated to 100 keV is about 0.03×10^{-8} cm and therefore such a microscope has a very high resolving power.

In the above analysis, we have assumed that the two object points are self-luminous so that the intensities can be added up. However, in actual practice, the objects are illuminated by the same source and, therefore, in general, there is some phase relationship between the waves emanating from the two object points; for such a case the intensities will not be strictly additive (see Sec. 14.6), nevertheless Eq. (18.43) will give the correct order for the limit of resolution.

18.6 TWO-SLIT FRAUNHOFER DIFFRACTION PATTERN

LO 6

In Sec. 18.3, we had studied the Fraunhofer diffraction pattern produced by a slit of width b and had found that the intensity distribution consisted of maxima and minima. In this section, we will study the Fraunhofer diffraction pattern produced by two parallel slits (each of width b) separated by a distance d. We would find that the resultant intensity distribution is a product of the single-slit diffraction pattern and the interference pattern produced by two point sources separated by a distance d.

In order to calculate the diffraction pattern we use a method similar to that used for the case of a single slit and assume that the slits consist of a large number of equally spaced point sources and that each point on the slit is a source of Huygens' secondary wavelets. Let the point sources be at $A_1, A_2, A_3, \ldots$ (in the first slit) and at $B_1, B_2, B_3, \ldots$ (in the second slit) [see Fig. 18.29]. As before, we assume that the distance between two consecutive points in either of the slits is Δ. If the diffracted rays make an angle θ with the normal to the plane of the slits, then the path difference between the disturbances reaching the point P from two consecutive points in a slit will be $\Delta \sin \theta$. The field produced by the first slit at the point P will, therefore, be given by [see Eq. (18.9)]

$$E_1 = A \frac{\sin \beta}{\beta} \cos (\omega t - \beta)$$

Similarly, the second slit will produce a field

$$E_2 = A \frac{\sin \beta}{\beta} \cos (\omega t - \beta - \Phi_1)$$

at the point P, where

$$\Phi_1 = \frac{2\pi}{\lambda} d \sin \theta$$

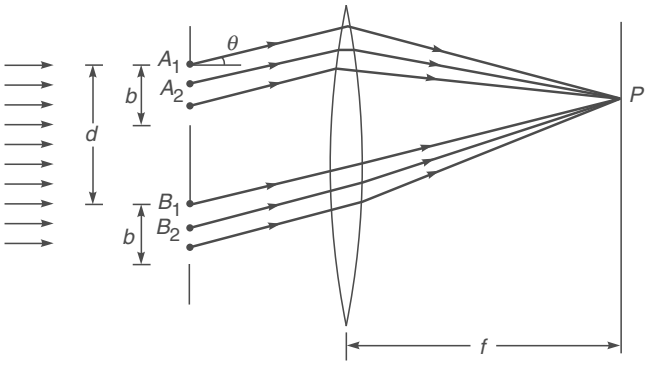

Fig. 18.29 Fraunhofer diffraction of a plane wave incident normally on a double slit.

represents the phase difference between the disturbances (reaching the point P) from two corresponding points on the slits; by corresponding points we imply pairs of points like (A_1, B_1), $(A_2, B_2), \ldots$ which are separated by a distance d. Hence the resultant field will be

$$E = E_1 + E_2$$

$$= A \frac{\sin \beta}{\beta} [\cos (\omega t - \beta) + \cos (\omega t - \beta - \Phi_1)]$$

which represents the interference of two waves, each of amplitude $A \frac{\sin \beta}{\beta}$ and differing in phase by Φ_1. The above equation can be rewritten in the form

$$E = A \frac{\sin \beta}{\beta} \cos \gamma \cos \left(\omega t - \frac{1}{2} \beta - \frac{1}{2} \Phi_1 \right)$$

where

$$\gamma = \frac{\Phi_1}{2} = \frac{\pi}{\lambda} d \sin \theta \qquad \textbf{(18.44)}$$

The intensity distribution will be of the form

$$I = 4 I_0 \frac{\sin^2 \beta}{\beta^2} \cos^2 \gamma \qquad \textbf{(18.45)}$$

where $I_0 \sin^2 \beta/\beta^2$ represents the intensity distribution produced by one of the slits. As can be seen, the intensity distribution is a product of two terms; the first term ($\sin^2 \beta/\beta^2$) represents the diffraction pattern produced by a single slit of width b and the second term ($\cos^2 \gamma$) represents the interference pattern produced by two point sources separated by a distance d. Indeed, if the slit widths are very small (so that there is almost no variation of the $\sin^2 \beta/\beta^2$ term with θ) then one simply obtains the Young's interference pattern (see Sec. 14.6).

In Fig. 18.30, we have shown the two slit diffraction patterns corresponding to d = 0, 0.0176, 0.035 and 0.070 cm with b = 0.0088 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The intensity distribution as predicted by Eq. (18.45) is shown in Fig. 18.31.

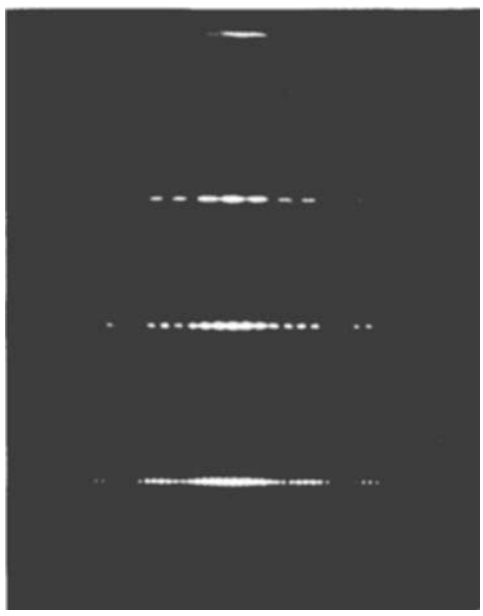

Fig. 18.30 The double-slit Fraunhofer diffraction pattern corresponding to b = 0.0088 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The values of d are 0, 0.0176, 0.035 and 0.070 cm respectively [Adapted from Ref. 17; used with permission].

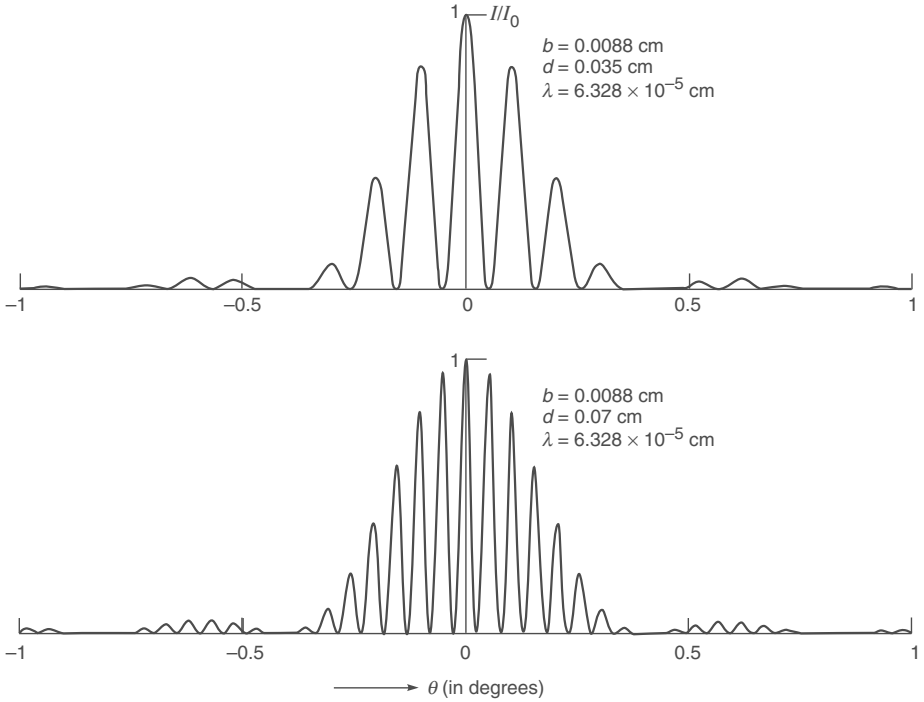

Fig. 18.31 The double-slit intensity distribution as predicted by Eq. (18.45) corresponding to $d = 0.035$ cm and 0.070 cm, respectively (b = 0.0088 cm and $\lambda = 6.328 \times 10^{-5}$ cm).

18.6.1 Positions of Maxima and Minima

Equation (45) tells us that the intensity is zero wherever

$$\beta = \pi, 2\pi, 3\pi, \ldots$$

or when

$$\gamma = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$$

The corresponding angles of diffraction will be given by the following equation:

$$\left.\begin{aligned} b \sin\theta &= m\lambda; \quad (m = 1, 2, 3, \ldots) \\ \text{and} \qquad d \sin\theta &= \left(n + \frac{1}{2}\right)\lambda; \quad (n = 1, 2, 3, \ldots) \end{aligned}\right\} \qquad \textbf{(18.46)}$$

The interference maxima occur when

$$\gamma = 0, \pi, 2\pi, \ldots$$

or when,

$$d \sin\theta = 0, \lambda, 2\lambda, 3\lambda, \ldots \qquad \textbf{(18.47)}$$

The actual positions of the maxima will approximately occur at the above angles provided the variation of the diffraction term is not too rapid. Further, a maximum may not occur at all if θ corresponds to a diffraction minimum, i.e., if $b \sin\theta = \lambda, 2\lambda, 3\lambda, \ldots$ These are usually referred to as missing orders. For example, in Fig. 18.31 we can see that for b = 0.0088 cm, the interference maxima are extremely weak around $\theta \simeq 0.41^\circ$; this is because of the fact that at

$$\theta = \sin^{-1}\left(\frac{\lambda}{b}\right)$$

$$= \sin^{-1}\left[\frac{6.328 \times 10^{-5}}{8.8 \times 10^{-3}}\right] = \sin^{-1}\left[7.19 \times 10^{-3}\right]$$

$$\simeq 0.00719 \text{ radians}$$

$$\simeq 0.412^\circ$$

the first minimum of the diffraction term occurs.

Example 18.9 Consider the case when $b = 8.8 \times 10^{-3}$ cm, $d = 7.0 \times 10^{-2}$ cm and $\lambda = 6.328 \times 10^{-5}$ cm (see Fig. 18.31). How many interference minima will occur between the two diffraction minima on either side of the central maximum? In the experimental arrangement corresponding to Fig. 18.30 the screen was placed at a distance of 15′. Calculate the fringe width.

Solution: The interference minima will occur when Eq. (18.46) is satisfied, i.e., when

$$\sin\theta = \left(n + \frac{1}{2}\right)\frac{\lambda}{d} = 0.904 \times 10^{-3}\left(n + \frac{1}{2}\right);$$

$$n = 0, 1, 2, \ldots$$

$$= 0.452 \times 10^{-3}, 1.356 \times 10^{-3}, 2.260 \times 10^{-3},$$

$$3.164 \times 10^{-3}, 4.068 \times 10^{-3}, 4.972 \times 10^{-3},$$

$$5.876 \times 10^{-3}, 6.780 \times 10^{-3}$$

Thus, there will be sixteen minima between the two first order diffraction minima.

The angular separation between two interference maxima is approximately given by (see Eq. 18.47)

$$\Delta\theta \simeq \frac{\lambda}{d} = 0.904 \times 10^{-4}$$

Thus, the fringe width will be

$$15 \times 12 \times 2.54 \times 0.904 \times 10^{-4} \simeq 0.0413 \text{ cm}$$

18.7 N-SLIT FRAUNHOFER DIFFRACTION PATTERN

LO 6

We next consider the diffraction pattern produced by N parallel slits, each of width b; the distance between two consecutive slits is assumed to be d.

As before, we assume that each slit consists of n equally spaced point sources with spacing Δ (see Fig. 18.32). Thus, the field at an arbitrary point P will essentially be a sum of N terms:

$$E = A\frac{\sin\beta}{\beta}\cos(\omega t - \beta) + A\frac{\sin\beta}{\beta}\cos(\omega t - \beta - \Phi_1) + \ldots + A\frac{\sin\beta}{\beta}\cos(\omega t - \beta - (N-1)\Phi_1) \quad \textbf{(18.48)}$$

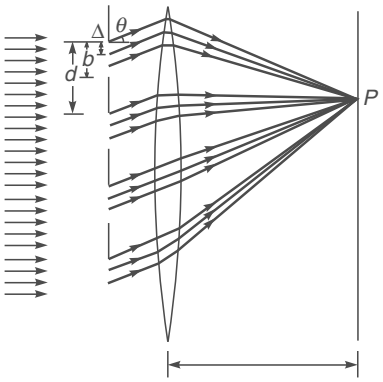

Fig. 18.32 Fraunhofer diffraction of a plane wave incident normally on a multiple slit.

where the first term represents the amplitude produced by the first slit, the second term by the second slit, etc., and the various symbols have the same meaning as in Sec. 18.5. Rewriting Eq. (18.48) we get

$$E = \frac{A\sin\beta}{\beta}\left[\cos(\omega t - \beta) + \cos(\omega t - \beta + \Phi_1) + \ldots + \cos(\omega t - \beta - (N-1)\Phi_1)\right]$$

$$= \frac{A\sin\beta}{\beta}\frac{\sin N\gamma}{\sin\gamma}\cos\left[\omega t - \beta - \frac{1}{2}(N-1)\Phi_1\right] \quad \textbf{(18.49)}$$

where

$$\gamma = \frac{\Phi_1}{2} = \frac{\pi}{\lambda} d\sin\theta$$

The corresponding intensity distribution will be

$$I = I_0\frac{\sin^2\beta}{\beta^2}\frac{\sin^2 N\gamma}{\sin^2\gamma} \quad \textbf{(18.50)}$$

where $I_0 \sin^2\beta/\beta^2$ represents the intensity distribution produced by a single slit. As can be seen, the intensity distribution is a product of two terms; the first term $\left(\frac{\sin^2\beta}{\beta^2}\right)$ represents the diffraction pattern produced by a single slit and the second term $\left(\frac{\sin^2 N\gamma}{\sin^2\gamma}\right)$ represents the interference pattern produced by N equally spaced point sources. For $N = 1$, Eq. (18.50) reduces to the single-slit diffraction pattern [see Eq. (18.10)] and for $N = 2$, to the double slit diffraction pattern [see Eq. (18.45)]. In Figs. 18.33 and 18.34, we have given a plot of the function

$$\frac{\sin^2 N\gamma}{\sin^2\gamma}$$

as a function of γ for $N = 5$ and $N = 12$. One can immediately see that as the value of N becomes very large, the above function would become very sharply peaked at $\gamma = 0, \pi, 2\pi, \ldots$ Between the two peaks, the function vanishes when

$$\gamma = \frac{p\pi}{N}; \quad p = \pm 1, \pm 2, \ldots \quad \text{but} \quad p \neq 0, \pm N, \pm 2N$$

which are referred to as secondary minima.

18.7.1 Positions of Maxima and Minima

When the value of N is very large, one obtains intense maxima at $\gamma \simeq m\pi$, i.e., when

$$d\sin\theta = m\lambda \quad (m = 0, 1, 2, \ldots) \quad \textbf{(18.51)}$$

This can be easily seen by noting that

$$\underset{\gamma\to m\pi}{Lt}\frac{\sin N\gamma}{\sin\gamma} = \underset{\gamma\to m\pi}{Lt}\frac{N\cos N\gamma}{\cos\gamma} = \pm N;$$

thus, the resultant amplitude and the corresponding intensity distributions are given by

$$E = N\frac{A\sin\beta}{\beta} \quad \textbf{(18.52)}$$

and

$$I = N^2 I_0\frac{\sin^2\beta}{\beta^2} \quad \textbf{(18.53)}$$

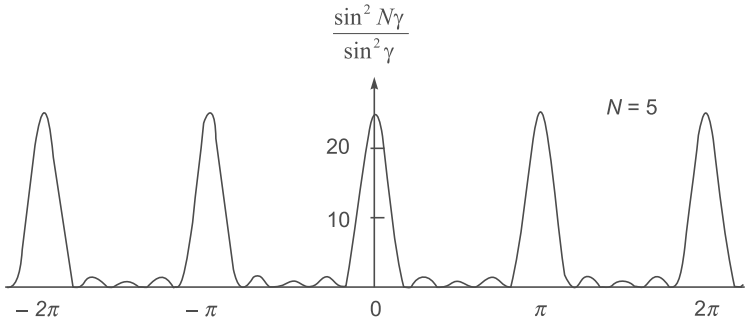

Fig. 18.33 The variation of the function $\sin^2(N\gamma)/\sin^2\gamma$ with γ for $N = 5$.

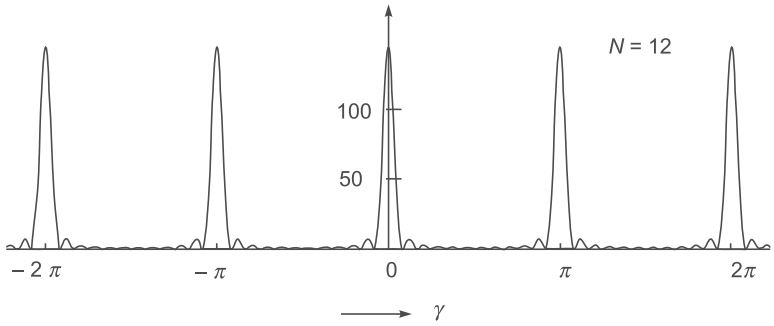

Fig. 18.34 The variation of the function $\sin^2(N\gamma)/\sin^2\gamma$ with γ for $N = 12$. As N becomes larger, the function would become more and more sharply peaked at $\gamma = 0, \pm\pi, \pm 2\pi, \pm 3\pi, \ldots$.

where

$$\beta = \frac{\pi b \sin\theta}{\lambda} = \frac{\pi b}{\lambda}\frac{m\lambda}{d} = \frac{\pi b m}{d} \qquad \textbf{(18.54)}$$

Such maxima are known as **principal maxima**. Physically, at these maxima the fields produced by each of the slits are in phase and, therefore, they add up and the resultant field is N times the field produced by each of the slits; consequently, the intensity has a large value unless $\frac{\sin^2\beta}{\beta^2}$ itself is very small. Since $|\sin\theta| \leq 1$, m cannot be greater than d/λ [see Eq. (18.51)]; thus, there will only be a finite number of principal maxima.

From Eq. (18.50) it can easily be seen that the intensity is zero when either

$$b \sin\theta = n\lambda,\ n = 1, 2, 3, \ldots \qquad \textbf{(18.55)}$$

or

$$N\gamma = p\pi,\ p \neq N, 2N, \ldots \qquad \textbf{(18.56)}$$

Equation (18.55) gives us the minima corresponding to the single slit diffraction pattern. The angles of diffraction corresponding to Eq. (18.56) are

$$d \sin\theta = \frac{\lambda}{N}, \frac{2\lambda}{N}, \ldots, \frac{(N-1)\lambda}{N}, \frac{(N+1)\lambda}{N}, \frac{(N+2)\lambda}{N},$$

$$\ldots, \frac{(2N-1)\lambda}{N}, \frac{(2N+1)\lambda}{N}, \frac{(2N+2)\lambda}{N}, \ldots \qquad \textbf{(18.57)}$$

Thus, between two principal maxima we have $(N-1)$ minima. Between two such consecutive minima the intensity has to have a maximum; these maxima are known as secondary maxima. Typical diffraction patterns for $N = 1$, 2, 3, and 4 are shown in Fig. 18.35 and the intensity distribution as predicted by Eq. (18.50) for $N = 4$ is shown in Fig. 18.36. When N is very large the principal maxima will be much more intense in comparison to the secondary maxima. We may mention here two points:

(a) A particular principal maximum may be absent if it corresponds to the angle which also determines the minimum of the single-slit diffraction pattern. This will happen when

$$d \sin\theta = m\lambda \qquad \textbf{(18.58)}$$

and

$$b \sin\theta = \lambda, 2\lambda, 3\lambda, \ldots \qquad \textbf{(18.59)}$$

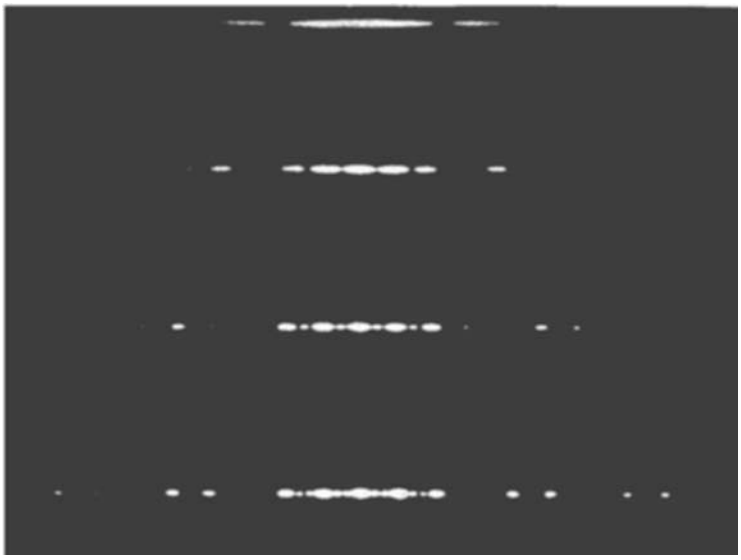

Fig. 18.35 The multiple-slit Fraunhofer diffraction patterns corresponding to b = 0.0044 cm, d = 0.0132 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The number of slits are 1, 2, 3 and 4 respectively (Adapted from Ref. 18.17; used with permission).

are satisfied simultaneously and is usually referred to as a missing order. Even when Eq. (18.59) does not hold exactly (i.e., if $b \sin\theta$ is close to an integral multiple of λ), the intensity of the corresponding principal maximum will be very weak (see, for example, Fig. 18.36 around $\theta \approx 0.8°$).

(b) In addition to the minima predicted by Eq. (18.56), we will also have the diffraction minima (see Eq. 18.55); however, when N is very large, the number of such minima will be very small.

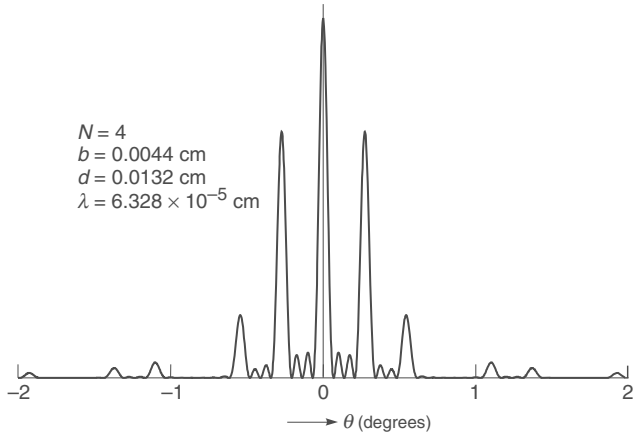

Fig. 18.36 The intensity distribution corresponding to the four-slit Fraunhofer diffraction pattern as predicted by Eq. (18.50) corresponding to b = 0.0044 cm, d = 0.0132 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The principle maxima occur at $\theta \approx 0.275°$, $0.55°$, $0.82°$, $1.1°$, ... Notice the (almost) absent third order.

18.7.2 Width of the Principal Maxima

We have shown above that in the diffraction pattern produced by N slits, the mth order principal maximum occurs at

$$d \sin\theta_m = m\lambda, \; m = 0, 1, 2, \ldots \tag{18.60}$$

Further, the minima occur at the angles given by Eq. (18.57). If $\theta_m + \Delta\theta_{1m}$ and $\theta_m - \Delta\theta_{2m}$ represent the angles of diffraction corresponding to the first minimum on either side of the principal maximum, then $\frac{1}{2}(\Delta\theta_{1m} + \Delta\theta_{2m})$ is known as the **angular half width of the *m*th order principal maximum**. For a large value of N, $\Delta\theta_{1m} \simeq \Delta\theta_{2m}$ which we write as $\Delta\theta_m$. Clearly,

$$d \sin(\theta_m \pm \Delta\theta_m) = m\lambda \pm \frac{\lambda}{N} \tag{18.61}$$

But

$$\sin(\theta_m \pm \Delta\theta_m) = \sin\theta_m \cos\Delta\theta_m \pm \cos\theta_m \sin\Delta\theta_m$$

$$\simeq \sin\theta_m \pm \Delta\theta_m \cos\theta_m \tag{18.62}$$

Thus Eq. (18.61) gives us

$$\Delta\theta_m \simeq \frac{\lambda}{Nd\cos\theta_m} \tag{18.63}$$

which shows that the principal maximum becomes sharper as N increases.

18.8 THE DIFFRACTION GRATING

LO 7

In Sec. 18.6, we have discussed the diffraction pattern produced by a system of parallel equidistant slits. An arrangement which essentially consists of a large number of equidistant slits is known as a **diffraction grating**; the corresponding diffraction pattern is known as the grating spectrum. Since the exact positions of the principal maxima in the diffraction pattern depend on the wavelength, the principal maxima corresponding to different spectral lines (associated with a source) will correspond to different angles of diffraction. Thus the grating spectrum provides us with an easily obtainable experimental set up for determination of wavelengths. From Eq. (18.63) we see that for narrow principal maxima (i.e., sharper spectral lines), a large value of N is required. A good quality grating, therefore, requires a large number of slits (typically about 15,000 per inch). This is achieved by ruling grooves with a diamond point on an optically transparent sheet of material; the grooves act as opaque spaces. After each groove is ruled, the machine lifts the diamond point and moves the sheet forward for the ruling of the next groove. Since the distance between two consecutive grooves is extremely small, the movement of the sheet is obtained with the help of the rotation of a screw which drives the carriage carrying it. Further, one of the important requirements of a good quality grating is that the lines should be as equally spaced as possible; consequently, the pitch of the screw must be constant, and it was not until the manufacture of a nearly perfect screw (which was achieved by Rowland in 1882) that the problem of construction of gratings was successfully solved. Rowland's arrangement gave 14,438 lines per inch, corresponding to $d = 2.54/14438 = 1.759 \times 10^{-4}$ cm. For such a grating, for $\lambda = 6 \times 10^{-5}$ cm, the maximum value of m would be 2, and, therefore, only the first two orders of the spectrum will be observed. However, for $\lambda = 5 \times 10^{-5}$ cm, the third order spectrum will also be visible.

Commercial gratings are produced by taking the cast of an actual grating on a transparent film like that of cellulose acetate. An appropriate strength solution of cellulose acetate is poured on the ruled surface and allowed to dry to form a strong thin film, detachable from the parent grating. These impressions of a grating are preserved by mounting the film between two glass sheets. Nowadays, gratings are also produced holographically, where one records the interference pattern between two plane or spherical waves (see Example 14.5). In contrast to ruled gratings, holographic gratings have a much larger number of lines/cm.

18.8.1 The Grating Spectrum

In Sec. 18.6, we have shown that the positions of the principal maxima are given by

$$d \sin \theta = m\lambda; \quad m = 0, 1, 2, \ldots \qquad \textbf{(18.64)}$$

This relation, which is also called the **grating equation**, can be used to study the dependence of the angle of diffraction θ on the wavelength λ. The zeroeth order principal maximum occurs at $\theta = 0$ irrespective of the wavelength. Thus, if we are using a polychromatic source (e.g., white light) then the central maximum will be of the same colour as the source itself. However, for $m \neq 0$, the angles of diffraction are different for different wavelengths and, therefore, various spectral components appear at different positions. Thus, by measuring the angles of diffraction for various colors one can (knowing the value of m) determine the values of the wavelengths. It may be mentioned that the intensity is maximum for the zeroeth order spectrum (where no dispersion occurs) and it falls off as the value of m increases.

If we differentiate Eq. (18.64), we would obtain

$$\frac{\Delta\theta}{\Delta\lambda} = \frac{m}{d \cos\theta} \qquad \textbf{(18.65)}$$

From this result we can deduce the following conclusions:

(a) Assuming θ to be very small (i.e., $\cos\theta \simeq 1$) we can see that the angle $\Delta\theta$ is directly proportional to the order of spectrum (m) for a given $\Delta\lambda$, so that for a given m, $\Delta\theta/\Delta\lambda$ is a constant. Such a spectrum is known as a **normal spectrum** and in this the difference in angle for two spectral lines is directly proportional to the difference in wavelengths. However, for large θ, it can easily be shown that the dispersion is greater at the red end of the spectrum.

(b) Equation (18.65) tells us that $\Delta\theta$ is inversely proportional to d, and therefore smaller the grating element, the larger will be the angular dispersion.

Figures 18.37 and 18.38 show schematic diagrams of the experimental arrangement for studying the grating spectrum of a polychromatic source. In Fig. 18.37, we have shown a small hole placed at the focal plane of the lens L_1. A parallel beam of white light emerging from L_1 falls on the grating and the diffraction pattern is observed on the focal plane of the lens L_2. If instead of a hole we have a slit at the focal plane of L_1 (see Fig. 18.38) — as it is indeed the case in a typical laboratory set up — we would have parallel beams propagating in different directions, and in the focal plane of the lens L_2 we will have a band spectrum as shown in Fig. 18.38.

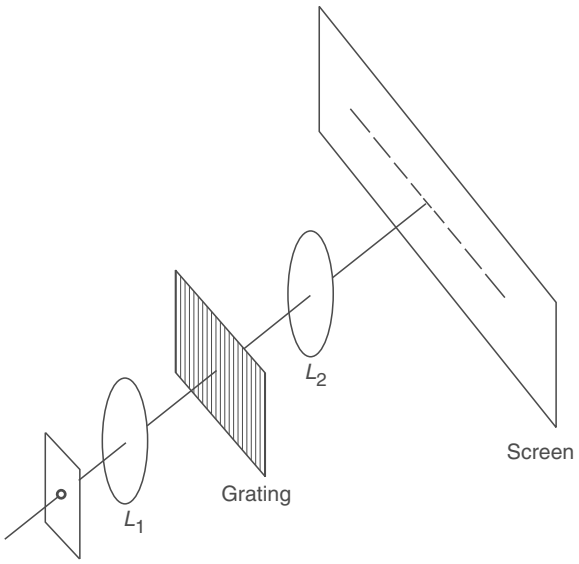

Fig. 18.37 Fraunhofer diffraction of a plane wave incident normally on a grating.

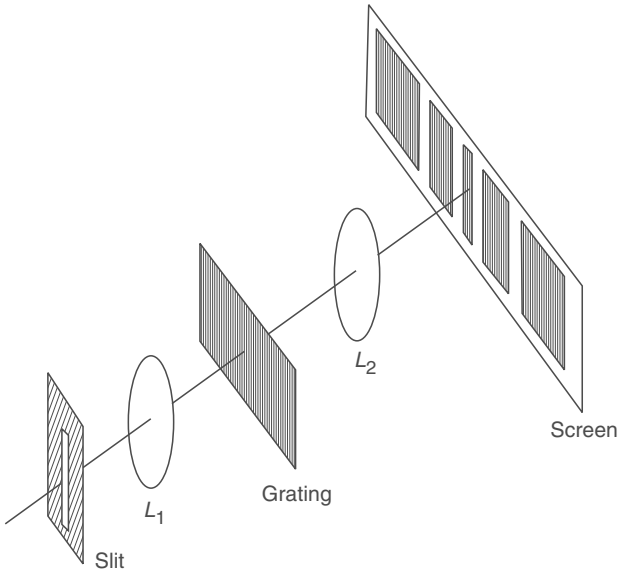

Fig. 18.38 If instead of a point source we have a slit in the focal plane of L_1 then one will obtain bands on the focal plane of L_2.

The lens L_2 is the objective of a telescope and the diffraction pattern is viewed through an eyepiece. The angles of diffraction for various orders of the grating spectrum can be measured and knowing the value of d, one can calculate the wavelength of different spectral lines.

Example 18.10 Consider a diffraction grating with 15,000 lines per inch. **(a)** Show that if we use a white light source the second and third order spectra overlap. **(b)** What will be the angular separation of the D_1 and D_2 lines of sodium in the second order spectra?

Solution: **(a)** The grating element is

$$d = \frac{2.54}{15000} = 1.69 \times 10^{-4} \text{ cm}$$

Let θ_{mv} and θ_{mr} represent the angles of diffraction for the mth order spectrum corresponding to the violet and red colors, respectively. Thus,

$$\theta_{2v} = \sin^{-1} \frac{2 \times 4 \times 10^{-5}}{1.69 \times 10^{-4}} \simeq \sin^{-1} 0.473 \simeq 28.2^\circ$$

$$\theta_{2r} = \sin^{-1} \frac{2 \times 7 \times 10^{-5}}{1.69 \times 10^{-4}} \simeq \sin^{-1} 0.828 \simeq 55.90^\circ$$

and

$$\theta_{3v} = \sin^{-1} \frac{3 \times 4 \times 10^{-5}}{1.69 \times 10^{-4}} \simeq \sin^{-1} 0.710 \simeq 45.23^\circ$$

where we have assumed the wavelengths of the violet and red colours to be 4×10^{-5} cm, and 7×10^{-5} cm, respectively. Since $\theta_{2r} > \theta_{3v}$, the second and third order spectra will overlap. Further since sin $\theta_{3r} > 1$, the third order spectrum for the red colour will not be observed.

(b) Since $d \sin \theta = m\lambda$, we have for small $\Delta\lambda$:

$$(d \cos \theta)\, \Delta\theta = m(\Delta\lambda)$$

or

$$\Delta\theta = \frac{m\Delta\lambda}{d\left\{1 - \left(\frac{m\lambda}{d}\right)^2\right\}^{1/2}}$$

$$\simeq \frac{2 \times 6 \times 10^{-8}}{1.69 \times 10^{-4}\left[1 - \left(\frac{2 \times 6 \times 10^{-5}}{1.69 \times 10^{-4}}\right)^2\right]^{1/2}}$$

$$\simeq 0.0010 \text{ radians} \simeq 3.47'$$

Thus, if we are using telescope of angular magnification 10, the two lines will appear to have an angular separation of 34.7′.

18.8.2 Resolving Power of a Grating

In the case of a grating the resolving power refers to the power of distinguishing two nearby spectral lines and is defined by the following equation:

$$R = \frac{\lambda}{\Delta\lambda} \qquad \textbf{(18.66)}$$

where $\Delta\lambda$ is the separation of two wavelengths which the grating can just resolve; the smaller the value of $\Delta\lambda$, the larger the resolving power.

The Rayleigh criterion (see Sec. 18.4) can again be used to define the limit of resolution. According to this criterion, if the principal maximum corresponding to the wavelength $\lambda + \Delta\lambda$ falls on the first minimum (on the either side of the principal maximum) of the wavelength λ, then the two wavelengths λ and $\lambda + \Delta\lambda$ are said to be just resolved (see Fig. 18.39). If this common diffraction angle is represented by θ and if we are looking at the mth order spectrum, then the two wavelengths λ and $\lambda + \Delta\lambda$ will be just resolved if the following two equations are simultaneously satisfied:

$$d \sin \theta = m(\lambda + \Delta\lambda) \tag{18.67}$$

and

$$d \sin \theta = m\lambda + \frac{\lambda}{N} \tag{18.68}$$

Thus,

$$R = \frac{\lambda}{\Delta\lambda} = mN \tag{18.69}$$

which implies that the resolving power depends on the total number of lines in the grating—obviously on only those lines which are exposed to the incident beam (see the derivation in Sec. 18.6). Further, the resolving power is proportional to the order of the spectrum. Thus, to resolve the D_1 and D_2 lines of sodium ($\Delta\lambda$ = 6 Å) in the first order, N must be at least $(5.89 \times 10^{-5})/(6 \times 10^{-8}) \approx 1{,}000$.

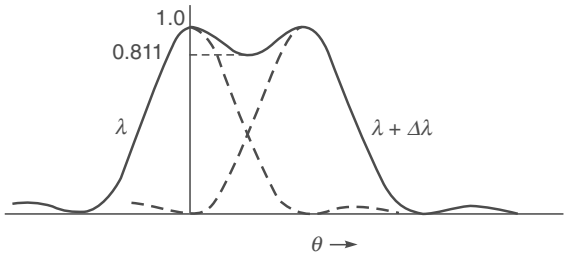

Fig. 18.39 The Rayleigh criterion for the resolution of two spectral lines.

From Eq. (18.69) it appears that the resolving power of the grating would increase indefinitely if N is increased; however, for a given width of the grating $D(= Nd)$, as N is increase, d decreases and therefore the maximum value of m also decreases. Thus if d becomes 2.5λ, only first and second order spectra will be seen and if it is further reduced to about 1.5λ then only the first order spectrum will be seen.

18.8.3 Resolving Power of a Prism

We conclude this section by calculating the resolving power of a prism. Figure 18.40 gives a schematic description of the experimental arrangement for observing the prism spectrum which is determined through the following formula:

$$n(\lambda) = \frac{\sin \dfrac{A + \delta(\lambda)}{2}}{\sin \dfrac{A}{2}} \tag{18.70}$$

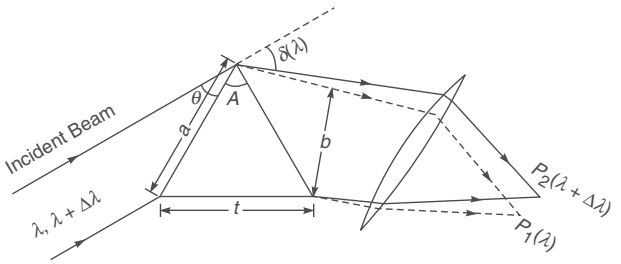

Fig. 18.40 The schematic of the experimental arrangement to observe the prism spectrum. P_1 and P_2 represent the images corresponding to λ and $\lambda + \Delta\lambda$, respectively.

where A represents the angle of the prism and δ the angle of minimum deviation. We assume that the refractive index decreases with λ (which is usually the case) so that δ also decreases with λ. In Fig. 18.40, the points P_1 and P_2 represent the images corresponding to λ and $\lambda + \Delta\lambda$, respectively. We are assuming that $\Delta\lambda$ is small so that the same position of the prism corresponds to the minimum deviation position for both wavelengths. In an actual experiment one usually has a slit source (perpendicular to the plane of the paper) forming line images at P_1 and P_2. Since the faces of the prism are rectangular, the intensity distribution will be similar to that produced by a slit of width b (see Sec. 18.2)*. For the lines to be just resolved the first diffraction minimum [m = 1 in Eq. (18.12)] of λ should fall at the central maximum of $\lambda + \Delta\lambda$, thus we must have

$$\Delta\delta \approx \frac{\lambda}{b} \tag{18.71}$$

In order to express $\Delta\delta$ in terms of $\Delta\lambda$, we differentiate Eq. (18.70):

$$\frac{dn}{d\lambda} = \frac{1}{\sin \dfrac{A}{2}} \cos\left[\frac{A + \delta(\lambda)}{2}\right] \frac{1}{2} \frac{d\delta}{d\lambda}$$

Thus,

$$\Delta\delta = \frac{2 \sin \dfrac{A}{2}}{\cos \dfrac{A + \delta(\lambda)}{2}} \frac{dn}{d\lambda} \Delta\lambda$$

* Since we have a slit source we need not consider the diffraction in a direction perpendicular to the plane of the diagram.

Chapter Twenty

FRESNEL DIFFRACTION

One of your commissioners, M. Poisson, had deduced from the integrals reported by the author [Fresnel] the singular result that the centre of the shadow of an opaque circular screen must, when the rays penetrate there at incidences which are only a little oblique, be just as illuminated as if the screen did not exist. The consequence have been submitted to the test of a direct experiment, and observation has perfectly confirmed the calculation.

—Dominique Arago to the French Academy of Sciences*

Learning Objectives

After reading this chapter, the reader should be able to:

LO 1: analyze Fresnel half-period zones and its applications.
LO 2: describe a more rigorous approach to Fresnel diffraction.
LO 3: describe propagation of Gaussian beams.
LO 4: study diffraction by a straight edge.
LO 5: evaluate the diffraction of a plane wave by a long narrow slit and transition to the Fraunhofer region.

Important Milestones

1816 *Augustin Fresnel developed the theory of diffraction using the wave theory of light.*
1817 *Using Fresnel's theory, Poisson predicted a bright spot at the center of the shadow of an opaque disc—this is usually referred to as the 'Poisson spot'.*
1818 *Fresnel and Arago carried out the experiment to demonstrate the existence of the Poisson spot validating the wave theory.*
1874 *Marie Cornu developed a graphical approach to study Fresnel diffraction—this came to be known as the Cornu's spiral.*

20.1 INTRODUCTION

In Chapter 18, we had mentioned that the phenomenon of diffraction can be broadly classified under two categories: under the first category comes the Fresnel class of diffraction in which either the source or the screen (or both) are at a finite distance from the diffracting aperture. In the second category comes the Fraunhofer class of diffraction (discussed in the last two chapters) in which the wave incident on the aperture is a plane wave and the diffraction pattern is observed on the focal plane of a convex lens, so that the screen is effectively at an infinite distance from the aperture.

In the present chapter, we will discuss the Fresnel class of diffraction and also study the transition to the Fraunhofer region. The underlying principle in the entire analysis is the Huygens–Fresnel principle according to which:

> *Each point on a wavefront is a source of secondary disturbance and the secondary wavelets emanating from different points mutually interfere.*

In order to appreciate the implications of this principle we consider the incidence of a plane wave on a circular hole of radius a as shown in Fig. 20.1. In Sec. 18.3, we had shown that the beam will undergo diffraction divergence and the angular spreading will be given by

* The author found this quotation in Ref. 20.1.

$$\Delta\theta \sim \frac{\lambda}{2a}$$

Thus, when $a >>> \lambda$ the intensity at a point R (which is deep inside the geometrical shadow) will be negligible; on the other hand, if $a \sim \lambda$ there will be almost uniform spreading out of the beam resulting in an (almost) uniform illumina-tion of the screen. This phenomenon is a manifestation of the fact that when $a >>> \lambda$, the secondary wavelets emanating from different points on the circular aperture so beautifully interfere to produce (almost) zero intensity in the geometrical shadow and a large intensity inside the circular region (see Fig. 20.1). However, if $a \sim \lambda$ then the aperture almost acts as a point source resulting in a uniform illumination of the screen (see Fig. 12.4 and Fig. 22 in prelims pages).

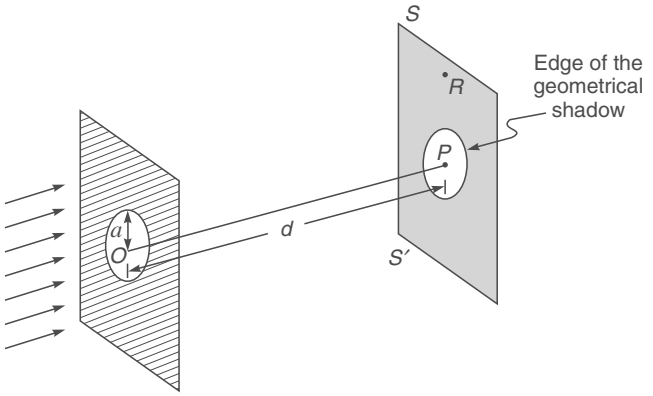

Fig. 20.1 Diffraction of a plane wave incident normally on a circular aperture of radius a.

We will first introduce the concept of Fresnel half-period zones to have a qualitative understanding of the Fresnel diffraction pattern; this will be followed by a more rigorous analysis of the Fresnel class of diffraction and its transition to the Fraunhofer region.

20.2 FRESNEL HALF-PERIOD ZONES

LO 1

Let us consider a plane wavefront WW′ propagating in the z-direction as shown in Fig. 20.2. In order to determine the field at an arbitrary point P due to the disturbances reaching from different portions of the wavefront, we make the following construction: from the point P we drop a perpendicular PO on the wavefront. If $PO = d$, then with point P as centre we draw spheres of radii $d + \lambda/2$, $d + 2\lambda/2$, $d + 3\lambda/2$,…; these spheres will intersect WW′ in circles as shown in Fig. 20.2. The radius of the nth circle will obviously be given by

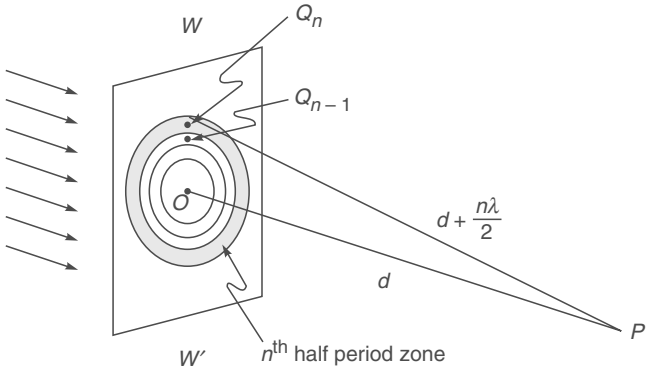

Fig. 20.2 Construction of Fresnel half-period zones.

$$r_n = \left[\left(d + n\frac{\lambda}{2}\right)^2 - d^2\right]^{1/2}$$

$$= \sqrt{n\lambda d}\left[1 + \frac{n\lambda}{4d}\right]^{1/2}$$

or

$$r_n \approx \sqrt{n\lambda d} \qquad \textbf{(20.1)}$$

where we have assumed $d >>> \lambda$, this is indeed justified for practical systems using visible light; of course, we are assuming that n is not a very large number. The annular region between the n^{th} circle and $(n-1)^{th}$ circle is known as the n^{th} **half-period zone**; the area of the n^{th} half-period zone is given by

$$A_n = \pi r_n^2 - \pi r_{n-1}^2$$

$$\approx \pi[n\lambda d - (n-1)\lambda d] = \pi\lambda d \qquad \textbf{(20.2)}$$

Thus the areas of all the half-period zones are approximately equal. Now the resultant disturbance produced by the nth zone will be π out-of-phase with the disturbance produced by the $(n-1)^{th}$ [or the $(n+1)^{th}$] zone. This can easily be seen from the following consideration: For infinitesimal area surrounding a point Q_n in the nth half period zone, there is a corresponding infinitesimal area surrounding the point Q_{n-1} in the $(n-1)^{th}$ half-period zone such that

$$Q_nP - Q_{n-1}P = \frac{\lambda}{2}$$

which corresponds to a phase difference of π. Since the areas of the zones are approximately equal, one can have a one-to-one correspondence between points in various zones. Thus, the resultant amplitude at the point P can be written as

$$u(P) = u_1 - u_2 + u_3 - u_4 + \ldots + (-1)^{m+1} u_m + \ldots. \qquad \textbf{(20.3)}$$

where u_n represents the net amplitude produced by the secondary wavelets emanating from the nth zone; the alternate negative and positive signs represent the fact that the result-

ant disturbances produced by two consecutive zones are π out-of-phase with respect to each other. The amplitude produced by a particular zone is proportional to the area of the zone and inversely proportional to the distance of the zone from the point P; further, it also depends on an obliquity factor which is proportional to $\frac{1}{2}(1+\cos\chi)$ where χ is the angle that the normal to the zone makes with the line QP; this obliquity factor comes out automatically from rigorous diffraction theory.* Thus, we may write

$$u_n = \text{constant}\ \frac{A_n}{Q_n P}\frac{(1+\cos\chi)}{2} \tag{20.4}$$

where A_n represents the area of the nth zone. It can be shown that if we use the exact expression for r_n, the area of the zones increase with n; however, this slight increase in the area is exactly compensated by the increased distance of the zone from the point P. In spite of this, the amplitudes $u_1, u_2, u_3 \ldots$ decrease monotonically because of increased obliquity. Thus, we may write

$$u_1 > u_2 > u_3 \ldots \tag{20.5}$$

The series expressed by Eq. (20.3) can be approximately summed due to a method by Schuster. We rewrite Eq. (20.3) as

$$u(P) = \frac{u_1}{2} + \left(\frac{u_1}{2} - u_2 + \frac{u_3}{2}\right) + \left(\frac{u_3}{2} - u_4 + \frac{u_5}{2}\right) + \ldots \tag{20.6}$$

where the last term would either $\frac{1}{2}u_m$ or $\left(\frac{1}{2}u_{m-1} - u_m\right)$ according to m being odd or even. If the obliquity factor is such that

$$u_n > \frac{1}{2}(u_{n-1} + u_{n+1}) \tag{20.7}$$

then the quantities inside the brackets in Eq. (20.6) will be negative; consequently,

$$\left.\begin{aligned} &u(P) < \frac{1}{2}u_1 + \frac{1}{2}u_m \ ;(m \text{ odd}) \\ \text{and}\quad &u(P) < \frac{1}{2}u_1 + \frac{1}{2}u_{m-1} - u_m \approx \frac{u_1}{2} - \frac{u_m}{2} \ ;(m \text{ even}) \end{aligned}\right\} \tag{20.8}$$

where we have assumed that the amplitude of the fields produced by consecutive zones differ only slightly. In order to obtain the upper limits, we rewrite Eq. (20.3) in the form

$$u(P) = u_1 - \frac{u_2}{2} - \left(\frac{u_2}{2} - u_3 + \frac{u_4}{2}\right) - \left(\frac{u_4}{2} - u_5 + \frac{u_6}{2}\right) \tag{20.9}$$

where the last term would now be $\left(-\frac{1}{2}u_{m-1} + u_m\right)$ when m is odd and $-\frac{1}{2}u_m$ when m is even. Since the quantities inside the brackets are negative, we obtain

$$\left.\begin{aligned} &u(P) > u_1 - \frac{u_2}{2} - \frac{u_{m-1}}{2} + u_m \approx \frac{u_1}{2} + \frac{u_m}{2} \\ &\qquad\qquad (\text{when } m \text{ is odd}) \\ \text{and}\quad &u(P) > u_1 - \frac{u_2}{2} - \frac{u_m}{2} \approx \frac{u_1}{2} - \frac{u_m}{2} \\ &\qquad\qquad (\text{when } m \text{ is even}) \end{aligned}\right\} \tag{20.10}$$

Using Eqs. (20.8) and (20.10) we may approximately write

$$\left.\begin{aligned} &u(P) \approx \frac{u_1}{2} + \frac{u_m}{2}\ \ (\text{when } m \text{ is odd}) \\ \text{and}\quad &u(P) \approx \frac{u_1}{2} - \frac{u_m}{2}\ \ (\text{when } m \text{ is even}) \end{aligned}\right\} \tag{20.11}$$

If we can neglect u_m in comparison to u_1 then the Eq. (20.11)** gives the remarkable result that

$$u(P) \approx \frac{u_1}{2} \tag{20.12}$$

implying that *the resultant amplitude produced by the entire wavefront is only one half of the amplitude produced by the first half-period zone.*

20.2.1 Diffraction by a Circular Aperture

We may use the above analysis to study the diffraction of a plane wave by a circular aperture. Let the point P be at a distance d from the circular aperture (see Fig. 20.1). We assume that the radius of the circular aperture, a, can be increased from zero onwards. As a increases, the intensity at the point P would also increase till the circular aperture contains the first half-period zone; this would happen when $a = \sqrt{\lambda d}$. The resultant amplitude at the point P would be u_1 which is twice the value of the amplitude for the unobstructed wavefront [see Eq. (20.12)]. The intensity would therefore be $4I_0$, where I_0 represents the intensity at the point P due to the unobstructed wavefront. If we further increase

* See, e.g., Ref. 20.3.

** If one assumes a form of the obliquity factor as given by Eq. (20.4) then it decreases from 1 to 1/2 as m increases from 1 to ∞; this implies that $|u_m|$ can never be smaller that $\frac{u_1}{2}$. However, when m is large, a slight shift of the point P on the axis will change the amplitude from $\frac{u_1}{2} + \frac{u_m}{2}$ to $\frac{u_1}{2} - \frac{u_m}{2}$, the changes will occur with such great rapidity that one can only observe the average value which will be $\frac{u_1}{2}$.

a then $u(P)$ would start decreasing and when the circular aperture contains the first two half-period zones (which would happen when $a = \sqrt{2\lambda d}$) the resultant amplitude ($= u_1 - u_2$) would be almost zero. *Thus, by increasing the hole diameter, the intensity at the point P decreases almost to zero*. This interesting result is once again due to the validity of the Huygens–Fresnel principle and hence would be valid for sound waves also. We may generalize the above result by noting that if

$$a = \sqrt{(2n+1)\lambda d}\ ; n = 0, 1, 2,\ldots \text{ [maxima]}$$

the aperture will contain an odd number of half-period zones and the intensity will be maximum; on the other hand, if

$$a = \sqrt{2n\lambda d}\ ; n = 1, 2,\ldots \text{ [minima]}$$

the aperture will contain an even number of half-period zones and the intensity will be minimum. In order to have a numerical appreciation, we note that for $d = 50$ cm and $\lambda = 5 \times 10^{-5}$ cm, the radii of the first, second and third zones would be 0.500 mm, 0.707 mm and 0.866 mm, respectively. As a corollary of the above analysis we can consider a circular aperture of a fixed radius a and study the intensity variation along the axis. Whenever the distance

$$d = \frac{a^2}{(2n+1)\lambda}\ ; n = 0, 1, 2,\ldots \text{ (maxima)}$$

the point P (see Fig. 20.1) will correspond to a maximum. Similarly, when

$$d = \frac{a^2}{2n\lambda}\ ; n = 1, 2,\ldots \text{ (minima)}$$

the point P will correspond to a minimum. The intensity distribution on a screen SS' at off-axis points can be approximately calculated by using the half-period zones, but such a calculation is fairly cumbersome. However, from the symmetry of the problem, one can deduce that the diffraction pattern has to be in the form of concentric circular rings with their centres at the point P.

20.2.2 Diffraction by an Opaque Disc–The Poisson Spot

If instead of the circular aperture we have a circular disc [see Fig. 20.3(a)] and if the disc obstructs the first p half-period zones then the field at the point P would be

$$u(P) = u_{p+1} - u_{p+2} + \ldots \approx \frac{u_{p+1}}{2} \qquad \textbf{(20.13)}$$

Thus, we should always obtain a bright spot on the axis behind a circular disc; (the more rigorous theory also predicts the same result — see Sec. 20.4.2). This is called the '**Poisson spot**'. We may mention here that it was in 1816 that the French physicist Augustin Fresnel developed the mathematical theory of diffraction using the wave theory of light. Simeon Poisson, the famous mathematician, used Fresnel's theory to predict a bright spot at the center of the shadow of an opaque disc. Poisson was a great supporter of the corpuscular theory of light and he said that since the bright spot is against common sense, the wave theory must be wrong. Shortly afterwards, Fresnel and Arago carried out the experiment to demonstrate the existence of the Poisson spot [see Fig. 20.3(b)], validating the wave theory.

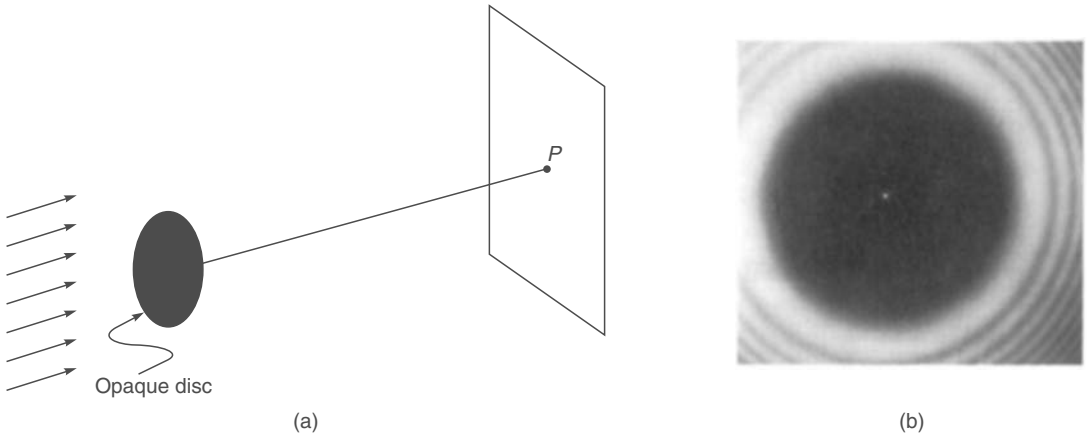

Fig. 20.3 (a) When a plane wave is incident normally on an opaque disc, a bright spot is always formed on an axial point. This spot is known as the Poisson spot. (b) The Poisson spot at the center of the shadow of a one cent coin; the screen is 20 m from the coin and the source of light is also 20 m from the coin. [Photograph reprinted with permission from P.M. Rinard, *Large scale diffraction patterns from circular objects*, American Journal of Physics, Vol 44, 70, 1976; Copyright (1976), American Association of Physics Teachers used with Permission].

20.3 THE ZONE-PLATE LO 1

A beautiful application of the concept of Fresnel half-period zones lies in the construction of the zone-plate which consists of a large number of concentric circles whose radii are proportional to the square root of natural numbers and the alternate annular regions of which are blackened (see Fig. 20.4). Let the radii of the circles be $\sqrt{1}\,K$, $\sqrt{2}\,K$, $\sqrt{3}\,K$, $\sqrt{4}\,K$,… where K is a constant and has the dimension of length. We consider a point P_1 which is at a distance K^2/λ from the zone plate; for this point the blackened rings correspond to the 2nd, 4th, 6th, … half-period zones. Thus, the even zones are obstructed and the resultant amplitude at P_1 [see Fig. 20.5 (a)] will be

$$u_1 + u_3 + u_5 + \ldots \qquad \textbf{(20.14)}$$

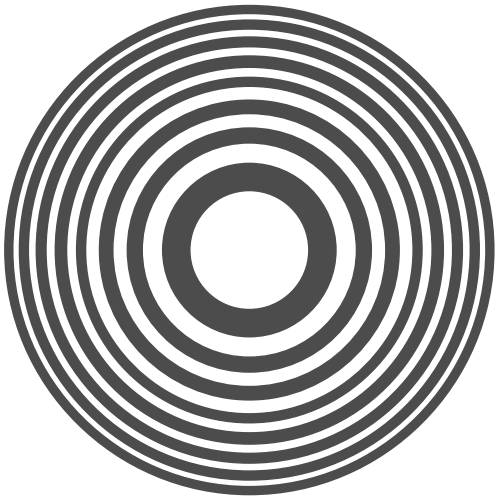

Fig. 20.4 The zone plate.

producing an intense maximum. For the point P_3 (which is at a distance $K^2/3\lambda$) the first blackened ring contains the 4th, 5th, 6th zones, the second blackened ring contains the 10th, 11th and 12th zones, etc.; thus the resultant amplitude would be

$$(u_1 - u_2 + u_3) + (u_7 - u_8 + u_9) + \ldots \qquad \textbf{(20.15)}$$

which would again correspond to a maximum, but it would not be as intense as the point P_1. Between the points P_1 and P_3 there will be a point P_2 (at a distance $K^2/2\lambda$) where the resultant amplitude would be

$$(u_1 - u_2) + (u_5 - u_6) + \ldots \qquad \textbf{(20.16)}$$

implying that corresponding to P_2 the first blackened ring contains the 3rd and 4th half-period zones, etc. Obviously, the point P_2 will correspond to a minimum. Thus, if a plane wave is incident normally on a zone-plate, then the corresponding focal points are at distances

$$\frac{K^2}{\lambda}, \frac{K^2}{3\lambda}, \frac{K^2}{5\lambda}, \ldots \qquad \textbf{(20.17)}$$

from the zone-plate. Elementary calculations will show that the zone-plate suffers from considerable chromatic aberrations [see Problem 20.5].

Example 20.1 Assume a plane wave ($\lambda = 5 \times 10^{-5}$ cm) to be incident on a circular aperture of radius 0.5 mm. We will calculate the positions of the brightest and darkest points on the axis. For the brightest point, the aperture should contain only the first zone and thus we must have (see Fig. 20.1)

$$(0.05)^2 = OP\,(5 \times 10^{-5})$$

Thus OP = 50 cm. Similarly, the darkest point would be at a distance

$$\frac{(0.05)^2}{2 \times 5 \times 10^{-5}} = 25 \text{ cm}$$

Example 20.2 Consider a zone-plate with radii

$$r_n = 0.1\sqrt{n} \text{ cm}$$

For $\lambda = 5 \times 10^{-5}$ cm, we will calculate the positions of various foci. The most intense focal point will be at a distance

$$\frac{r_1^2}{\lambda} = \frac{0.01}{5 \times 10^{-5}} = 200 \text{ cm}$$

The other focal points will be at distances of 200/3, 200/5, 200/7 cm, etc. Between any two consecutive foci there will be dark points on the axis corresponding to which the first circle will contain an even number of half-period zones.

The zone-plate can also be used for imaging points on the axis, e.g., if we have a point source at S then a bright image will be formed at P, where the point P should be such that (see Fig. 20.5(b)):

$$SL + LP - SP = \frac{\lambda}{2} \qquad \textbf{(20.18)}$$

the point L being on the periphery of the first circle of the zone plate [see Fig. 20.5(b)]. If the radius of the first circle is r_1, then

$$\begin{aligned} SL + LP - SP &= \sqrt{a^2 + r_1^2} + \sqrt{b^2 + r_1^2} - (a + b) \\ &\approx a\left[1 + \frac{r_1^2}{2a^2}\right] + b\left[1 + \frac{r_1^2}{2b^2}\right] - (a + b) \\ &\approx \frac{r_1^2}{2}\left(\frac{1}{a} + \frac{1}{b}\right) \end{aligned} \qquad \textbf{(20.19)}$$

Thus Eq. (20.18) becomes

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{f} \qquad \textbf{(20.20)}$$

where $f = r_1^2/\lambda$ represents the focal length. Equation (20.20) resembles the lens law. A very interesting demonstration experiment of the zone-plate can be carried out by using

From the above equation it follows that a phase variation of the type

$$\exp\left[i\frac{k}{2R}(x^2+y^2)\right]$$

(on the x-y plane) represents a *diverging* spherical wave of radius R. If we now compare the above expression with Eqs. (20.36) and (20.37) we can say that the radius of curvature of the phase front of the propagating Gaussian beam is given by Eq. (20.37). Thus as the beam propagates, the phase front which was plane at $z = 0$ becomes curved. In Sec. 27.5, we will use the above analysis to determine conditions for a Gaussian beam to resonate between two spherical mirrors. Here, as a simple application, we will consider a resonator configuration consisting of a plane mirror and a spherical mirror separated by a distance d (see Fig. 20.10); indeed such a configuration is used to produce a single transverse mode oscillation in a ruby laser. On the plane $z = 0$, the phase front is plane and for the beam to resonate, the phase front must have a radius of curvature equal to the radius of curvature of the mirror:

$$R = d\left[1+\frac{\pi^2 w_0^4}{\lambda^2 d^2}\right] \quad \Rightarrow \quad w_0^2 = \frac{\lambda d}{\pi}\sqrt{\left(\frac{R}{d}-1\right)} \qquad \textbf{(20.41)}$$

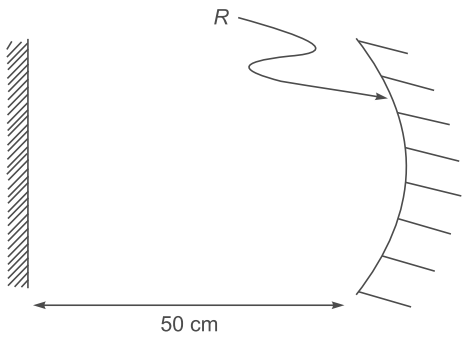

Fig. 20.10 A simple resonator consisting of a plane mirror and a concave mirror of radius R.

Example 20.3 We assume $\lambda \approx 0.6328\ \mu$m, $d \approx 50$ cm and $R \approx 100$ cm; which are typical parameters for a He-Ne laser. Simple calculations will show that $w_0 \approx 0.32$ mm. If we increase R to 200 cm, we will get $w_0 \approx 0.38$ mm. Obviously R should be greater than d. In Sec. 27.5, we will consider the more general case of a Gaussian beam resonating between two mirrors of radii of curvatures R_1 and R_2 as shown in Fig. 27.26.

20.6 DIFFRACTION AT A STRAIGHT EDGE LO 4

Before we discuss the straight edge diffraction pattern, we introduce the Fresnel integrals.

Fresnel Integrals: Fresnel integrals are defined by the following equations:

$$C(\tau) = \int_0^{\tau} \cos\left(\frac{1}{2}\pi u^2\right) du \qquad \textbf{(20.42)}$$

and

$$S(\tau) = \int_0^{\tau} \sin\left(\frac{1}{2}\pi u^2\right) du \qquad \textbf{(20.43)}$$

Since the integrands are even functions of τ, the Fresnel integrals $C(\tau)$ and $S(\tau)$ are odd functions of τ:

$$C(-\tau) = -C(\tau) \quad \text{and} \quad S(-\tau) = -S(\tau) \qquad \textbf{(20.44)}$$

Further, since

$$\int_{-\infty}^{\infty} e^{-\alpha x^2}\, dx = \sqrt{\frac{\pi}{\alpha}} \qquad \textbf{(20.45)}$$

we have

$$\int_{-\infty}^{\infty} e^{i\pi u^2/2}\, du = \sqrt{\frac{\pi}{-i\pi/2}} = \sqrt{2}\, e^{i\pi/4} = (1+i) \qquad \textbf{(20.46)}$$

Now,

$$\int_{-\infty}^{+\infty} \exp\left[i\frac{\pi u^2}{2}\right] du$$

$$= 2\left[\int_0^{\infty} \cos\left(\frac{1}{2}\pi u^2\right) du + i\int_0^{\infty} \sin\left(\frac{1}{2}\pi u^2\right) du\right]$$

$$= 2\,[C(\infty) + i\,S(\infty)]$$

Thus, using Eq. (20.46), we get $C(\infty) = \frac{1}{2} = S(\infty)$.

To summarize, the Fresnel integrals have the following important properties:

$$C(\infty) = S(\infty) = \frac{1}{2};\ C(0) = S(0) = 0 \qquad \textbf{(20.47)}$$

$$C(-\tau) = -C(\tau) \quad \text{and} \quad S(-\tau) = -S(\tau) \qquad \textbf{(20.48)}$$

The values of the Fresnel integrals for typical values of τ are tabulated in Table 20.1.

Table 20.1 Table of Fresnel Integrals*

$$C(\tau)=\int_0^{\tau}\cos\left(\frac{\pi}{2}v^2\right)dv;\quad S(\tau)=\int_0^{\tau}\sin\left(\frac{\pi}{2}v^2\right)dv$$

τ	$C(\tau)$	$S(\tau)$	τ	$C(\tau)$	$S(\tau)$
0.0	0.00000	0.00000	2.6	0.38894	0.54999
0.2	0.19992	0.00419	2.8	0.46749	0.39153
0.4	0.39748	0.03336	3.0	0.60572	0.49631
0.6	0.58110	0.11054	3.2	0.46632	0.59335
0.8	0.72284	0.24934	3.4	0.43849	0.42965
1.0	0.77989	0.43826	3.6	0.58795	0.49231
1.2	0.71544	0.62340	3.8	0.44809	0.56562
1.4	0.54310	0.71353	4.0	0.49843	0.42052
1.6	0.36546	0.63889	4.2	0.54172	0.56320
1.8	0.33363	0.45094	4.4	0.43833	0.46227
2.0	0.48825	0.34342	4.6	0.56724	0.51619
2.2	0.63629	0.45570	4.8	0.43380	0.49675
2.4	0.55496	0.61969	5.0	0.56363	0.49919
			∞	0.5	0.5

Figure 20.11 gives a parametric representation of the Fresnel integrals and is known as the **Cornu's spiral**. The horizontal and the vertical axes represent $C(\tau)$ and $S(\tau)$, respectively and the numbers written on the spiral are the values of τ. For example, as can be seen from the figure, for $\tau = 1.0$, $C(\tau) \approx 0.77989$ and $S(\tau) \approx 0.43826$.

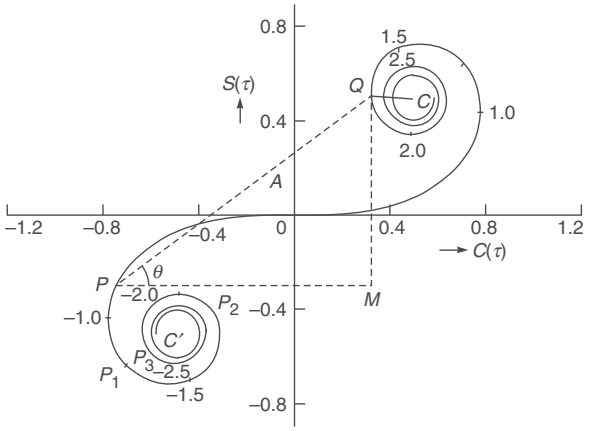

Fig. 20.11 The Cornu's spiral which is a parametric plot of $C(\tau)$ and $S(\tau)$.

We will now use Eq. (20.23) to calculate the diffraction pattern of a plane wave incident normally on a straight edge (see Fig. 20.12). Obviously, there will be no variation of intensity along the x-axis and, therefore, without any loss of generality, we may assume the coordinates of an arbitrary point P (on the screen) to be $(0, y)$, where the origin has been assumed to be on the edge of the geometrical shadow. If the x and y coordinates of an arbitrary point M on the plane of the straight edge are denoted by ξ and η, then

$$\begin{aligned} r = MP &= [\xi^2 + (\eta - y)^2 + d^2]^{1/2} \\ &= d\left[1 + \frac{\xi^2 + (\eta - y)^2}{d^2}\right]^{1/2} \\ &\approx d + \frac{\xi^2 + (\eta - y)^2}{2d} \end{aligned} \qquad \textbf{(20.49)}$$

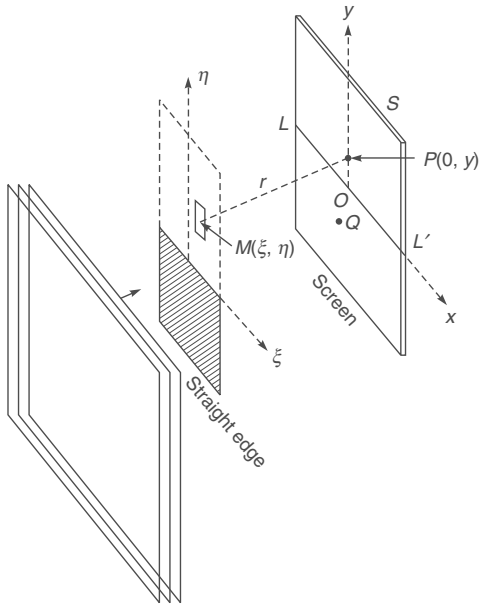

Fig. 20.12 Diffraction of a plane wave incident normally on a straight edge.

where d is the distance between the straight edge and the screen. On substituting the expression for r from Eq. (20.49) in Eq. (20.21), we obtain

$$u(P) \approx -\frac{i}{\lambda}\frac{A}{d}\int_{-\infty}^{\infty} d\xi \int_0^{\infty} d\eta \exp\left[ik\left\{d + \frac{\xi^2 + (\eta - y)^2}{2d}\right\}\right] \qquad \textbf{(20.50)}$$

where, in the denominator of the integrand, we have replaced r by its minimum value**, d. In order to express the above expression in terms of the Fresnel integrals, we introduce two dimensionless variables u and v such that

* Table adapted from Ref. 20.5; a more detailed table (with greater accuracy) has been given there.

** This is justified because in carrying out the integration, only a small region around the point $r = d$ contributes; the contribution due to far-off points is small because of the rapid oscillations of the exponential term in the integrand (see also footnote in Sec. 19.2).

and
$$\left.\begin{aligned}\frac{1}{2}\pi u^2 &= \frac{k}{2d}\xi^2 = \frac{\pi}{\lambda d}\xi^2 \\ \frac{1}{2}\pi v^2 &= \frac{k}{2d}(\eta - y)^2 = \frac{\pi}{\lambda d}(\eta - y)^2\end{aligned}\right\}$$

Thus we may assume u and v to be *defined* by the following equations:

and
$$\left.\begin{aligned}u &= \sqrt{\frac{2}{\lambda d}}\,\xi \\ v &= \sqrt{\frac{2}{\lambda d}}\,(\eta - y)\end{aligned}\right\} \qquad \textbf{(20.51)}$$

With these substitutions, Eq. (20.51) becomes

$$u(P) = -\frac{i}{2}u_0\int_{-\infty}^{+\infty}\exp\left(\frac{i\pi u^2}{2}\right)du\int_{v_0}^{\infty}\exp\left(\frac{i\pi v^2}{2}\right)dv \qquad \textbf{(20.52)}$$

where

$$v_0 = -\sqrt{\frac{2}{\lambda d}}\,y \qquad \textbf{(20.53)}$$

and

$$u_0 = A\,e^{ikd}$$

represents the field at the point P in the absence of the straight edge. In order to calculate the intensity distribution we use the Fresnel integrals; thus,

$$\int_{-\infty}^{+\infty}\exp\left[i\frac{\pi u^2}{2}\right]du = 2\,[C(\infty) + S(\infty)]$$
$$= (1 + i) \qquad \textbf{(20.54)}$$

Further,

$$\int_{v_0}^{\infty}\exp\left(\frac{i\pi v^2}{2}\right)dv = \left[\int_0^{\infty}\cos\left(\frac{\pi}{2}v^2\right)dv - \int_0^{v_0}\cos\left(\frac{\pi}{2}v^2\right)dv\right]$$
$$+ i\left[\int_0^{\infty}\sin\left(\frac{\pi}{2}v^2\right)dv - \int_0^{v_0}\sin\left(\frac{\pi}{2}v^2\right)dv\right]$$
$$= \left[\frac{1}{2} - C(v_0)\right] + i\left[\frac{1}{2} - S(v_0)\right] \qquad \textbf{(20.55)}$$

Substituting in Eq. (20.52), we obtain

$$u(P) = -\frac{i}{2}u_0(1+i)\left[\left\{\frac{1}{2} - C(v_0)\right\} + i\left\{\frac{1}{2} - S(v_0)\right\}\right]$$
$$= \frac{1-i}{2}u_0\left[\left\{\frac{1}{2} - C(v_0)\right\} + i\left\{\frac{1}{2} - S(v_0)\right\}\right] \qquad \textbf{(20.56)}$$

It is of interest to note that a large value of y corresponds to a point which is very far above the edge of the geometrical shadow. For such a point v_0 would tend to $-\infty$ [see Eq. (20.53)] and we would obtain

$$u(P) = \frac{1-i}{2}u_0\left[\left(\frac{1}{2} + \frac{1}{2}\right) + i\left(\frac{1}{2} + \frac{1}{2}\right)\right]$$
$$= u_0 \qquad \textbf{(20.57)}$$

Thus, as expected, the amplitude at such a point is the same as that in the absence of the edge. This also justifies the value of the constant given by Eq. (19.3). On the other hand, when the point P is deep inside the geometrical shadow (i.e., when $y \to -\infty$ and hence $v_0 \to \infty$), we obtain

$$C(v_0) = S(v_0) \to \tfrac{1}{2}$$

giving
$$u(P) \to 0$$

as it should indeed be. The intensity distribution corresponding to Eq. (20.56) would be given by

$$I(P) = \frac{1}{2}I_0\left[\left\{\frac{1}{2} - C(v_0)\right\}^2 + \left\{\frac{1}{2} - S(v_0)\right\}^2\right] \qquad \textbf{(20.58)}$$

If the point P is such that it lies on the edge of the geometrical shadow (i.e., on the line LL' (see Fig. 20.12) then $y = 0$ and hence $v_0 = 0$; thus

$$I(P) = \frac{1}{2}I_0\left[\frac{1}{4} + \frac{1}{4}\right] = \frac{1}{4}I_0 \qquad \textbf{(20.59)}$$

where we have used the fact that $C(0) = S(0) = 0$. Thus the intensity on the edge of the geometrical shadow is $1/4^{\text{th}}$ of the intensity that would have been in the absence of the edge. In order to determine the field at an arbitrary point P, we may use Table 20.1 to calculate the RHS of Eq. (20.58). The intensity variation is plotted in Fig. 20.16 from which one can make the following observations:

(i) Figure 20.13 represents a universal curve, i.e., for given values of λ and d, one simply has to calculate v_0 as the observation point moves along the y-axis. For example, the first three maxima occur at

and
$$\left.\begin{aligned}v_0 &\approx -1.22 \;\text{ with }\; I \approx 1.37\,I_0 \\ v_0 &\approx -2.34 \;\text{ with }\; I \approx 1.20\,I_0 \\ v_0 &\approx -3.08 \;\text{ with }\; I \approx 1.15\,I_0\end{aligned}\right\} \text{ maxima}$$

Similarly, the first three minima occur at

and
$$\left.\begin{aligned}v_0 &\approx -1.87 \;\text{ with }\; I \approx 0.778\,I_0 \\ v_0 &\approx -2.74 \;\text{ with }\; I \approx 0.843\,I_0 \\ v_0 &\approx -3.39 \;\text{ with }\; I \approx 0.872\,I_0\end{aligned}\right\} \text{ minima}$$

Thus, as we go inside the geometrical shadow, the intensity modulation decreases (see Fig. 20.13).

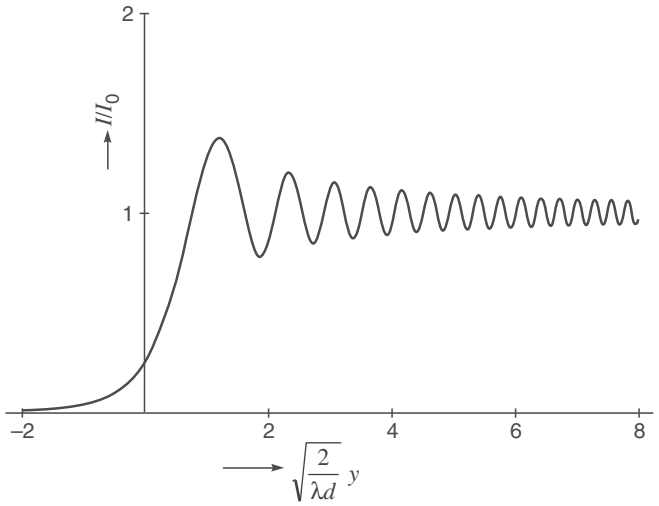

Fig. 20.13 The intensity variation corresponding to the straight edge diffraction pattern.

(ii) For a given experimental set-up, the determination of the positions of maxima and minima is quite straightforward. For example, for $\lambda = 6 \times 10^{-5}$ cm and $d = 120$ cm

$$y = -\sqrt{\frac{\lambda d}{2}}\, v_0 = -0.06\, v_0 \text{ cm}$$

Thus, the first three maxima will occur at

$$y \approx 0.732,\ 1.404 \quad \text{and} \quad 1.848 \text{ mm}$$

respectively. Similarly, the first three minima will occur at

$$y \approx 1.122,\ 1.644 \text{ and } 2.034 \text{ mm}$$

respectively.

(iii) As we go inside the geometrical shadow the intensity monotonically decreases to zero.

(iv) One could have also studied the intensity variation directly from the Cornu's spiral (see Fig. 20.11). This is due to the fact that associated with the Cornu's spiral, we have the following interesting property: let us write

$$[C(\tau_2) - C(\tau_1)] + i[S(\tau_2) - S(\tau_1)] \equiv A\, e^{i\theta} \qquad \textbf{(20.60)}$$

Thus,

$$[C(\tau_2) - C(\tau_1)] = A\cos\theta$$

and

$$[S(\tau_2) - S(\tau_1)] = A\sin\theta$$

Let the points P and Q on the Cornu's spiral (see Fig. 20.11) correspond to $\tau = \tau_1$ and $\tau = \tau_2$ respectively. It is obvious that

$$PM = [C(\tau_2) - C(\tau_1)] = A\cos\theta$$

and

$$QM = [S(\tau_2) - S(\tau_1)] = A\sin\theta$$

Thus the length of the line joining the points P and Q will be A and the angle that the line makes with the abscissa will be θ. In order to use the Cornu's spiral we rewrite Eq. (20.56):

$$u = \frac{1-i}{2}\, u_0 \left[\left\{\frac{1}{2} - C(v_0)\right\} + i\left\{\frac{1}{2} - S(v_0)\right\}\right]$$

Let us first consider a point of observation Q in the geometrical shadow region. Consequently v_0 will be positive. Let the point Q on the spiral (see Fig. 20.11) correspond to $\tau = v_0$. Since the point C in the curve corresponds to $\tau = \infty$, we have

$$\left\{\frac{1}{2} - C(v_0)\right\} + i\left\{\frac{1}{2} - S(v_0)\right\} = (QC)\, e^{i\psi}$$

where ψ is the angle that QC makes with the abscissa [see Eq. (20.60)]. Thus,

$$u(Q) = \frac{1-i}{2}\,(QC)\, e^{i\psi}\, u_0$$

or

$$I(Q) = \frac{1}{2}(QC)^2\, I_0 \qquad \textbf{(20.61)}$$

We can easily see that as the point of observation moves into the shadow region, the value of v_0 increases. Thus the point Q keeps on moving on the spiral towards the point C and the length QC decreases uniformly. Hence in the shadow region the intensity uniformly decreases to zero (see Figs. 20.13 and 20.14).

As we move away from the edge of the geometrical shadow to the illuminated region, the value of v_0 becomes negative and the corresponding point P (on the Cornu's spiral) lies in the third quadrant as shown in Fig. 20.11. The intensity is again given by

$$I(P) = \frac{1}{2}(PC)^2\, I_0$$

As the value of v_0 becomes more and more negative, the length PC keeps on increasing till the point P reaches P_1 which corresponds to $v_0 \approx -1.22$. The intensity at this point is maximum and the length $P_1C \approx 1.66$. Thus, the corresponding intensity is

$$I(P_1) \approx \frac{1}{2}(1.66)^2\, I_0 \approx 1.37\, I_0 \qquad \textbf{(20.62)}$$

As the value of v_0 becomes further negative the length PC starts decreasing till it reaches the point P_2. Thus,